MATHEMATICAL METHODS

Sergei Treil - Linear Algebra Done Wrong

Syllabus-only edition

Only the portions listed in the supplied course outline are retained.

Included syllabus portions

The sequence below follows the course-outline categories. Original textbook pages are preserved; material outside the listed sections is omitted.

Course block	Included textbook material
Foundation of LA	Chapter 1, Sections 1-7
Systems of linear equations	Chapter 2, Sections 1-8
Inner Product Spaces	Chapter 5, Sections 1-5
Spectral Theory	Chapter 4, Sections 1-2
Quadratic Forms	Chapter 7, Sections 1-4
Reference only	Chapter 3, Determinants, Sections 1-6

FOUNDATION OF LINEAR ALGEBRA

Treil Chapter 1, Sections 1-7

Chapter 1

Basic Notions

1. Vector spaces

A vector space V is a (non-empty) collection of objects, called vectors (denoted in this book by lowercase bold letters, like $\mathbf{v}$), along with two operations, addition of vectors and multiplication by a number (scalar)[1], such that the results of both operations are always in V, and the following 8 properties below (the so-called *axioms* of a vector space) hold:

The first 4 properties deal with the addition:

1. Commutativity: $\mathbf{v}+\mathbf{w}=\mathbf{w}+\mathbf{v}$ for all $\mathbf{v},\mathbf{w}\in V$;
2. Associativity: $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$ for all $\mathbf{u},\mathbf{v},\mathbf{w}\in V$;
3. Zero vector: there exists a special vector, denoted by $\mathbf{0}$ such that $\mathbf{v}+\mathbf{0}=\mathbf{v}$ for all $\mathbf{v}\in V$;
4. Additive inverse: For every vector $\mathbf{v}\in V$ there exists a vector $\mathbf{w}\in V$ such that $\mathbf{v}+\mathbf{w}=\mathbf{0}$. Such additive inverse is usually denoted as $-\mathbf{v}$;

   The next two properties concern multiplication:

5. Multiplicative identity: $1\mathbf{v}=\mathbf{v}$ for all $\mathbf{v}\in V$;

A question arises, "How one can memorize the above properties?" And the answer is that one does not need to, see below!

[1] We need some visual distinction between vectors and other objects, so in this book we use bold lowercase letters for vectors and regular lowercase letters for numbers (scalars). In some (more advanced) books Latin letters are reserved for vectors, while Greek letters are used for scalars; in even more advanced texts any letter can be used for anything and the reader must understand from the context what each symbol means. I think it is helpful, especially for a beginner to have some visual distinction between different objects, so a bold lowercase letters will always denote a vector. And on a blackboard an arrow (like in $\vec{v}$) is used to identify a vector.

6. Multiplicative associativity: $(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v})$ for all $\mathbf{v} \in V$ and all scalars α, β;

   And finally, two distributive properties, which connect multiplication and addition:

7. $\alpha(\mathbf{u}+\mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars α;

8. $(\alpha+\beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$ for all $\mathbf{v} \in V$ and all scalars α, β.

Remark. The above properties seem hard to memorize, but it is not necessary. They are simply the familiar rules of algebraic manipulations with numbers, that you know from high school. The only new twist here is that you have to understand what operations you can apply to what objects. You can add vectors, and you can multiply a vector by a number (scalar). Of course, you can do with number all possible manipulations that you have learned before. But, you cannot multiply two vectors, or add a number to a vector.

Remark. It is easy to prove that zero vector $\mathbf{0}$ is unique, and that given $\mathbf{v} \in V$ its additive inverse $-\mathbf{v}$ is also unique.

It is also not hard to show using properties 5, 6 and 8 that $\mathbf{0} = 0\mathbf{v}$ for any $\mathbf{v} \in V$, and that $-\mathbf{v} = (-1)\mathbf{v}$. Note, that to do this one still needs to use other properties of a vector space in the proofs, in particular properties 3 and 4.

If the scalars are the usual real numbers, we call the space V a *real* vector space. If the scalars are the complex numbers, i.e. if we can multiply vectors by complex numbers, we call the space V a *complex* vector space.

Note, that any complex vector space is a real vector space as well (if we can multiply by complex numbers, we can multiply by real numbers), but not the other way around.

If you do not know what a field is, do not worry, since in this book we consider only the case of real and complex spaces.

It is also possible to consider a situation when the scalars are elements of an arbitrary field $\mathbb{F}$. In this case we say that V is a vector space over the field $\mathbb{F}$. Although many of the constructions in the book (in particular, everything in Chapters 1–3) work for general fields, in this text we are considering only real and complex vector spaces.

If we do not specify the set of scalars, or use a letter $\mathbb{F}$ for it, then the results are true for both real and complex spaces. If we need to distinguish real and complex cases, we will explicitly say which case we are considering.

Note, that in the definition of a vector space over an arbitrary field, we *require* the set of scalars to be a field, so we can always divide (without a remainder) by a non-zero scalar. Thus, it is possible to consider vector space over rationals, but not over the integers.

1.1. Examples.

Example. The space $\mathbb{R}^n$ consists of all columns of size n,

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

whose entries are real numbers. Addition and multiplication are defined entrywise, i.e.

$$\alpha \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \\ \alpha v_n \end{pmatrix}, \qquad \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{pmatrix}$$

Example. The space $\mathbb{C}^n$ also consists of columns of size n, only the entries now are complex numbers. Addition and multiplication are defined exactly as in the case of $\mathbb{R}^n$, the only difference is that we can now multiply vectors by *complex* numbers, i.e. $\mathbb{C}^n$ is a *complex* vector space.

Many results in this text are true for both $\mathbb{R}^n$ and $\mathbb{C}^n$. In such cases we will use notation $\mathbb{F}^n$.

Example. The space $M_{m\times n}$ (also denoted as $M_{m,n}$) of $m \times n$ matrices: the addition and multiplication by scalars are defined entrywise. If we allow only real entries (and so only multiplication only by reals), then we have a real vector space; if we allow complex entries and multiplication by complex numbers, we then have a complex vector space.

Formally, we have to distinguish between between real and complex cases, i.e. write something like $M^{\mathbb{R}}_{m,n}$ or $M^{\mathbb{C}}_{m,n}$. However, in most situations there is no difference between real and complex case, and there is no need to specify which case we are considering. If there is a difference we say explicitly which case we are considering.

Remark. As we mentioned above, the axioms of a vector space are just the familiar rules of algebraic manipulations with (real or complex) numbers, so if we put scalars (numbers) for the vectors, all axioms will be satisfied. Thus, the set $\mathbb{R}$ of real numbers is a real vector space, and the set $\mathbb{C}$ of complex numbers is a complex vector space.

More importantly, since in the above examples all vector operations (addition and multiplication by a scalar) are performed entrywise, for these examples the axioms of a vector space are automatically satisfied because they are satisfied for scalars (can you see why?). So, we do not have to

check the axioms, we get the fact that the above examples are indeed vector spaces for free!

The same can be applied to the next example, the coefficients of the polynomials play the role of entries there.

Example. The space $\mathbb{P}_n$ of polynomials of degree at most n, consists of all polynomials p of form

$$p(t) = a_0 + a_1 t + a_2 t^2 + \ldots + a_n t^n,$$

where t is the independent variable. Note, that some, or even all, coefficients a_k can be 0.

In the case of real coefficients a_k we have a real vector space, complex coefficient give us a complex vector space. Again, we will specify whether we treating real or complex case only when it is essential; otherwise everything applies to both cases.

Question: What are zero vectors in each of the above examples?

1.2. Matrix notation. An $m \times n$ matrix is a rectangular array with m rows and n columns. Elements of the array are called *entries* of the matrix.

It is often convenient to denote matrix entries by indexed letters: the first index denotes the number of the row, where the entry is, and the second one is the number of the column. For example

$$A = (a_{j,k})_{j=1,\ k=1}^{m,\ n} = \begin{pmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & \ldots & a_{m,n} \end{pmatrix} \tag{1.1}$$

is a general way to write an $m \times n$ matrix.

Very often for a matrix A the entry in row number j and column number k is denoted by $A_{j,k}$ or $(A)_{j,k}$, and sometimes as in example (1.1) above the same letter but in lowercase is used for the matrix entries.

Given a matrix A, its *transpose* (or transposed matrix) A^T, is defined by transforming the rows of A into the columns. For example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

So, the columns of A^T are the rows of A and vice versa, the rows of A^T are the columns of A.

The formal definition is as follows: $(A^T)_{j,k} = (A)_{k,j}$ meaning that the entry of A^T in the row number j and column number k equals the entry of A in the row number k and column number j.

The transpose of a matrix has a very nice interpretation in terms of linear transformations, namely it gives the so-called *adjoint* transformation. We will study this in detail later, but for now transposition will be just a useful formal operation.

One of the first uses of the transpose is that we can write a column vector $\mathbf{x} \in \mathbb{F}^n$ (recall that $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$) as $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$. If we put the column vertically, it will use significantly more space.

Exercises.

1.1. Let $\mathbf{x} = (1, 2, 3)^T$, $\mathbf{y} = (y_1, y_2, y_3)^T$, $\mathbf{z} = (4, 2, 1)^T$. Compute $2\mathbf{x}$, $3\mathbf{y}$, $\mathbf{x} + 2\mathbf{y} - 3\mathbf{z}$.

1.2. Which of the following sets (with natural addition and multiplication by a scalar) are vector spaces. Justify your answer.

a) The set of all continuous functions on the interval $[0, 1]$;

b) The set of all non-negative functions on the interval $[0, 1]$;

c) The set of all polynomials of degree *exactly* n;

d) The set of all symmetric $n \times n$ matrices, i.e. the set of matrices $A = \{a_{j,k}\}_{j,k=1}^n$ such that $A^T = A$.

1.3. True or false:

a) Every vector space contains a zero vector;

b) A vector space can have more than one zero vector;

c) An $m \times n$ matrix has m rows and n columns;

d) If f and g are polynomials of degree n, then $f + g$ is also a polynomial of degree n;

e) If f and g are polynomials of degree at most n, then $f + g$ is also a polynomial of degree at most n

1.4. Prove that a zero vector $\mathbf{0}$ of a vector space V is unique.

1.5. What matrix is the zero vector of the space $M_{2\times 3}$?

1.6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

1.7. Prove that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v} \in V$.

1.8. Prove that for any vector $\mathbf{v}$ its additive inverse $-\mathbf{v}$ is given by $(-1)\mathbf{v}$.

2. Linear combinations, bases.

Let V be a vector space, and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in V$ be a collection of vectors. A *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ is a sum of form

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_p\mathbf{v}_p = \sum_{k=1}^{p} \alpha_k\mathbf{v}_k.$$

Definition 2.1. A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots \mathbf{v}_n \in V$ is called a basis (for the vector space V) if any vector $\mathbf{v} \in V$ admits a *unique* representation as a linear combination

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n = \sum_{k=1}^{n} \alpha_k\mathbf{v}_k.$$

The coefficients $\alpha_1, \alpha_2, \dots, \alpha_n$ are called *coordinates* of the vector $\mathbf{v}$ (in the basis, or with respect to the basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$).

Another way to say that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis is to say that for any possible choice of the right side $\mathbf{v}$, the equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_m\mathbf{v}_n = \mathbf{v}$ (with unknowns x_k) has a unique solution.

Before discussing any properties of bases[2], let us give a few examples, showing that such objects exist, and that it makes sense to study them.

Example 2.2. In the first example the space V is $\mathbb{F}^n$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. Consider vectors

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

(the vector $\mathbf{e}_k$ has all entries 0 except the entry number k, which is 1). The system of vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is a basis in $\mathbb{F}^n$. Indeed, any vector

$$\mathbf{v} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n$$

can be represented as the linear combination

$$\mathbf{v} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots x_n\mathbf{e}_n = \sum_{k=1}^{n} x_k\mathbf{e}_k$$

[2]the plural for the "basis" is *bases*, the same as the plural for "base"

and this representation is unique. The system $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n \in \mathbb{F}^n$ is called *the standard basis* in $\mathbb{F}^n$

Example 2.3. In this example the space is the space $\mathbb{P}_n$ of the polynomials of degree at most n. Consider vectors (polynomials) $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n \in \mathbb{P}_n$ defined by

$$\mathbf{e}_0 := 1, \quad \mathbf{e}_1 := t, \quad \mathbf{e}_2 := t^2, \quad \mathbf{e}_3 := t^3, \quad \ldots, \quad \mathbf{e}_n := t^n.$$

Clearly, any polynomial p, $p(t) = a_0 + a_1 t + a_2 t^2 + \ldots + a_n t^n$ admits a unique representation

$$p = a_0\mathbf{e}_0 + a_1\mathbf{e}_1 + \ldots + a_n\mathbf{e}_n.$$

So the system $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n \in \mathbb{P}_n$ is a basis in $\mathbb{P}_n$. We will call it the standard basis in $\mathbb{P}_n$.

Remark 2.4. If a vector space V has a basis $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, then any vector $\mathbf{v}$ is uniquely defined by its coefficients in the decomposition $\mathbf{v} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$. So, if we stack the coefficients α_k in a column, we can operate with them as if they were column vectors, i.e. as with elements of $\mathbb{F}^n$ (again here $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$, but everything also works for an abstract field $\mathbb{F}$).

This is a very important remark, that will be used throughout the book. It allows us to translate any statement about the standard column space $\mathbb{F}^n$ to a vector space V with a basis $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$

Namely, if $\mathbf{v} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$ and $\mathbf{w} = \sum_{k=1}^n \beta_k \mathbf{v}_k$, then

$$\mathbf{v} + \mathbf{w} = \sum_{k=1}^n \alpha_k \mathbf{v}_k + \sum_{k=1}^n \beta_k \mathbf{v}_k = \sum_{k=1}^n (\alpha_k + \beta_k)\mathbf{v}_k,$$

i.e. to get the column of coordinates of the sum one just need to add the columns of coordinates of the summands. Similarly, to get the coordinates of $\alpha\mathbf{v}$ we need simply to multiply the column of coordinates of $\mathbf{v}$ by α.

2.1. Generating and linearly independent systems. The definition of a basis says that any vector admits a unique representation as a linear combination. This statement is in fact two statements, namely that the representation exists and that it is unique. Let us analyze these two statements separately.

If we only consider the existence we get the following notion

Definition 2.5. A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p \in V$ is called a generating system (also a *spanning system*, or a *complete system*) in V if any vector $\mathbf{v} \in V$ admits representation as a linear combination

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_p\mathbf{v}_p = \sum_{k=1}^p \alpha_k \mathbf{v}_k.$$

The only difference from the definition of a basis is that we do not assume that the representation above is unique.

The words *generating*, *spanning* and *complete* here are synonyms. I personally prefer the term complete, because of my operator theory background. Generating and spanning are more often used in linear algebra textbooks.

Clearly, any basis is a generating (complete) system. Also, if we have a basis, say $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, and we add to it several vectors, say $\mathbf{v}_{n+1}, \ldots, \mathbf{v}_p$, then the new system will be a generating (complete) system. Indeed, we can represent any vector as a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, and just ignore the new ones (by putting corresponding coefficients $\alpha_k = 0$).

Now, let us turn our attention to the uniqueness. We do not want to worry about existence, so let us consider the zero vector $\mathbf{0}$, which always admits a representation as a linear combination.

Definition. A linear combination $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_p\mathbf{v}_p$ is called *trivial* if $\alpha_k = 0\ \forall k$.

A trivial linear combination is always (for all choices of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$) equal to $\mathbf{0}$, and that is probably the reason for the name.

Definition. A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p \in V$ is called *linearly independent* if only the trivial linear combination ($\sum_{k=1}^{p} \alpha_k\mathbf{v}_k$ with $\alpha_k = \mathbf{0}\ \forall k$) of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ equals $\mathbf{0}$.

In other words, the system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ is linearly independent iff the equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_p\mathbf{v}_p = \mathbf{0}$ (with unknowns x_k) has only trivial solution $x_1 = x_2 = \ldots = x_p = 0$.

If a system is not linearly independent, it is called *linearly dependent*. By negating the definition of linear independence, we get the following

Definition. A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ is called *linearly dependent* if $\mathbf{0}$ can be represented as a nontrivial linear combination, $\mathbf{0} = \sum_{k=1}^{p} \alpha_k\mathbf{v}_k$. Non-trivial here means that at least one of the coefficient α_k is non-zero. This can be (and usually is) written as $\sum_{k=1}^{p} |\alpha_k| \neq 0$.

So, restating the definition we can say, that a system is linearly dependent if and only if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_p$, $\sum_{k=1}^{p} |\alpha_k| \neq 0$ such that

$$\sum_{k=1}^{p} \alpha_k\mathbf{v}_k = \mathbf{0}.$$

An alternative definition (in terms of equations) is that a system $\mathbf{v}_1$, $\mathbf{v}_2, \ldots, \mathbf{v}_p$ is linearly dependent iff the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_p\mathbf{v}_p = \mathbf{0}$$

(with unknowns x_k) has a non-trivial solution. Non-trivial, once again means that at least one of x_k is different from 0, and it can be written as $\sum_{k=1}^{p} |x_k| \neq 0$.

The following proposition gives an alternative description of linearly dependent systems.

Proposition 2.6. *A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in V$ is linearly dependent if and only if one of the vectors $\mathbf{v}_k$ can be represented as a linear combination of the other vectors,*

$$\mathbf{v}_k = \sum_{\substack{j=1 \\ j\neq k}}^{p} \beta_j \mathbf{v}_j. \tag{2.1}$$

Proof. Suppose the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ is linearly dependent. Then there exist scalars α_k, $\sum_{k=1}^{p} |\alpha_k| \neq 0$ such that

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_p \mathbf{v}_p = \mathbf{0}.$$

Let k be the index such that $\alpha_k \neq 0$. Then, moving all terms except $\alpha_k \mathbf{v}_k$ to the right side we get

$$\alpha_k \mathbf{v}_k = -\sum_{\substack{j=1 \\ j\neq k}}^{p} \alpha_j \mathbf{v}_j.$$

Dividing both sides by α_k we get (2.1) with $\beta_j = -\alpha_j/\alpha_k$.

On the other hand, if (2.1) holds, $\mathbf{0}$ can be represented as a non-trivial linear combination

$$\mathbf{v}_k - \sum_{\substack{j=1 \\ j\neq k}}^{p} \beta_j \mathbf{v}_j = \mathbf{0}.$$

□

Obviously, any basis is a linearly independent system. Indeed, if a system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis, $\mathbf{0}$ admits a unique representation

$$\mathbf{0} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_n \mathbf{v}_n = \sum_{k=1}^{n} \alpha_k \mathbf{v}_k.$$

Since the trivial linear combination always gives $\mathbf{0}$, the trivial linear combination must be the *only one* giving $\mathbf{0}$.

So, as we already discussed, if a system is a basis it is a complete (generating) and linearly independent system. The following proposition shows that the converse implication is also true.

In many textbooks a basis is defined as a complete and linearly independent system. By Proposition 2.7 this definition is equivalent to ours.

Proposition 2.7. *A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in V$ is a basis if and only if it is linearly independent and complete (generating).*

Proof. We already know that a basis is always linearly independent and complete, so in one direction the proposition is already proved.

Let us prove the other direction. Suppose a system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is linearly independent and complete. Take an arbitrary vector $\mathbf{v} \in V$. Since the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is linearly complete (generating), $\mathbf{v}$ can be represented as

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_n \mathbf{v}_n = \sum_{k=1}^{n} \alpha_k \mathbf{v}_k.$$

We only need to show that this representation is unique.

Suppose $\mathbf{v}$ admits another representation

$$\mathbf{v} = \sum_{k=1}^{n} \widetilde{\alpha}_k \mathbf{v}_k.$$

Then

$$\sum_{k=1}^{n} (\alpha_k - \widetilde{\alpha}_k)\mathbf{v}_k = \sum_{k=1}^{n} \alpha_k \mathbf{v}_k - \sum_{k=1}^{n} \widetilde{\alpha}_k \mathbf{v}_k = \mathbf{v} - \mathbf{v} = \mathbf{0}.$$

Since the system is linearly independent, $\alpha_k - \widetilde{\alpha}_k = 0 \ \forall k$, and thus the representation $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_n \mathbf{v}_n$ is unique. □

Remark. In many textbooks a basis is defined as a complete and linearly independent system (by Proposition 2.7 this definition is equivalent to ours). Although this definition is more common than one presented in this text, I prefer the latter. It emphasizes the main property of a basis, namely that any vector admits a unique representation as a linear combination.

Proposition 2.8. *Any (finite) generating system contains a basis.*

Proof. Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in V$ is a generating (complete) set. If it is linearly independent, it is a basis, and we are done.

Suppose it is not linearly independent, i.e. it is linearly dependent. Then there exists a vector $\mathbf{v}_k$ which can be represented as a linear combination of the vectors $\mathbf{v}_j$, $j \neq k$.

Since $\mathbf{v}_k$ can be represented as a linear combination of vectors $\mathbf{v}_j$, $j \neq k$, any linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ can be represented as a linear combination of the same vectors without $\mathbf{v}_k$ (i.e. the vectors $\mathbf{v}_j$, $1 \leq j \leq p$, $j \neq k$). So, if we delete the vector $\mathbf{v}_k$, the new system will still be a complete one.

If the new system is linearly independent, we are done. If not, we repeat the procedure.

Repeating this procedure finitely many times we arrive to a linearly independent and complete system, because otherwise we delete all vectors and end up with an empty set.

So, any finite complete (generating) set contains a complete linearly independent subset, i.e. a basis. □

Exercises.

2.1. Find a basis in the space of 3×2 matrices $M_{3\times 2}$.

2.2. True or false:

a) Any set containing a zero vector is linearly dependent

b) A basis must contain $\mathbf{0}$;

c) subsets of linearly dependent sets are linearly dependent;

d) subsets of linearly independent sets are linearly independent;

e) If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n = \mathbf{0}$ then all scalars α_k are zero;

2.3. Recall, that a matrix is called *symmetric* if $A^T = A$. Write down a basis in the space of *symmetric* 2×2 matrices (there are many possible answers). How many elements are in the basis?

2.4. Write down a basis for the space of

a) 3×3 symmetric matrices;

b) $n \times n$ symmetric matrices;

c) $n \times n$ *antisymmetric* ($A^T = -A$) matrices;

2.5. Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1}$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent. **Hint:** Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^{r} \alpha_k\mathbf{v}_k$ and show that the system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

2.6. Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly *independent*?

3. Linear Transformations. Matrix–vector multiplication

The words "transformation", "transform", "mapping", "map", "operator", "function" all denote the same object.

A *transformation* T from a set X to a set Y is a rule that for each argument (input) $x \in X$ assigns a value (output) $y = T(x) \in Y$.

The set X is called the *domain* of T, and the set Y is called the *target space* or *codomain* of T.

We write $T : X \to Y$ to say that T is a transformation with the domain X and the target space Y.

Definition. Let V, W be vector spaces (over the same field $\mathbb{F}$). A transformation $T : V \to W$ is called linear if

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ $\forall \mathbf{u}, \mathbf{v} \in V$;
2. $T(\alpha \mathbf{v}) = \alpha T(\mathbf{v})$ for all $\mathbf{v} \in V$ and for all scalars $\alpha \in \mathbb{F}$.

Properties 1 and 2 together are equivalent to the following one:

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) \qquad \text{for all } \mathbf{u}, \mathbf{v} \in V \quad \text{and for all scalars } \alpha, \beta.$$

3.1. Examples. You dealt with linear transformation before, may be without even suspecting it, as the examples below show.

Example. Differentiation: Let $V = \mathbb{P}_n$ (the set of polynomials of degree at most n), $W = \mathbb{P}_{n-1}$, and let $T : \mathbb{P}_n \to \mathbb{P}_{n-1}$ be the differentiation operator,

$$T(p) := p' \qquad \forall p \in \mathbb{P}_n.$$

Since $(f + g)' = f' + g'$ and $(\alpha f)' = \alpha f'$, this is a linear transformation.

Example. Rotation: in this example $V = W = \mathbb{R}^2$ (the usual coordinate plane), and a transformation $T_\gamma : \mathbb{R}^2 \to \mathbb{R}^2$ takes a vector in $\mathbb{R}^2$ and rotates it counterclockwise by γ radians. Since T_γ rotates the plane as a whole, it rotates as a whole the parallelogram used to define a sum of two vectors (parallelogram law). Therefore the property 1 of linear transformation holds. It is also easy to see that the property 2 is also true.

Example. Reflection: in this example again $V = W = \mathbb{R}^2$, and the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is the reflection in the first coordinate axis, see the fig. It can also be shown geometrically, that this transformation is linear, but we will use another way to show that.

Namely, it is easy to write a formula for T,

$$T\Big(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\Big) = \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix}$$

and from this formula it is easy to check that the transformation is linear.

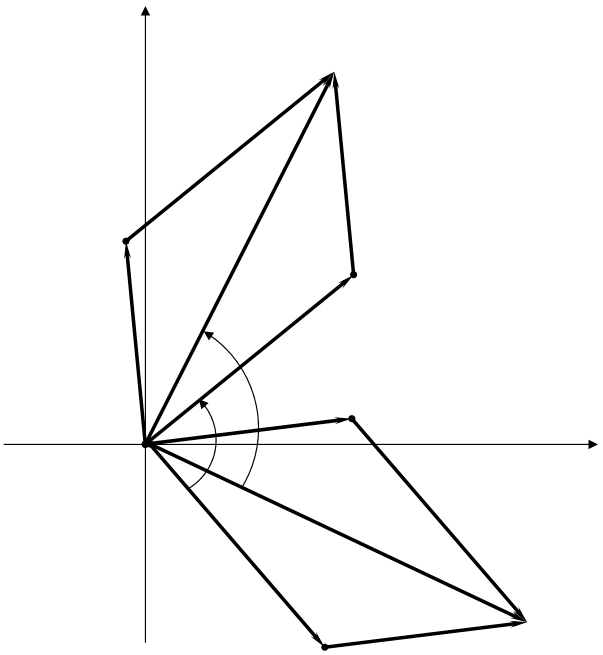


Figure 1. Rotation

Example. Let us investigate linear transformations $T : \mathbb{R} \to \mathbb{R}$. Any such transformation is given by the formula

$$T(x) = ax \qquad \text{where } a = T(1).$$

Indeed,

$$T(x) = T(x \times 1) = xT(1) = xa = ax.$$

So, any linear transformation of $\mathbb{R}$ is just a multiplication by a constant.

3.2. Linear transformations $\mathbb{F}^n \to \mathbb{F}^m$. Matrix–column multiplication. It turns out that a linear transformation $T : \mathbb{F}^n \to \mathbb{F}^m$ also can be represented as a multiplication, not by a scalar, but by a matrix.

Let us see how. Let $T : \mathbb{F}^n \to \mathbb{F}^m$ be a linear transformation. What information do we need to compute $T(\mathbf{x})$ for all vectors $\mathbf{x} \in \mathbb{F}^n$? My claim is that it is sufficient to know how T acts on the standard basis $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ of $\mathbb{F}^n$. Namely, it is sufficient to know n vectors in $\mathbb{F}^m$ (i.e. the vectors of size m),

$$\mathbf{a}_1 = T(\mathbf{e}_1), \quad \mathbf{a}_2 := T(\mathbf{e}_2), \quad \dots\ , \quad \mathbf{a}_n := T(\mathbf{e}_n).$$

Indeed, let

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Then $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \ldots + x_n\mathbf{e}_n = \sum_{k=1}^{n} x_k\mathbf{e}_k$ and

$$T(\mathbf{x}) = T(\sum_{k=1}^{n} x_k\mathbf{e}_k) = \sum_{k=1}^{n} T(x_k\mathbf{e}_k) = \sum_{k=1}^{n} x_k T(\mathbf{e}_k) = \sum_{k=1}^{n} x_k\mathbf{a}_k.$$

So, if we join the vectors (columns) $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ together in a matrix $A = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n]$ ($\mathbf{a}_k$ being the kth column of A, $k = 1, 2, \ldots, n$), this matrix contains all the information about T.

Let us show how one should define the product of a matrix and a vector (column) to represent the transformation T as a product, $T(\mathbf{x}) = A\mathbf{x}$. Let

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & \ldots & a_{m,n} \end{pmatrix}.$$

Recall, that the column number k of A is the vector $\mathbf{a}_k$, i.e.

$$\mathbf{a}_k = \begin{pmatrix} a_{1,k} \\ a_{2,k} \\ \vdots \\ a_{m,k} \end{pmatrix}.$$

Then if we want $A\mathbf{x} = T(\mathbf{x})$ we get

$$A\mathbf{x} = \sum_{k=1}^{n} x_k\mathbf{a}_k = x_1 \begin{pmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{m,1} \end{pmatrix} + x_2 \begin{pmatrix} a_{1,2} \\ a_{2,2} \\ \vdots \\ a_{m,2} \end{pmatrix} + \ldots + x_n \begin{pmatrix} a_{1,n} \\ a_{2,n} \\ \vdots \\ a_{m,n} \end{pmatrix}.$$

So, the matrix–vector multiplication should be performed by the following *column by coordinate* rule:

multiply each column of the matrix by the corresponding coordinate of the vector.

Example.

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 14 \\ 10 \end{pmatrix}.$$

The "column by coordinate" rule is very well adapted for parallel computing. It will be also very important in different theoretical constructions later.

However, when doing computations manually, it is more convenient to compute the result one entry at a time. This can be expressed as the following *row by column* rule:

> To get the entry number k of the result, one need to multiply row number k of the matrix by the vector, that is, if $A\mathbf{x} = \mathbf{y}$, then $y_k = \sum_{j=1}^{n} a_{k,j}x_j$, $k = 1, 2, \dots m$;

here x_j and y_k are coordinates of the vectors $\mathbf{x}$ and $\mathbf{y}$ respectively, and $a_{j,k}$ are the entries of the matrix A.

Example.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 2\cdot 2 + 3\cdot 3 \\ 4\cdot 1 + 5\cdot 2 + 6\cdot 3 \end{pmatrix} = \begin{pmatrix} 14 \\ 32 \end{pmatrix}$$

3.3. Linear transformations and generating sets. As we discussed above, linear transformation T (acting from $\mathbb{F}^n$ to $\mathbb{F}^m$) is completely defined by its values on the standard basis in $\mathbb{F}^n$.

The fact that we consider the standard basis is not essential, one can consider any basis, even any generating (spanning) set. Namely,

> A linear transformation $T : V \to W$ is completely defined by its values on a generating set (in particular by its values on a basis).

So, if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a generating set (in particular, if it is a basis) in V, and T and T_1 are linear transformations $T, T_1 : V \to W$ such that

$$T\mathbf{v}_k = T_1\mathbf{v}_k, \qquad k = 1, 2, \dots, n$$

then $T = T_1$.

The proof of this statement is trivial and left as an exercise.

3.4. Conclusions.

- To get the matrix of a linear transformation $T : \mathbb{F}^n \to \mathbb{F}^m$ one needs to join the vectors $\mathbf{a}_k = T\mathbf{e}_k$ (where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is the standard basis in $\mathbb{F}^n$) into a matrix: kth column of the matrix is $\mathbf{a}_k$, $k = 1, 2, \dots, n$.
- If the matrix A of the linear transformation T is known, then $T(\mathbf{x})$ can be found by the matrix–vector multiplication, $T(\mathbf{x}) = A\mathbf{x}$. To perform matrix–vector multiplication one can use either "column by coordinate" or "row by column" rule.

The latter seems more appropriate for manual computations. The former is well adapted for parallel computers, and will be used in different theoretical constructions.

For a linear transformation $T : \mathbb{F}^n \to \mathbb{F}^m$, its matrix is usually denoted as $[T]$. However, very often people do not distinguish between a linear transformation and its matrix, and use the same symbol for both. When it does not lead to confusion, we will also use the same symbol for a transformation and its matrix.

The notation $T\mathbf{v}$ is often used instead of $T(\mathbf{v})$.

Since a linear transformation is essentially a multiplication, the notation $T\mathbf{v}$ is often used instead of $T(\mathbf{v})$. We will also use this notation. Note that the usual order of algebraic operations apply, i.e. $T\mathbf{v} + \mathbf{u}$ means $T(\mathbf{v}) + \mathbf{u}$, not $T(\mathbf{v} + \mathbf{u})$.

In the matrix vector multiplication using the "row by column" rule be sure that you have the same number of entries in the row and in the column. The entries in the row and in the column should end simultaneously: if not, the multiplication is not defined.

Remark. In the matrix–vector multiplication $A\mathbf{x}$ the number of columns of the matrix A matrix must coincide with the size of the vector $\mathbf{x}$, i.e. a vector in $\mathbb{F}^n$ can only be multiplied by an $m \times n$ matrix.

It makes sense, since an $m \times n$ matrix defines a linear transformation $\mathbb{F}^n \to \mathbb{F}^m$, so vector $\mathbf{x}$ must belong to $\mathbb{F}^n$.

The easiest way to remember this is to remember that if performing multiplication you run out of some elements faster, then the multiplication is not defined. For example, if using the "row by column" rule you run out of row entries, but still have some unused entries in the vector, the multiplication is not defined. It is also not defined if you run out of vector's entries, but still have unused entries in the row.

Remark. One does not have to restrict himself to the case of $\mathbb{F}^n$ with standard basis: everything described in this section works for transformation between arbitrary vector spaces as long as there is a basis in the domain and in the target space. Of course, if one changes a basis, the matrix of the linear transformation will be different. This will be discussed later in Section 8.

Exercises.

3.1. Multiply:

a) $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$;

b) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$;

c) $\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$;

d) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

3.2. Let a linear transformation in $\mathbb{R}^2$ be the reflection in the line $x_1 = x_2$. Find its matrix.

3.3. For each linear transformation below find it matrix

a) $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x, y)^T = (x + 2y, 2x - 5y, 7y)^T$;

b) $T : \mathbb{R}^4 \to \mathbb{R}^3$ defined by $T(x_1, x_2, x_3, x_4)^T = (x_1+x_2+x_3+x_4, x_2-x_4, x_1+3x_2+6x_4)^T$;

c) $T : \mathbb{P}_n \to \mathbb{P}_n$, $Tf(t) = f'(t)$ (find the matrix with respect to the standard basis $1, t, t^2, \dots, t^n$);

d) $T : \mathbb{P}_n \to \mathbb{P}_n$, $Tf(t) = 2f(t) + 3f'(t) - 4f''(t)$ (again with respect to the standard basis $1, t, t^2, \dots, t^n$).

3.4. Find 3×3 matrices representing the transformations of $\mathbb{R}^3$ which:

a) project every vector onto x-y plane;

b) reflect every vector through x-y plane;

c) rotate the x-y plane through $30°$, leaving z-axis alone.

3.5. Let A be a linear transformation. If $\mathbf{z}$ is the center of the straight interval $[\mathbf{x}, \mathbf{y}]$, show that $A\mathbf{z}$ is the center of the interval $[A\mathbf{x}, A\mathbf{y}]$. **Hint:** What does it mean that $\mathbf{z}$ is the center of the interval $[\mathbf{x}, \mathbf{y}]$?

3.6. The set $\mathbb{C}$ of complex numbers can be canonically identified with the space $\mathbb{R}^2$ by treating each $z = x + iy \in \mathbb{C}$ as a column $(x, y)^T \in \mathbb{R}^2$.

a) Treating $\mathbb{C}$ as a complex vector space, show that the multiplication by $\alpha = a + ib \in \mathbb{C}$ is a linear transformation in $\mathbb{C}$. What is its matrix?

b) Treating $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ show that the multiplication by $\alpha = a + ib$ defines a linear transformation there. What is its matrix?

c) Define $T(x+iy) = 2x-y+i(x-3y)$. Show that this transformation is not a linear transformation in the complex vectors space $\mathbb{C}$, but if we treat $\mathbb{C}$ as the real vector space $\mathbb{R}^2$ then it is a linear transformation there (i.e. that T is a *real linear* but not a *complex linear* transformation).
Find the matrix of the real linear transformation T.

3.7. Show that any linear transformation in $\mathbb{C}$ (treated as a complex vector space) is a multiplication by $\alpha \in \mathbb{C}$.

4. Linear transformations as a vector space

What operations can we perform with linear transformations? We can always multiply a linear transformation for a scalar, i.e. if we have a linear

transformation $T : V \to W$ and a scalar α we can define a new transformation αT by

$$(\alpha T)\mathbf{v} = \alpha(T\mathbf{v}) \qquad \forall \mathbf{v} \in V.$$

It is easy to check that αT is also a linear transformation:

$$\begin{aligned}(\alpha T)(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2) &= \alpha(T(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2)) && \text{by the definition of } \alpha T \\ &= \alpha(\alpha_1 T\mathbf{v}_1 + \alpha_2 T\mathbf{v}_2) && \text{by the linearity of } T \\ &= \alpha_1\alpha T\mathbf{v}_1 + \alpha_2\alpha T\mathbf{v}_2 = \alpha_1(\alpha T)\mathbf{v}_1 + \alpha_2(\alpha T)\mathbf{v}_2\end{aligned}$$

If T_1 and T_2 are linear transformations with the same domain and target space ($T_1 : V \to W$ and $T_2 : V \to W$, or in short $T_1, T_2 : V \to W$), then we can add these transformations, i.e. define a new transformation $T = (T_1 + T_2) : V \to W$ by

$$(T_1 + T_2)\mathbf{v} = T_1\mathbf{v} + T_2\mathbf{v} \qquad \forall \mathbf{v} \in V.$$

It is easy to check that the transformation $T_1 + T_2$ is a linear one, one just needs to repeat the above reasoning for the linearity of αT.

So, if we fix vector spaces V and W and consider the collection of all linear transformations from V to W (let us denote it by $\mathcal{L}(V, W)$), we can define 2 operations on $\mathcal{L}(V, W)$: multiplication by a scalar and addition. It can be easily shown that these operations satisfy the axioms of a vector space, defined in Section 1.

This should come as no surprise for the reader, since axioms of a vector space essentially mean that operation on vectors follow standard rules of algebra. And the operations on linear transformations are defined as to satisfy these rules!

As an illustration, let us write down a formal proof of the first distributive law (axiom 7) of a vector space. We want to show that $\alpha(T_1 + T_2) = \alpha T_1 + \alpha T_2$. For any $\mathbf{v} \in V$

$$\begin{aligned}\alpha(T_1 + T_2)\mathbf{v} &= \alpha((T_1 + T_2)\mathbf{v}) && \text{by the definition of multiplication} \\ &= \alpha(T_1\mathbf{v} + T_2\mathbf{v}) && \text{by the definition of the sum} \\ &= \alpha T_1\mathbf{v} + \alpha T_2\mathbf{v} && \text{by Axiom 7 for } W \\ &= (\alpha T_1 + \alpha T_2)\mathbf{v} && \text{by the definition of the sum}\end{aligned}$$

So indeed $\alpha(T_1 + T_2) = \alpha T_1 + \alpha T_2$.

Remark. Linear operations (addition and multiplication by a scalar) on linear transformations $T : \mathbb{F}^n \to \mathbb{F}^m$ correspond to the respective operations on their matrices. Since we know that the set of $m \times n$ matrices is a vector space, this immediately implies that $\mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ is a vector space.

We presented the abstract proof above, first of all because it work for general spaces, for example, for spaces without a basis, where we cannot

work with coordinates. Secondly, the reasonings similar to the abstract one presented here, are used in many places, so the reader will benefit from understanding it.

And as the reader gains some mathematical sophistication, he/she will see that this abstract reasoning is indeed a very simple one, that can be performed almost automatically.

5. Composition of linear transformations and matrix multiplication.

5.1. Definition of the matrix multiplication. Knowing matrix–vector multiplication, one can easily guess what is the natural way to define the product AB of two matrices: Let us multiply by A each column of B (matrix-vector multiplication) and join the resulting column-vectors into a matrix. Formally,

if $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_r$ are the columns of B, then $A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_r$ are the columns of the matrix AB.

Recalling the *row by column* rule for the matrix–vector multiplication we get the following *row by column rule for the matrices*

the entry $(AB)_{j,k}$ (the entry in the row j and column k) of the product AB is defined by $(AB)_{j,k} = (\text{row } \#j \text{ of } A) \cdot (\text{column } \#k \text{ of } B)$

Formally it can be rewritten as

$$(AB)_{j,k} = \sum_l a_{j,l} b_{l,k},$$

if $a_{j,k}$ and $b_{j,k}$ are entries of the matrices A and B respectively.

I intentionally did not speak about sizes of the matrices A and B, but if we recall the row by column rule for the matrix–vector multiplication, we can see that in order for the multiplication to be defined, the size of a row of A should be equal to the size of a column of B.

In other words the product AB is defined if and only if A is an $m \times n$ and B is $n \times r$ matrix.

5.2. Motivation: composition of linear transformations. One can ask yourself here: Why are we using such a complicated rule of multiplication? Why don't we just multiply matrices entrywise?

And the answer is, that the multiplication, as it is defined above, arises naturally from the composition of linear transformations.

Suppose we have two linear transformations, $T_1 : \mathbb{F}^n \to \mathbb{F}^m$ and $T_2 : \mathbb{F}^r \to \mathbb{F}^n$. Define the composition $T = T_1 \circ T_2$ of the transformations T_1, T_2 as

$$T(\mathbf{x}) = T_1(T_2(\mathbf{x})) \qquad \forall x \in \mathbb{F}^r.$$

We will usually identify a linear transformation and its matrix, but in the next few paragraphs we will distinguish them

Note that $T_2(\mathbf{x}) \in \mathbb{F}^n$. Since $T_1 : \mathbb{F}^n \to \mathbb{F}^m$, the expression $T_1(T_2(\mathbf{x}))$ is well defined and the result belongs to $\mathbb{F}^m$. So, $T : \mathbb{F}^r \to \mathbb{F}^m$.

It is easy to show that T is a linear transformation (exercise), so it is defined by an $m \times r$ matrix. How one can find this matrix, knowing the matrices of T_1 and T_2?

Let A be the matrix of T_1 and B be the matrix of T_2. As we discussed in the previous section, the columns of T are vectors $T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_r)$, where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r$ is the standard basis in $\mathbb{F}^r$. For $k = 1, 2, \dots, r$ we have

$$T(\mathbf{e}_k) = T_1(T_2(\mathbf{e}_k)) = T_1(B\mathbf{e}_k) = T_1(\mathbf{b}_k) = A\mathbf{b}_k$$

(operators T_2 and T_1 are simply the multiplication by B and A respectively).

So, the columns of the matrix of T are $A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_r$, and that is exactly how the matrix AB was defined!

Let us return to identifying again a linear transformation with its matrix. Since the matrix multiplication agrees with the composition, we can (and will) write T_1T_2 instead of $T_1 \circ T_2$ and $T_1T_2\mathbf{x}$ instead of $T_1(T_2(\mathbf{x}))$.

Note: order of transformations!

Note that in the composition T_1T_2 the transformation T_2 is applied first! The way to remember this is to see that in $T_1T_2\mathbf{x}$ the transformation T_2 meets $\mathbf{x}$ fist.

Remark. There is another way of checking the dimensions of matrices in a product, different form the row by column rule: for a composition T_1T_2 to be defined it is necessary that $T_2\mathbf{x}$ belongs to the domain of T_1. If T_2 acts from some space, say $\mathbb{F}^r$ to $\mathbb{F}^n$, then T_1 must act from $\mathbb{F}^n$ to some space, say $\mathbb{F}^m$. So, in order for T_1T_2 to be defined the matrices of T_1 and T_2 should be of sizes $m \times n$ and $n \times r$ respectively—the same condition as obtained from the *row by column* rule.

Example. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the reflection in the line $x_1 = 3x_2$. It is a linear transformation, so let us find its matrix. To find the matrix, we need to compute $T\mathbf{e}_1$ and $T\mathbf{e}_2$. However, the direct computation of $T\mathbf{e}_1$ and $T\mathbf{e}_2$ involves significantly more trigonometry than a sane person is willing to remember.

An easier way to find the matrix of T is to represent it as a composition of simple linear transformation. Namely, let γ be the angle between the x_1 axis and the line $x_1 = 3x_2$, and let T_0 be the reflection in the x_1-axis. Then to get the reflection T we can first rotate the plane by the angle $-\gamma$, moving the line $x_1 = 3x_2$ to the x_1-axis, then reflect everything in the x_1

axis, and then rotate the plane by γ, taking everything back. Formally it can be written as

$$T = R_\gamma T_0 R_{-\gamma}$$

(note the order of terms!), where R_γ is the rotation by γ. The matrix of T_0 is easy to compute,

$$T_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the rotation matrices are known

$$R_\gamma = \begin{pmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma, \end{pmatrix},$$
$$R_{-\gamma} = \begin{pmatrix} \cos(-\gamma) & -\sin(-\gamma) \\ \sin(-\gamma) & \cos(-\gamma), \end{pmatrix} = \begin{pmatrix} \cos\gamma & \sin\gamma \\ -\sin\gamma & \cos\gamma, \end{pmatrix}$$

To compute $\sin\gamma$ and $\cos\gamma$ take a vector in the line $x_1 = 3x_2$, say a vector $(3,1)^T$. Then

$$\cos\gamma = \frac{\text{first coordinate}}{\text{length}} = \frac{3}{\sqrt{3^2+1^2}} = \frac{3}{\sqrt{10}}$$

and similarly

$$\sin\gamma = \frac{\text{second coordinate}}{\text{length}} = \frac{1}{\sqrt{3^2+1^2}} = \frac{1}{\sqrt{10}}$$

Gathering everything together we get

$$T = R_\gamma T_0 R_{-\gamma} = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\frac{1}{\sqrt{10}}\begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix}$$
$$= \frac{1}{10}\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix}$$

It remains only to perform matrix multiplication here to get the final result. □

5.3. Properties of matrix multiplication. Matrix multiplication enjoys a lot of properties, familiar to us from high school algebra:

1. Associativity: $A(BC) = (AB)C$, provided that either left or right side is well defined; we therefore can (and will) simply write ABC in this case.
2. Distributivity: $A(B+C) = AB + AC$, $(A+B)C = AC + BC$, provided either left or right side of each equation is well defined.
3. One can take scalar multiplies out: $A(\alpha B) = (\alpha A)B = \alpha(AB) = \alpha AB$.

These properties are easy to prove. One should prove the corresponding properties for linear transformations, and they almost trivially follow from the definitions. The properties of linear transformations then imply the properties for the matrix multiplication.

The new twist here is that the commutativity fails:

> matrix multiplication is non-commutative, i.e. generally for matrices $AB \neq BA$.

One can see easily it would be unreasonable to expect the commutativity of matrix multiplication. Indeed, let A and B be matrices of sizes $m \times n$ and $n \times r$ respectively. Then the product AB is well defined, but if $m \neq r$, BA is not defined.

Even when both products are well defined, for example, when A and B are $n \times n$ (square) matrices, the multiplication is still non-commutative. If we just pick the matrices A and B at random, the chances are that $AB \neq BA$: we have to be very lucky to get $AB = BA$.

5.4. Transposed matrices and multiplication. Given a matrix A, its *transpose* (or transposed matrix) A^T is defined by transforming the rows of A into the columns. For example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

So, the columns of A^T are the rows of A and vice versa, the rows of A^T are the columns of A.

The formal definition is as follows: $(A^T)_{j,k} = (A)_{k,j}$ meaning that the entry of A^T in the row number j and column number k equals the entry of A in the row number k and row number j.

The transpose of a matrix has a very nice interpretation in terms of linear transformations, namely it gives the so-called *adjoint* transformation. We will study this in detail later, but for now transposition will be just a useful formal operation.

One of the first uses of the transpose is that we can write a column vector $\mathbf{x} \in \mathbb{F}^n$ as $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$. If we put the column vertically, it will use significantly more space.

A simple analysis of the row by columns rule shows that

$$(AB)^T = B^T A^T,$$

i.e. when you take the transpose of the product, you change the order of the terms.

5.5. Trace and matrix multiplication. For a square $(n \times n)$ matrix $A = (a_{j,k})$ its trace (denoted by $\operatorname{trace} A$) is the sum of the diagonal entries

$$\operatorname{trace} A = \sum_{k=1}^{n} a_{k,k}.$$

Theorem 5.1. *Let A and B be matrices of size $m \times n$ and $n \times m$ respectively (so the both products AB and BA are well defined). Then*

$$\operatorname{trace}(AB) = \operatorname{trace}(BA)$$

We leave the proof of this theorem as an exercise, see Problem 5.6 below. There are essentially two ways of proving this theorem. One is to compute the diagonal entries of AB and of BA and compare their sums. This method requires some proficiency in manipulating sums in $\sum$ notation.

If you are not comfortable with algebraic manipulations, there is another way. We can consider two linear transformations, T and T_1, acting from $M_{n\times m}$ to $\mathbb{F} = \mathbb{F}^1$ defined by

$$T(X) = \operatorname{trace}(AX), \qquad T_1(X) = \operatorname{trace}(XA)$$

To prove the theorem it is sufficient to show that $T = T_1$; the equality for $X = B$ gives the theorem.

Since a linear transformation is completely defined by its values on a generating system, we need just to check the equality on some simple matrices, for example on matrices $X_{j,k}$, which has all entries 0 except the entry 1 in the intersection of jth column and kth row.

Exercises.

5.1. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \; B = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & -2 \end{pmatrix}, \; C = \begin{pmatrix} 1 & -2 & 3 \\ -2 & 1 & -1 \end{pmatrix}, \; D = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

a) Mark all the products that are defined, and give the dimensions of the result: AB, BA, ABC, ABD, BC, BC^T, B^TC, DC, D^TC^T.

b) Compute AB, $A(3B + C)$, B^TA, $A(BD)$, $(AB)D$.

5.2. Let T_γ be the matrix of rotation by γ in $\mathbb{R}^2$. Check by matrix multiplication that $T_\gamma T_{-\gamma} = T_{-\gamma} T_\gamma = I$

5.3. Multiply two rotation matrices T_α and T_β (it is a rare case when the multiplication is commutative, i.e. $T_\alpha T_\beta = T_\beta T_\alpha$, so the order is not essential). Deduce formulas for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$ from here.

5.4. Find the matrix of the orthogonal projection in $\mathbb{R}^2$ onto the line $x_1 = -2x_2$. **Hint:** What is the matrix of the projection onto the coordinate axis x_1?

5.5. Find linear transformations $A, B : \mathbb{R}^2 \to \mathbb{R}^2$ such that $AB = \mathbf{0}$ but $BA \neq \mathbf{0}$.

5.6. Prove Theorem 5.1, i.e. prove that $\operatorname{trace}(AB) = \operatorname{trace}(BA)$.

5.7. Construct a non-zero matrix A such that $A^2 = \mathbf{0}$.

5.8. Find the matrix of the reflection through the line $y = -2x/3$. Perform all the multiplications.

6. Invertible transformations and matrices. Isomorphisms

6.1. Identity transformation and identity matrix. Among all linear transformations, there is a special one, the identity transformation (operator) I, $I\mathbf{x} = \mathbf{x}$, $\forall \mathbf{x}$.

To be precise, there are infinitely many identity transformations: for any vector space V, there is the identity transformation $I = I_V : V \to V$, $I_V \mathbf{x} = \mathbf{x}$, $\forall \mathbf{x} \in V$. However, when it is does not lead to the confusion we will use the same symbol I for all identity operators (transformations). We will use the notation I_V only we want to emphasize in what space the transformation is acting.

Often, the symbol E is used in Linear Algebra textbooks for the identity matrix. I prefer I, since it is used in operator theory.

Clearly, if $I : \mathbb{F}^n \to \mathbb{F}^n$ is the identity transformation in $\mathbb{F}^n$, its matrix is the $n \times n$ matrix

$$I = I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

(1 on the main diagonal and 0 everywhere else). When we want to emphasize the size of the matrix, we use the notation I_n; otherwise we just use I.

Clearly, for an arbitrary linear transformation A, the equalities

$$AI = A, \qquad IA = A$$

hold (whenever the product is defined).

6.2. Invertible transformations.

Definition. Let $A : V \to W$ be a linear transformation. We say that the transformation A is *left invertible* if there exist a linear transformation $B : W \to V$ such that

$$BA = I \qquad (I = I_V \text{ here}).$$

The transformation A is called *right invertible* if there exists a linear transformation $C : W \to V$ such that

$$AC = I \qquad (\text{here } I = I_W).$$

The transformations B and C are called *left* and *right inverses* of A. Note, that we did not assume the uniqueness of B or C here, and generally left and right inverses are not unique.

Definition. A linear transformation $A : V \to W$ is called *invertible* if it is both right and left invertible.

Theorem 6.1. *If a linear transformation $A : V \to W$ is invertible, then its left and right inverses B and C are unique and coincide.*

Corollary. *A transformation $A : V \to W$ is invertible if and only if there exists a unique linear transformation (denoted A^{-1}), $A^{-1} : W \to V$ such that*

$$A^{-1}A = I_V, \qquad AA^{-1} = I_W.$$

Very often this property is used as the definition of an invertible transformation

The transformation A^{-1} is called the *inverse* of A.

Proof of Theorem 6.1. Let $BA = I$ and $AC = I$. Then

$$BAC = B(AC) = BI = B.$$

On the other hand

$$BAC = (BA)C = IC = C,$$

and therefore $B = C$.

Suppose for some transformation B_1 we have $B_1A = I$. Repeating the above reasoning with B_1 instead of B we get $B_1 = C$. Therefore the left inverse B is unique. The uniqueness of C is proved similarly. □

Definition. A matrix is called *invertible* (resp. *left invertible, right invertible*) if the corresponding linear transformation is invertible (resp. left invertible, right invertible).

Theorem 6.1 asserts that a matrix A is invertible if there exists a unique matrix A^{-1} such that $A^{-1}A = I$, $AA^{-1} = I$. The matrix A^{-1} is called (surprise) the *inverse* of A.

Examples.

1. The identity transformation (matrix) is invertible, $I^{-1} = I$;
2. The rotation R_γ

$$R_\gamma = \begin{pmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{pmatrix}$$

is invertible, and the inverse is given by $(R_\gamma)^{-1} = R_{-\gamma}$. This equality is clear from the geometric description of R_γ, and it also can be checked by the matrix multiplication;

3. The column $(1,1)^T$ is left invertible but not right invertible. One of the possible left inverses in the row $(1/2, 1/2)$.

   To show that this matrix is not right invertible, we just notice that there are more than one left inverse. **Exercise:** describe all left inverses of this matrix.

4. The row $(1,1)$ is right invertible, but not left invertible. The column $(1/2, 1/2)^T$ is a possible right inverse.

An invertible matrix must be square (to be proved later)

Remark 6.2. An invertible matrix *must* be square ($n \times n$). Moreover, if a square matrix A has either left or right inverse, it is invertible. So, it is sufficient to check only one of the identities $AA^{-1} = I$, $A^{-1}A = I$.

This fact will be proved later. Until we prove this fact, we will not use it. I presented it here only to stop students from trying wrong directions.

6.2.1. *Properties of the inverse transformation.*

Inverse of a product: $(AB)^{-1} = B^{-1}A^{-1}$. Note the change of order

Theorem 6.3 (Inverse of the product)**.** *If linear transformations A and B are invertible (and such that the product AB is defined), then the product AB is invertible and*

$$(AB)^{-1} = B^{-1}A^{-1}$$

(note the change of the order!)

Proof. Direct computation shows:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

and similarly

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

□

Remark 6.4. The invertibility of the product AB does not imply the invertibility of the factors A and B (can you think of an example?). However, if one of the factors (either A or B) and the product AB are invertible, then the second factor is also invertible.

We leave the proof of this fact as an exercise.

Theorem 6.5 (Inverse of A^T)**.** *If a matrix A is invertible, then A^T is also invertible and*

$$(A^T)^{-1} = (A^{-1})^T$$

Proof. Using $(AB)^T = B^T A^T$ we get

$$(A^{-1})^T A^T = (AA^{-1})^T = I^T = I,$$

and similarly

$$A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I.$$

□

And finally, if A is invertible, then A^{-1} is also invertible, $(A^{-1})^{-1} = A$.

So, let us summarize the main properties of the inverse:

1. If A is invertible, then A^{-1} is also invertible, $(A^{-1})^{-1} = A$;
2. If A and B are invertible and the product AB is defined, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is invertible, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

6.3. Isomorphism. Isomorphic spaces. An invertible linear transformation $A : V \to W$ is called an *isomorphism.* We did not introduce anything new here, it is just another name for the object we already studied.

Two vector spaces V and W are called *isomorphic* (denoted $V \cong W$) if there is an isomorphism $A : V \to W$.

Isomorphic spaces can be considered as different representation of the same space, meaning that all properties and constructions involving vector space operations are preserved under isomorphism.

The theorem below illustrates this statement.

Theorem 6.6. *Let $A : V \to W$ be an isomorphism, and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in V. Then the system $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ is a basis in W.*

We leave the proof of the theorem as an exercise.

Remark. In the above theorem one can replace "basis" by "linearly independent", or "generating", or "linearly dependent"—all these properties are preserved under isomorphisms.

Remark. If A is an isomorphism, then so is A^{-1}. Therefore in the above theorem we can state that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis if and only if $A\mathbf{v}_1, A\mathbf{v}_2$, $\dots, A\mathbf{v}_n$ is a basis.

The converse to the Theorem 6.6 is also true

Theorem 6.7. *Let $A : V \to W$ be a linear map,and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ be bases in V and W respectively. If $A\mathbf{v}_k = \mathbf{w}_k$, $k = 1, 2, \dots, n$, then A is an isomorphism.*

Proof. Define the inverse transformation A^{-1} by $A^{-1}\mathbf{w}_k = \mathbf{v}_k$, $k = 1, 2, \dots, n$ (as we know, a linear transformation is defined by its values on a basis). □

Examples.

1. Let $A : \mathbb{F}^{n+1} \to \mathbb{P}_n^{\mathbb{F}}$ ($\mathbb{P}_n^{\mathbb{F}}$ is the set of polynomials $\sum_{k=0}^{n} a_k t^k$, $\alpha_k \in \mathbb{F}$ of degree at most n) is defined by

$$A\mathbf{e}_1 = 1,\ A\mathbf{e}_2 = t, \dots, A\mathbf{e}_n = t^{n-1},\ A\mathbf{e}_{n+1} = t^n$$

By Theorem 6.7 A is an isomorphism, so $\mathbb{P}_n^{\mathbb{F}} \cong \mathbb{F}^{n+1}$.

2. Let V be a vector space (over $\mathbb{F}$) with a basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Define transformation $A : \mathbb{F}^n \to V$ by
$$A\mathbf{e}_k = \mathbf{v}_k, \qquad k = 1, 2, \dots, n,$$
where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is the standard basis in $\mathbb{F}^n$. Again by Theorem 6.7 A is an isomorphism, so $V \cong \mathbb{F}^n$.

Any real vector space with a basis is isomorphic to $\mathbb{R}^n$ (for some n). Similarly, any complex vector space with a basis is isomorphic to $\mathbb{C}^n$.

3. The space $M_{2\times 3}^{\mathbb{F}}$ of 2×3 matrices with entries in $\mathbb{F}$ is isomorphic to $\mathbb{R}^6$;

4. More generally, $M_{m\times n}^{\mathbb{F}} \cong \mathbb{F}^{m\cdot n}$

6.4. Invertibility and equations.

Doesn't this remind you of a basis?

Theorem 6.8. *Let $A : X \to Y$ be a linear transformation. Then A is invertible if and only if for any right side $\mathbf{b} \in Y$ the equation*
$$A\mathbf{x} = \mathbf{b}$$
has a unique solution $\mathbf{x} \in X$.

Proof. Suppose A is invertible. Then $\mathbf{x} = A^{-1}\mathbf{b}$ solves the equation $A\mathbf{x} = \mathbf{b}$. To show that the solution is unique, suppose that for some other vector $\mathbf{x}_1 \in X$
$$A\mathbf{x}_1 = \mathbf{b}$$
Multiplying this identity by A^{-1} from the left we get
$$A^{-1}A\mathbf{x}_1 = A^{-1}\mathbf{b},$$
and therefore $\mathbf{x}_1 = A^{-1}\mathbf{b} = \mathbf{x}$. Note that both identities, $AA^{-1} = I$ and $A^{-1}A = I$ were used here.

Let us now suppose that the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x}$ for any $\mathbf{b} \in Y$. Let us use symbol $\mathbf{y}$ instead of $\mathbf{b}$. We know that given $\mathbf{y} \in Y$ the equation
$$A\mathbf{x} = \mathbf{y}$$
has a unique solution $\mathbf{x} \in X$. Let us call this solution $B(\mathbf{y})$.

Note that $B(\mathbf{y})$ is defined for all $\mathbf{y} \in Y$, so we defined a transformation $B : Y \to X$.

Let us check that B is a linear transformation. We need to show that $B(\alpha\mathbf{y}_1+\beta\mathbf{y}_2) = \alpha B(\mathbf{y}_1)+\beta B(\mathbf{y}_2)$. Let $\mathbf{x}_k := B(\mathbf{y}_k)$, $k = 1, 2$, i.e. $A\mathbf{x}_k = \mathbf{y}_k$, $k = 1, 2$. Then
$$A(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha A\mathbf{x}_1 + \beta A\mathbf{x}_2 = \alpha\mathbf{y}_1 + \beta\mathbf{y}_2,$$
which means
$$B(\alpha\mathbf{y}_1 + \beta\mathbf{y}_2) = \alpha B(\mathbf{y}_1) + \beta B(\mathbf{y}_2).$$

And finally, let us show that B is indeed the inverse of A. Take $\mathbf{x} \in X$ and let $\mathbf{y} = A\mathbf{x}$, so by the definition of B we have $\mathbf{x} = B\mathbf{y}$. Then for all $\mathbf{x} \in X$

$$BA\mathbf{x} = B\mathbf{y} = \mathbf{x},$$

so $BA = I$. Similarly, for arbitrary $\mathbf{y} \in Y$ let $\mathbf{x} = B\mathbf{y}$, so $\mathbf{y} = A\mathbf{x}$. Then for all $\mathbf{y} \in Y$

$$AB\mathbf{y} = A\mathbf{x} = \mathbf{y}$$

so $AB = I$. □

Recalling the definition of a basis we get the following corollary of Theorems 6.6 and 6.7.

Corollary 6.9. *An $m \times n$ matrix is invertible if and only if its columns form a basis in $\mathbb{F}^m$.*

Exercises.

6.1. Prove, that if $A : V \to W$ is an isomorphism (i.e. an invertible linear transformation) and $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V, then $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n$ is a basis in W.

6.2. Find all right inverses to the 1×2 matrix (row) $A = (1, 1)$. Conclude from here that the row A is not left invertible.

6.3. Find all left inverses of the column $(1, 2, 3)^T$

6.4. Is the column $(1, 2, 3)^T$ right invertible? Justify

6.5. Find two matrices A and B that AB is invertible, but A and B are not. **Hint:** square matrices A and B would not work. **Remark:** It is easy to construct such A and B in the case when AB is a 1×1 matrix (a scalar). But can you get 2×2 matrix AB? 3×3? $n \times n$?

6.6. Suppose the product AB is invertible. Show that A is right invertible and B is left invertible. **Hint:** you can just write formulas for right and left inverses.

6.7. Let A and AB be invertible (assuming that the product AB is well defined). Prove that B is invertible.

6.8. Let A be $n \times n$ matrix. Prove that if $A^2 = \mathbf{0}$ then A is not invertible

6.9. Suppose $AB = \mathbf{0}$ for some non-zero matrix B. Can A be invertible? Justify.

6.10. Write matrices of the linear transformations T_1 and T_2 in $\mathbb{F}^5$, defined as follows: T_1 interchanges the coordinates x_2 and x_4 of the vector $\mathbf{x}$, and T_2 just adds to the coordinate x_2 a times the coordinate x_4, and does not change other coordinates, i.e.

$$T_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \\ x_5 \end{pmatrix}, \qquad T_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 + ax_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix};$$

here a is some fixed number.

Show that T_1 and T_2 are invertible transformations, and write the matrices of the inverses. **Hint:** it may be simpler, if you first describe the inverse transformation, and then find its matrix, rather than trying to guess (or compute) the inverses of the matrices T_1, T_2.

6.11. Find the matrix of the rotation in $\mathbb{R}^3$ through the angle α around the vector $(1, 2, 3)^T$. We assume that the rotation is counterclockwise if we sit at the tip of the vector and looking at the origin.

You can present the answer as a product of several matrices: you don't have to perform the multiplication.

6.12. Give examples of matrices (say 2×2) such that:

a) $A + B$ is not invertible although both A and B are invertible;

b) $A + B$ is invertible although both A and B are not invertible;

c) All of A, B and $A + B$ are invertible

6.13. Let A be an invertible symmetric ($A^T = A$) matrix. Is the inverse of A symmetric? Justify.

7. Subspaces.

A *subspace* of a vector space V is a non-empty subset $V_0 \subset V$ of V which is closed under the vector addition and multiplication by scalars, i.e.

1. If $\mathbf{v} \in V_0$ then $\alpha\mathbf{v} \in V_0$ for all scalars α;
2. For any $\mathbf{u}, \mathbf{v} \in V_0$ the sum $\mathbf{u} + \mathbf{v} \in V_0$;

Again, the conditions 1 and 2 can be replaced by the following one:

$$\alpha\mathbf{u} + \beta\mathbf{v} \in V_0 \qquad \text{for all } \mathbf{u}, \mathbf{v} \in V_0, \text{ and for all scalars } \alpha, \beta.$$

Note, that a subspace $V_0 \subset V$ with the operations (vector addition and multiplication by scalars) inherited from V, is a vector space. Indeed, since V is non-empty, it contain at least 1 vector $\mathbf{v}$ and since $\mathbf{0} = 0\mathbf{v}$, the above condition 1. imples that the zero vector $\mathbf{0}$ is in V. Also, for any $\mathbf{v} \in V$ its additive inverse $-\mathbf{v}$ in given by $-\mathbf{v} = (-1)\mathbf{v}$, so again by property 1. $-\mathbf{v} \in V$. The rest of the axiom of the vector space are satisfied because all operations are inherited from the vector space V. The only thing that could possibly go wrong, is that the result of some operation does not belong to V_0. But the definition of a subspace prohibits this!

Now let us consider some examples:

1. *Trivial* subspaces of a space V, namely V itself and $\{\mathbf{0}\}$ (the subspace consisting only of zero vector). Note, that the empty set $\varnothing$ is not a vector space, since it does not contain a zero vector, so it is not a subspace.

SYSTEMS OF LINEAR EQUATIONS

Treil Chapter 2, Sections 1-8

Chapter 2

Systems of linear equations

1. Different faces of linear systems.

There exist several points of view on what a system of linear equations, or in short a *linear system* is. The first, naïve one is, that it is simply a collection of m linear equations with n unknowns $x_1, x_2, \ldots, x_n$,

$$\begin{cases} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \ldots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \ldots & + & a_{2,n}x_n & = & b_2 \\ \quad\ldots & & & & & & & & \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \ldots & + & a_{m,n}x_n & = & b_m \end{cases} .$$

To solve the system is to find *all* n-tuples of numbers $x_1, x_2, \ldots, x_n$ which satisfy all m equations simultaneously.

If we denote $\mathbf{x} := (x_1, x_2, \ldots, x_n)^T \in \mathbb{F}^n$, $\mathbf{b} = (b_1, b_2, \ldots, b_m)^T \in \mathbb{F}^m$, and

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & \ldots & a_{m,n} \end{pmatrix},$$

then the above linear system can be written in the *matrix form* (as a *matrix-vector equation*)

$$A\mathbf{x} = \mathbf{b}.$$

To solve the above equation is to find all vectors $\mathbf{x} \in \mathbb{F}^n$ satisfying $A\mathbf{x} = \mathbf{b}$.

And finally, recalling the "column by coordinate" rule of the matrix-vector multiplication, we can write the system as a *vector equation*

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n = \mathbf{b},$$

where $\mathbf{a}_k$ is the kth column of the matrix A, $\mathbf{a}_k = (a_{1,k}, a_{2,k}, \ldots, a_{m,k})^T$, $k = 1, 2, \ldots, n$.

Note, these three examples are essentially just different representations of the same mathematical object.

Before explaining how to solve a linear system, let us notice that it does not matter what we call the unknowns, x_k, y_k or something else. So, all the information necessary to solve the system is contained in the matrix A, which is called *the coefficient matrix* of the system and in the vector (right side) $\mathbf{b}$. Hence, all the information we need is contained in the following matrix

$$\left(\begin{array}{cccc|c} a_{1,1} & a_{1,2} & \ldots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \ldots & a_{m,n} & b_m \end{array}\right)$$

which is obtained by attaching the column $\mathbf{b}$ to the matrix A. This matrix is called the *augmented matrix* of the system. We will usually put the vertical line separating A and $\mathbf{b}$ to distinguish between the augmented matrix and the coefficient matrix.

2. Solution of a linear system. Echelon and reduced echelon forms

Linear system are solved by the *Gauss–Jordan elimination* (which is sometimes called *row reduction*). By performing operations on rows of the augmented matrix of the system (i.e. on the equations), we reduce it to a simple form, the so-called *echelon form*. When the system is in the *echelon form*, one can easily write the solution.

2.1. Row operations. There are three types of row operations we use:

1. Row exchange: interchange two rows of the matrix;
2. Scaling: multiply a row by a non-zero scalar a;
3. Row replacement: replace a row # k by its sum with a constant multiple of a row # j; all other rows remain intact;

It is clear that the operations 1 and 2 do not change the set of solutions of the system; they essentially do not change the system.

As for the operation 3, one can easily see that it does not lose solutions. Namely, let a "new" system be obtained from an "old" one by a row operation of type 3. Then any solution of the "old" system is a solution of the "new" one.

To see that we do not gain anything extra, i.e. that any solution of the "new" system is also a solution of the "old" one, we just notice that row operation of type 3 are *reversible*, i.e. the "old' system also can be obtained from the "new" one by applying a row operation of type 3 (can you say which one?)

2.1.1. *Row operations and multiplication by elementary matrices.* There is another, more "advanced" explanation why the above row operations are legal. Namely, every row operation is equivalent to the multiplication of the matrix from the left by one of the special elementary matrices.

Namely, the multiplication by the matrix

$$
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j \\
\\
\\
\\
k \\
\\
\\
\end{array}
\begin{array}{c}
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& & & 1 & \vdots & & & & \vdots & & \\
\dots & \dots & \dots & 0 & \dots & \dots & \dots & 1 & & & \\
& & & \vdots & 1 & & & \vdots & & & \\
& & & \vdots & & \ddots & & \vdots & & & \\
& & & \vdots & & & 1 & \vdots & & & \\
\dots & \dots & \dots & 1 & \dots & \dots & \dots & 0 & & & \\
& & & & & & & & 1 & & \\
& & 0 & & & & & & & \ddots & \\
& & & & & & & & & & 1
\end{array}\right)
\end{array}
$$

just interchanges the rows number j and number k. Multiplication by the matrix

$$
\begin{array}{c}
\\
\\
\\
k \\
\\
\\
\end{array}
\begin{array}{c}
\left(\begin{array}{ccccccc}
1 & & & \vdots & & & \\
& \ddots & & \vdots & & 0 & \\
& & 1 & 0 & & & \\
\dots & \dots & 0 & a & 0 & & \\
& & & 0 & 1 & & \\
& 0 & & & & \ddots & 0 \\
& & & & & 0 & 1
\end{array}\right)
\end{array}
$$

multiplies the row number k by a. Finally, multiplication by the matrix

$$\begin{array}{c} \\ \\ j \\ \\ k \\ \\ \end{array} \begin{pmatrix} 1 & & \vdots & & \vdots & & \\ & \ddots & \vdots & & \vdots & 0 & \\ \dots & \dots & 1 & \dots & 0 & & \\ & & \vdots & \ddots & \vdots & & \\ \dots & \dots & a & \dots & 1 & & \\ & 0 & & & & \ddots & \\ & & & & & & 1 \end{pmatrix}$$

adds to the row $\#k$ row $\#j$ multiplied by a, and leaves all other rows intact.

A way to describe (or to remember) these elementary matrices: they are obtained from I by applying the corresponding row operation to it

To see, that the multiplication by these matrices works as advertised, one can just see how the multiplications act on vectors (columns).

Note that all these matrices are invertible (compare with reversibility of row operations). The inverse of the first matrix is the matrix itself. To get the inverse of the second one, one just replaces a by $1/a$. And finally, the inverse of the third matrix is obtained by replacing a by $-a$. To see that the inverses are indeed obtained this way, one again can simply check how they act on columns.

So, performing a row operation on the augmented matrix of the system $A\mathbf{x} = \mathbf{b}$ is equivalent to the multiplication of the system (from the left) by a special invertible matrix E. Left multiplying the equality $A\mathbf{x} = \mathbf{b}$ by E we get that any solution of the equation

$$A\mathbf{x} = \mathbf{b}$$

is also a solution of

$$EA\mathbf{x} = E\mathbf{b}.$$

Multiplying this equation (from the left) by E^{-1} we get that any of its solutions is a solution of the equation

$$E^{-1}EA\mathbf{x} = E^{-1}E\mathbf{b},$$

which is the original equation $A\mathbf{x} = \mathbf{b}$. So, a row operation does not change the solution set of a system.

2.2. Row reduction. The main step of row reduction consists of three sub-steps:

1. Find the leftmost non-zero column of the matrix;
2. Make sure, by applying row operations of type 1 (row exchange), if necessary, that the first (the upper) entry of this column is non-zero. This entry will be called the *pivot entry* or simply the *pivot*;

3. "Kill" (i.e. make them 0) all non-zero entries below the pivot by adding (subtracting) an appropriate multiple of the first row from the rows number $2, 3, \dots, m$.

We apply the main step to a matrix, then we leave the first row alone and apply the main step to rows $2, \dots, m$, then to rows $3, \dots, m$, etc.

The point to remember is that after we subtract a multiple of a row from all rows below it (step 3), we leave it alone and do not change it in any way, not even interchange it with another row.

After applying the main step finitely many times (at most m), we get what is called the *echelon form* of the matrix.

2.2.1. *An example of row reduction.* Let us consider the following linear system:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 3x_1 + 2x_2 + x_3 = 7 \\ 2x_1 + x_2 + 2x_3 = 1 \end{cases}$$

The augmented matrix of the system is

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 3 & 2 & 1 & 7 \\ 2 & 1 & 2 & 1 \end{array}\right)$$

Subtracting 3·Row#1 from the second row, and subtracting 2·Row#1 from the third one we get:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 3 & 2 & 1 & 7 \\ 2 & 1 & 2 & 1 \end{array}\right) \begin{array}{l} \\ -3R_1 \\ -2R_1 \end{array} \quad \sim \quad \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -4 & -8 & 4 \\ 0 & -3 & -4 & -1 \end{array}\right)$$

Multiplying the second row by $-1/4$ we get

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & -3 & -4 & -1 \end{array}\right)$$

Adding 3·Row#2 to the third row we obtain

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & -3 & -4 & -1 \end{array}\right) \begin{array}{l} \\ \\ +3R_2 \end{array} \quad \sim \quad \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 2 & -4 \end{array}\right)$$

Now we can use the so called *back substitution* to solve the system. Namely, from the last row (equation) we get $x_3 = -2$. Then from the second equation we get

$$x_2 = -1 - 2x_3 = -1 - 2(-2) = 3,$$

and finally, from the first row (equation)

$$x_1 = 1 - 2x_2 - 3x_3 = 1 - 6 + 6 = 1.$$

So, the solution is

$$\begin{cases} x_1 = 1 \\ x_2 = 3, \\ x_3 = -2, \end{cases}$$

or in vector form

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

or $\mathbf{x} = (1, 3, -2)^T$. We can check the solution by multiplying $A\mathbf{x}$, where A is the coefficient matrix.

Instead of using back substitution, we can do row reduction from bottom to top, killing all the entries above the main diagonal of the coefficient matrix: we start by multiplying the last row by $1/2$, and the rest is pretty self-explanatory:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -2 \end{array}\right) \begin{matrix} \\ \\ -2R_3 \end{matrix} \begin{matrix} -3R_3 \\ \\ \end{matrix} \sim \left(\begin{array}{ccc|c} 1 & 2 & 0 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{array}\right) -2R_2$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{array}\right)$$

and we just read the solution $\mathbf{x} = (1, 3, -2)^T$ off the augmented matrix.

We leave it as an exercise to the reader to formulate the algorithm for the backward phase of the row reduction.

2.3. Echelon form. A matrix is in *echelon form* if it satisfies the following two conditions:

1. All zero rows (i.e. the rows with all entries equal 0), if any, are below all non-zero entries.

For a non-zero row, let us call the leftmost non-zero entry the *leading entry*. Then the second property of the echelon form can be formulated as follows:

2. For any non-zero row its leading entry is strictly to the right of the leading entry in the previous row.

Pivots: leading (leftmost non-zero entries) in a row.

The leading entry in each row in echelon form is also called *pivot entry*, or simply *pivot*, because these entries are exactly the *pivots* we used in the row reduction.

A particular case of the echelon form is the so-called *triangular* form. We got this form in our example above. In this form the coefficient matrix is square ($n \times n$), all its entries on the main diagonal are non-zero, and all the

entries below the main diagonal are zero. The right side, i.e. the rightmost column of the augmented matrix can be arbitrary.

After the backward phase of the row reduction, we get what the so-called *reduced echelon form* of the matrix: coefficient matrix equal I, as in the above example, is a particular case of the reduced echelon form.

The general definition is as follows: we say that a matrix is in the *reduced echelon form*, if it is in the echelon form and

3. All pivot entries are equal 1;
4. All entries above the pivots are 0. Note, that all entries below the pivots are also 0 because of the echelon form.

To get reduced echelon form from echelon form, we work from the bottom to the top and from the right to the left, using row replacement to kill all entries above the pivots.

An example of the reduced echelon form is the system with the coefficient matrix equal I. In this case, one just reads the solution from the reduced echelon form. In general case, one can also easily read the solution from the reduced echelon form. For example, let the reduced echelon form of the system (augmented matrix) be

$$\left(\begin{array}{ccccc|c} \boxed{1} & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & \boxed{1} & 5 & 0 & 2 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \end{array}\right);$$

here we boxed the pivots. The idea is to move the variables, corresponding to the columns without pivot (the so-called *free variables*) to the right side. Then we can just write the solution.

$$\begin{cases} x_1 = 1 - 2x_2 \\ x_2 \text{ is free} \\ x_3 = 2 - 5x_4 \\ x_4 \text{ is free} \\ x_5 = 3 \end{cases}$$

or in the vector form

$$\mathbf{x} = \begin{pmatrix} 1 - 2x_2 \\ x_2 \\ 2 - 5x_4 \\ x_4 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \\ 3 \end{pmatrix} + x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 0 \\ -5 \\ 1 \\ 0 \end{pmatrix}, \quad x_2, x_4 \in \mathbb{F}.$$

One can also find the solution from the echelon form by using back substitution: the idea is to work from bottom to top, moving all free variables to the right side.

Exercises.

2.1. Write the systems of equations below in matrix form and as vector equations:

a) $\begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases}$

b) $\begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases}$

c) $\begin{cases} x_1 + 2x_2 + 2x_4 = 6 \\ 3x_1 + 5x_2 - x_3 + 6x_4 = 17 \\ 2x_1 + 4x_2 + x_3 + 2x_4 = 12 \\ 2x_1 - 7x_3 + 11x_4 = 7 \end{cases}$

d) $\begin{cases} x_1 - 4x_2 - x_3 + x_4 = 3 \\ 2x_1 - 8x_2 + x_3 - 4x_4 = 9 \\ -x_1 + 4x_2 - 2x_3 + 5x_4 = -6 \end{cases}$

e) $\begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = 2 \\ 2x_1 + 4x_2 - x_3 + 6x_4 = 5 \\ x_2 + 2x_4 = 3 \end{cases}$

f) $\begin{cases} 2x_1 - 2x_2 - x_3 + 6x_4 - 2x_5 = 1 \\ x_1 - x_2 + x_3 + 2x_4 - x_5 = 2 \\ 4x_1 - 4x_2 + 5x_3 + 7x_4 - x_5 = 6 \end{cases}$

g) $\begin{cases} 3x_1 - x_2 + x_3 - x_4 + 2x_5 = 5 \\ x_1 - x_2 - x_3 - 2x_4 - x_5 = 2 \\ 5x_1 - 2x_2 + x_3 - 3x_4 + 3x_5 = 10 \\ 2x_1 - x_2 - 2x_4 + x_5 = 5 \end{cases}$

Solve the systems. Write the answers in the vector form.

2.2. Find *all* solutions of the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3 = \mathbf{0},$$

where $\mathbf{v}_1 = (1,1,0)^T$, $\mathbf{v}_2 = (0,1,1)^T$ and $\mathbf{v}_3 = (1,0,1)^T$. What conclusion can you make about linear independence (dependence) of the system of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$?

3. Analyzing the pivots.

All questions about existence of a solution and it uniqueness can be answered by analyzing pivots in the echelon (reduced echelon) form of the augmented matrix of the system. First of all, let us investigate the question of when is the equation $A\mathbf{x} = \mathbf{b}$ *inconsistent*, i.e. when it does not have a solution. The answer follows immediately, if one just thinks about it:

a system is inconsistent (does not have a solution) if and only if there is a pivot in the last column of an echelon form of the *augmented* matrix, i.e. iff an echelon form of the augmented matrix has a row $\left(\begin{array}{ccccc|c} 0 & 0 & \dots & 0 & b \end{array}\right)$, $b \neq 0$ in it.

Indeed, such a row correspond to the equation $0x_1 + 0x_2 + \ldots + 0x_n = b \neq 0$ that does not have a solution. If we don't have such a row, we just make the reduced echelon form and then read the solution off it.

Now, three more statements. Note, they all deal with the *coefficient matrix*, and not with the augmented matrix of the system.

1. A solution (if it exists) is unique iff there are no free variables, that is if and only if the echelon form of the coefficient matrix has a pivot in every column;
2. Equation $A\mathbf{x} = \mathbf{b}$ is consistent for all right sides $\mathbf{b}$ if and only if the echelon form of the coefficient matrix has a pivot in every row.
3. Equation $A\mathbf{x} = \mathbf{b}$ has a *unique solution* for any right side $\mathbf{b}$ if and only if echelon form of the coefficient matrix A has a pivot in every column and every row.

The first statement is trivial, because free variables are responsible for all non-uniqueness. I should only emphasize that this statement *does not say anything* about the existence.

The second statement is a tiny bit more complicated. If we have a pivot in every row of the coefficient matrix, we cannot have the pivot in the last column of the *augmented* matrix, so the system is always consistent, no matter what the right side $\mathbf{b}$ is.

Let us show that if we have a zero row in the echelon form of the coefficient matrix A, then we can pick a right side $\mathbf{b}$ such that the system $A\mathbf{x} = \mathbf{b}$ is not consistent. Let A_e echelon form of the coefficient matrix A. Then

$$A_e = EA,$$

where E is the product of elementary matrices, corresponding to the row operations, $E = E_N \ldots E_2 E_1$. If A_e has a zero row, then the last row is also zero. Therefore, if we put $\mathbf{b}_e = (0, \ldots, 0, 1)^T$ (all entries are 0, except the last one), then the equation

$$A_e\mathbf{x} = \mathbf{b}_e$$

does not have a solution. Multiplying this equation by E^{-1} from the left, and recalling that $E^{-1}A_e = A$, we get that the equation

$$A\mathbf{x} = E^{-1}\mathbf{b}_e$$

does not have a solution.

Finally, statement 3 immediately follows from statements 1 and 2. □

From the above analysis of pivots we get several very important corollaries. The main observation we use is

In echelon form, any row and any column have no more than 1 pivot in it (it can have 0 pivots)

3.1. Corollaries about linear independence and bases. Dimension. Questions as to when a system of vectors in $\mathbb{F}^n$ is a basis, a linearly independent or a spanning system, can be easily answered by the row reduction.

Proposition 3.1. *Let us have a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{F}^n$, and let $A = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m]$ be an $n \times m$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$. Then*

1. *The system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is linearly independent iff echelon form of A has a pivot in every column;*
2. *The system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is complete in $\mathbb{F}^n$ (spanning, generating) iff echelon form of A has a pivot in every row;*
3. *The system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ is a basis in $\mathbb{F}^n$ iff echelon form of A has a pivot in every column and in every row.*

Proof. The system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{F}^n$ is linearly independent if and only if the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_m\mathbf{v}_m = \mathbf{0}$$

has the unique (trivial) solution $x_1 = x_2 = \ldots = x_m = 0$, or equivalently, the equation $A\mathbf{x} = \mathbf{0}$ has unique solution $\mathbf{x} = \mathbf{0}$. By statement 1 above, it happens if and only if there is a pivot in every column of the matrix.

Similarly, the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{F}^n$ is complete in $\mathbb{F}^n$ if and only if the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_m\mathbf{v}_m = \mathbf{b}$$

has a solution for any right side $\mathbf{b} \in \mathbb{F}^n$. By statement 2 above, it happens if and only if there is a pivot in every row in echelon form of the matrix.

And finally, the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{F}^n$ is a basis in $\mathbb{F}^n$ if and only if the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_m\mathbf{v}_m = \mathbf{b}$$

has unique solution for any right side $\mathbf{b} \in \mathbb{F}^n$. By statement 3 this happens if and only if there is a pivot in every column and in every row of echelon form of A. □

Proposition 3.2. *Any linearly independent system of vectors in $\mathbb{F}^n$ cannot have more than n vectors in it.*

Proof. Let a system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in \mathbb{F}^n$ be linearly independent, and let $A = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m]$ be the $n \times m$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$. By Proposition 3.1 echelon form of A must have a pivot in every column, which is impossible if $m > n$ (number of pivots cannot be more than number of rows). □

Proposition 3.3. *Any two bases in a vector space V have the same number of vectors in them.*

Proof. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m$ be two different bases in V. Without loss of generality we can assume that $n \le m$. Consider an isomorphism $A : \mathbb{F}^n \to V$ defined by

$$A\mathbf{e}_k = \mathbf{v}_k, \qquad k = 1, 2, \dots n,$$

where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is the standard basis in $\mathbb{R}^n$.

Since A^{-1} is also an isomorphism, the system

$$A^{-1}\mathbf{w}_1, A^{-1}\mathbf{w}_2, \dots, A^{-1}\mathbf{w}_m$$

is a basis (see Theorem 6.6 in Chapter 1). So it is linearly independent, and by Proposition 3.2, $m \le n$. Together with the assumption $n \le m$ this implies that $m = n$. □

The statement below is a particular case of the above proposition.

Proposition 3.4. *Any basis in $\mathbb{F}^n$ must have exactly n vectors in it.*

Proof. This fact follows immediately from the previous proposition, but there is also a direct proof. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ be a basis in $\mathbb{F}^n$ and let A be the $n \times m$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$. The fact that the system is a basis, means that the equation

$$A\mathbf{x} = \mathbf{b}$$

has a unique solution for any (all possible) right side $\mathbf{b}$. The existence means that there is a pivot in every row (of a reduced echelon form of the matrix), hence the number of pivots is exactly n. The uniqueness mean that there is pivot in every column of the coefficient matrix (its echelon form), so

$$m = \text{number of columns} = \text{number of pivots} = n$$

□

Proposition 3.5. *Any spanning (generating) set in $\mathbb{F}^n$ must have at least n vectors.*

Proof. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ be a complete system in $\mathbb{F}^n$, and let A be $n \times m$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$. Statement 2 of Proposition 3.1 implies

that echelon form of A has a pivot in every row. Since number of pivots cannot exceed the number of columns, $n \leq m$. □

3.2. Corollaries about invertible matrices.

Proposition 3.6. *A matrix A is invertible if and only if its echelon form has pivot in every column and every row.*

Proof. As it was discussed in the beginning of the section, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for any right side $\mathbf{b}$ if and only if the echelon form of A has pivot in every row and every column. But, we know, see Theorem 6.8 in Chapter 1, that the matrix (linear transformation) A is invertible if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for any possible right side $\mathbf{b}$.

There is also an alternative proof. We know that a matrix is invertible if and only if its columns form a basis in (see Corollary 6.9 in Section 6.4, Chapter 1). Proposition 3.4 above states that it happens if and only if there is a pivot in every row and every column. □

The above proposition immediately implies the following

Corollary 3.7. *An invertible matrix must be square ($n \times n$).*

Proposition 3.8. *If a square ($n \times n$) matrix is left invertible, or if it is right invertible, then it is invertible. In other words, to check the invertibility of a square matrix A it is sufficient to check only one of the conditions $AA^{-1} = I$, $A^{-1}A = I$.*

Note, that this proposition applies only to square matrices!

Proof. We know that matrix A is invertible if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for any right side $\mathbf{b}$. This happens if and only if the echelon form of the matrix A has pivots in every row and in every column.

If a matrix A is left invertible, the equation $A\mathbf{x} = \mathbf{0}$ has unique solution $\mathbf{x} = \mathbf{0}$. Indeed, if B is a left inverse of A (i.e. $BA = I$), and $\mathbf{x}$ satisfies

$$A\mathbf{x} = \mathbf{0},$$

then multiplying this identity by B from the left we get $\mathbf{x} = \mathbf{0}$, so the solution is unique. Therefore, the echelon form of A has pivots in every column (no free variables). If the matrix A is square ($n \times n$), the echelon form also has pivots in every row (n pivots, and a row can have no more than 1 pivot), so the matrix is invertible.

If a matrix A is right invertible, and C is its right inverse ($AC = I$), then for $\mathbf{x} = C\mathbf{b}$, $\mathbf{b} \in \mathbb{F}^n$

$$A\mathbf{x} = AC\mathbf{b} = I\mathbf{b} = \mathbf{b}.$$

Therefore, for any right side $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x} = C\mathbf{b}$. Thus, echelon form of A has a pivot in every row. If A is square, it also has a pivot in every column, so A is invertible. □

Exercises.

3.1. For what value of b the system

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ b \end{pmatrix}$$

has a solution. Find the general solution of the system for this value of b.

3.2. Determine, if the vectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

are linearly independent or not.

Do these four vectors span $\mathbb{R}^4$? (In other words, is it a generating system?) What about $\mathbb{C}^4$?

3.3. Determine, which of the following systems of vectors are bases in $\mathbb{R}^3$:

a) $(1, 2, -1)^T$, $(1, 0, 2)^T$, $(2, 1, 1)^T$;

b) $(-1, 3, 2)^T$, $(-3, 1, 3)^T$, $(2, 10, 2)^T$;

c) $(67, 13, -47)^T$, $(\pi, -7.84, 0)^T$, $(3, 0, 0)^T$.

Which of the systems are bases in $\mathbb{C}^3$?

3.4. Do the polynomials $x^3 + 2x$, $x^2 + x + 1$, $x^3 + 5$ generate (span) $\mathbb{P}_3$? Justify your answer.

3.5. Can 5 vectors in $\mathbb{F}^4$ be linearly independent? Justify your answer.

3.6. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the columns of $A^2 = AA$.

3.7. Prove or disprove: If the columns of a square $(n \times n)$ matrix A are linearly independent, so are the rows of $A^3 = AAA$.

3.8. Show that if the equation $A\mathbf{x} = \mathbf{0}$ has unique solution (i.e. if echelon form of A has pivot in every column), then A is left invertible. **Hint:** elementary matrices may help.
Note: It was shown in the text that if A is left invertible, then the equation $A\mathbf{x} = \mathbf{0}$ has unique solution. But here you are asked to prove the converse of this statement, which was not proved.

Remark: This can be a very hard problem, for it requires deep understanding of the subject. However, when you understand what to do, the problem becomes almost trivial.

3.9. Is the reduced echelon form of a matrix unique? Justify your conclusion.

Namely, suppose that by performing some row operations (not necessarily following any algorithm) we end up with a reduced echelon matrix. Do we always end up with the same matrix, or can we get different ones? Note that we are only allowed to perform row operations, the "column operations"' are forbidden.

Hint: What happens if you start with an invertible matrix? Also, are the pivots always in the same columns, or can it be different depending on what row operations you perform? If you can tell what the pivot columns are without reverting to row operations, then the positions of pivot columns do not depend on them.

4. Finding A^{-1} by row reduction.

As it was discussed above, an invertible matrix must be square, and its echelon form must have pivots in every row and every column. Therefore reduced echelon form of an invertible matrix is the identity matrix I. Therefore,

Any invertible matrix is row equivalent (i.e. can be reduced by row operations) to the identity matrix.

Now let us state a simple algorithm of finding the inverse of an $n \times n$ matrix:

1. Form an *augmented* $n \times 2n$ matrix $(A \,|\, I)$ by writing the $n \times n$ identity matrix right of A;
2. Performing row operations on the augmented matrix transform A to the identity matrix I;
3. The matrix I that we added will be automatically transformed to A^{-1};
4. If it is impossible to transform A to the identity by row operations, A is not invertible.

There are several possible explanations of the above algorithm. The first, a naïve one, is as follows: we know that (for an invertible A) the vector $A^{-1}\mathbf{b}$ is the solution of the equation $A\mathbf{x} = \mathbf{b}$. So to find the column number k of A^{-1} we need to find the solution of $A\mathbf{x} = \mathbf{e}_k$, where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ is the standard basis in $\mathbb{R}^n$. The above algorithm just solves the equations

$$A\mathbf{x} = \mathbf{e}_k, \qquad k = 1, 2, \dots, n$$

simultaneously!

Let us also present another, more "advanced" explanation. As we discussed above, every row operation can be realized as a left multiplication

by an elementary matrix. Let $E_1, E_2, \dots, E_N$ be the elementary matrices corresponding to the row operation we performed, and let $E = E_N \cdots E_2 E_1$ be their product.[1] We know that the row operations transform A to identity, i.e. $EA = I$, so $E = A^{-1}$. But the same row operations transform the augmented matrix $(A \,|\, I)$ to $(EA \,|\, E) = (I \,|\, A^{-1})$. □

This "advanced" explanation using elementary matrices implies an important proposition that will be often used later.

Theorem 4.1. *Any invertible matrix can be represented as a product of elementary matrices.*

Proof. As we discussed in the previous paragraph, $A^{-1} = E_N \cdots E_2 E_1$, so

$$A = (A^{-1})^{-1} = E_1^{-1} E_2^{-1} \cdots E_N^{-1}.$$

□

An Example. Suppose we want to find the inverse of the matrix

$$\begin{pmatrix} 1 & 4 & -2 \\ -2 & -7 & 7 \\ 3 & 11 & -6 \end{pmatrix}.$$

Augmenting the identity matrix to it and performing row reduction we get

$$\left(\begin{array}{ccc|ccc} 1 & 4 & -2 & 1 & 0 & 0 \\ -2 & -7 & 7 & 0 & 1 & 0 \\ 3 & 11 & -6 & 0 & 0 & 1 \end{array}\right)\begin{matrix} \\ +2R_1 \\ -3R_1 \end{matrix} \sim \left(\begin{array}{ccc|ccc} 1 & 4 & -2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & -1 & 0 & -3 & 0 & 1 \end{array}\right)\begin{matrix} \\ \\ +R_2 \end{matrix} \sim$$

$$\left(\begin{array}{ccc|ccc} 1 & 4 & -2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 3 & -1 & 1 & 1 \end{array}\right)\times 3 \sim \left(\begin{array}{ccc|ccc} 3 & 12 & -6 & 3 & 0 & 0 \\ 0 & 1 & 3 & 2 & 1 & 0 \\ 0 & 0 & 3 & -1 & 1 & 1 \end{array}\right)\begin{matrix} +2R_3 \\ -R_3 \\ \\ \end{matrix} \sim$$

Here in the last row operation we multiplied the first row by 3 to avoid fractions in the backward phase of row reduction. Continuing with the row reduction we get

$$\left(\begin{array}{ccc|ccc} 3 & 12 & 0 & 1 & 2 & 2 \\ 0 & 1 & 0 & 3 & 0 & -1 \\ 0 & 0 & 3 & -1 & 1 & 1 \end{array}\right)-12R_2 \sim \left(\begin{array}{ccc|ccc} 3 & 0 & 0 & -35 & 2 & 14 \\ 0 & 1 & 0 & 3 & 0 & -1 \\ 0 & 0 & 3 & -1 & 1 & 1 \end{array}\right)$$

Dividing the first and the last row by 3 we get the inverse matrix

$$\begin{pmatrix} -35/3 & 2/3 & 14/3 \\ 3 & 0 & -1 \\ -1/3 & 1/3 & 1/3 \end{pmatrix}$$

[1] Although it does not matter here, but please notice, that if the row operation E_1 was performed first, E_1 must be the rightmost term in the product

Exercises.

4.1. Find the inverse of the matrices

$$\begin{pmatrix} 1 & 2 & 1 \\ 3 & 7 & 3 \\ 2 & 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 1 & -1 & 2 \\ 1 & 1 & -2 \\ 1 & 1 & 4 \end{pmatrix}.$$

Show all steps

5. Dimension. Finite-dimensional spaces.

Definition. The dimension $\dim V$ of a vector space V is the number of vectors in a basis.

For a vector space consisting only of zero vector $\mathbf{0}$ we put $\dim V = 0$. If V does not have a (finite) basis, we put $\dim V = \infty$.

If $\dim V$ is finite, we call the space V *finite-dimensional*; otherwise we call it *infinite-dimensional*.

Proposition 3.3 asserts that the dimension is well defined, i.e. that it does not depend on the choice of a basis.

Proposition 2.8 from Chapter 1 states that any finite spanning system in a vector space V contains a basis. This immediately implies the following

Proposition 5.1. *A vector space V is finite-dimensional if and only if it has a finite spanning system.*

Suppose, that we have a system of vectors in a finite-dimensional vector space, and we want to check if it is a basis (or if it is linearly independent, or if it is complete)? Probably the simplest way is to use an isomorphism $A : V \to \mathbb{R}^n$, $n = \dim E$ to move the problem to $\mathbb{R}^n$, where all such questions can be answered by row reduction (studying pivots).

Note, that if $\dim V = n$, then there always exists an isomorphism $A : V \to \mathbb{R}^n$. Indeed, if $\dim V = n$ then there exists a basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in V$, and one can define an isomorphism $A : V \to \mathbb{R}^n$ by

$$A\mathbf{v}_k = \mathbf{e}_k, \qquad k = 1, 2, \dots, n.$$

As an example, let us give the following two corollaries of the above Propositions 3.2, 3.5:

Proposition 5.2. *Any linearly independent system in a finite-dimensional vector space V cannot have more than* $\dim V$ *vectors in it.*

Proof. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in V$ be a linearly independent system, and let $A : V \to \mathbb{R}^n$ be an isomorphism. Then $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_m$ is a linearly independent system in $\mathbb{R}^n$, and by Proposition 3.2 $m \le n$. □

Proposition 5.3. *Any generating system in a finite-dimensional vector space V must have at least* $\dim V$ *vectors in it.*

Proof. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m \in V$ be a complete system in V, and let $A : V \to \mathbb{R}^n$ be an isomorphism. Then $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_m$ is a complete system in $\mathbb{R}^n$, and by Proposition 3.5 $m \geq n$. □

5.1. Completing a linearly independent system to a basis. The following statement will play an important role later.

Proposition 5.4 (Completion to a basis)**.** *A linearly independent system of vectors in a finite-dimensional space can be completed to a basis, i.e. if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are linearly independent vectors in a finite-dimensional vector space V then one can find vectors $\mathbf{v}_{r+1}, \mathbf{v}_{r+2} \dots, \mathbf{v}_n$ such that the system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V.*

Proof. Let $n = \dim V$.Take any vector not belonging to $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$ and call it $\mathbf{v}_{r+1}$ (one can always do that because the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ is not generating). By Exercise 2.5 from Chapter 1 the system $\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent (notice that in this case $r < n$ by Proposition 5.2). Repeat the procedure with the new system to get vector $\mathbf{v}_{r+2}$, and so on.

We will stop the process when we get a generating system. Note, that the process cannot continue infinitely, because a linearly independent system of vectors in V cannot have more than $n = \dim V$ vectors. □

5.2. Subspaces of finite dimensional spaces.

Theorem 5.5. *Let V be a subspace of a vector space W,* $\dim W < \infty$*. Then V is finite dimensional and* $\dim V \leq \dim W$*.*

Moreover, if $\dim V = \dim W$ *then $V = W$ (we are still assuming that V is a subspace of W here).*

Remark. This theorem looks like a complete banality, like an easy corollary of Proposition 5.2. But we can apply Proposition 5.2 only if we already have a basis in V. And we only have a basis in W, and we cannot say how many vectors in this basis belong to V; in fact, it is easy to construct an example where none of the vectors in the basis of W belongs to V.

Proof of Theorem 5.5. If $V = \{\mathbf{0}\}$ then the theorem is trivial, so let us assume otherwise.

We want to find a basis in V. Take a non-zero vector $\mathbf{v}_1 \in V$. If $V = \operatorname{span}\{\mathbf{v}_1\}$, we got our basis (consisting of the one vector $\mathbf{v}_1$).

If not, we continue by induction. Suppose we constructed r linearly independent vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$. If $V = \operatorname{span}\{\mathbf{v}_k : 1 \leq k \leq r\}$,

then we have found a basis in V. If not, there exists a vector $\mathbf{v}_{r+1} \in V$, $\mathbf{v}_{r+1} \notin \operatorname{span}\{\mathbf{v}_k : 1 \le k \le r\}$. By Exercise 2.5 from Chapter 1 the system $\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

□

Exercises.

5.1. True or false:

a) Every vector space that is generated by a finite set has a basis;

b) Every vector space has a (finite) basis;

c) A vector space cannot have more than one basis;

d) If a vector space has a finite basis, then the number of vectors in every basis is the same.

e) The dimension of $\mathbb{P}_n$ is n;

f) The dimension on $M_{m\times n}$ is $m+n$;

g) If vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ generate (span) the vector space V, then every vector in V can be written as a linear combination of vector $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in only one way;

h) Every subspace of a finite-dimensional space is finite-dimensional;

i) If V is a vector space having dimension n, then V has exactly one subspace of dimension 0 and exactly one subspace of dimension n.

5.2. Prove that if V is a vector space having dimension n, then a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in V is linearly independent if and only if it spans V.

5.3. Prove that a linearly independent system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V is a basis if and only if $n = \dim V$.

5.4. (An old exercise revisited: now this problem should be easy) Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly *independent*? **Hint:** What dimension can the subspace $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ have?

5.5. Let vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be a basis in V. Show that $\mathbf{u}+\mathbf{v}+\mathbf{w}$, $\mathbf{v}+\mathbf{w}$, $\mathbf{w}$ is also a basis in V.

5.6. Consider in the space $\mathbb{R}^5$ vectors $\mathbf{v}_1 = (2, -1, 1, 5, -3)^T$, $\mathbf{v}_2 = (3, -2, 0, 0, 0)^T$, $\mathbf{v}_3 = (1, 1, 50, -921, 0)^T$.

a) Prove that these vectors are linearly independent.

b) Complete this system of vectors to a basis.

If you do part b) first you can do everything without any computations.

6. General solution of a linear system.

In this short section we discuss the structure of the general solution (i.e. of the solution set) of a linear system.

We call a system $A\mathbf{x} = \mathbf{b}$ homogeneous if the right side $\mathbf{b} = \mathbf{0}$, i.e. a homogeneous system is a system of form $A\mathbf{x} = \mathbf{0}$.

With each system

$$A\mathbf{x} = \mathbf{b}$$

we can associate a homogeneous system just by putting $\mathbf{b} = \mathbf{0}$.

Theorem 6.1 (General solution of a linear equation)**.** *Let a vector $\mathbf{x}_1$ satisfy the equation $A\mathbf{x} = \mathbf{b}$, and let H be the set of all solutions of the associated homogeneous system*

$$A\mathbf{x} = \mathbf{0}.$$

Then the set

$$\{\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_\mathrm{h} : \mathbf{x}_\mathrm{h} \in H\}$$

is the set of all solutions of the equation $A\mathbf{x} = \mathbf{b}$.

In other words, this theorem can be stated as

$$\boxed{\begin{array}{l}\text{General solution}\\ \text{of } A\mathbf{x} = \mathbf{b}\end{array}} = \boxed{\begin{array}{l}\text{A particular solu-}\\ \text{tion of } A\mathbf{x} = \mathbf{b}\end{array}} + \boxed{\begin{array}{l}\text{General solution}\\ \text{of } A\mathbf{x} = \mathbf{0}\end{array}}.$$

Proof. Fix a vector $\mathbf{x}_1$ satisfying $A\mathbf{x}_1 = \mathbf{b}$. Let a vector $\mathbf{x}_\mathrm{h}$ satisfy $A\mathbf{x}_h = \mathbf{0}$. Then for $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_\mathrm{h}$ we have

$$A\mathbf{x} = A(\mathbf{x}_1 + \mathbf{x}_\mathrm{h}) = A\mathbf{x}_1 + A\mathbf{x}_\mathrm{h} = \mathbf{b} + \mathbf{0} = \mathbf{b},$$

so any $\mathbf{x}$ of form

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_\mathrm{h}, \qquad \mathbf{x}_\mathrm{h} \in H$$

is a solution of $A\mathbf{x} = \mathbf{b}$.

Now let $\mathbf{x}$ satisfy $A\mathbf{x} = \mathbf{b}$. Then for $\mathbf{x}_\mathrm{h} := \mathbf{x} - \mathbf{x}_1$ we get

$$A\mathbf{x}_\mathrm{h} = A(\mathbf{x} - \mathbf{x}_1) = A\mathbf{x} - A\mathbf{x}_1 = \mathbf{b} - \mathbf{b} = \mathbf{0},$$

so $\mathbf{x}_\mathrm{h} \in H$. Therefore *any* solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ can be represented as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_\mathrm{h}$ with some $\mathbf{x}_\mathrm{h} \in H$. □

The power of this theorem is in its generality. It applies to all linear equations, we do not have to assume here that vector spaces are finite-dimensional. You will meet this theorem in differential equations, integral equations, partial differential equations, etc. Besides showing the structure of the solution set, this theorem allows one to separate investigation of uniqueness from the study of existence. Namely, to study uniqueness, we only need to analyze uniqueness of the homogeneous equation $A\mathbf{x} = \mathbf{0}$, which always has a solution.

There is an immediate application in this course: this theorem allows us to check a solution of a system $A\mathbf{x} = \mathbf{b}$. For example, consider a system

$$\begin{pmatrix} 2 & 3 & 1 & 4 & -9 \\ 1 & 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & 2 & -5 \\ 2 & 2 & 2 & 3 & -8 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 17 \\ 6 \\ 8 \\ 14 \end{pmatrix}.$$

Performing row reduction one can find the solution of this system

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \qquad x_3, x_5 \in \mathbb{F}. \tag{6.1}$$

The parameters x_3, x_5 can be denoted here by any other letters, t and s, for example; we are keeping notation x_3 and x_5 here only to remind us that the parameters came from the corresponding free variables.

Now, let us suppose, that we are just given this solution, and we want to check whether or not it is correct. Of course, we can repeat the row operations, but this is too time consuming. Moreover, if the solution was obtained by some non-standard method, it can look differently from what we get from the row reduction. For example, the formula

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{F} \tag{6.2}$$

gives the same set as (6.1) (can you say why?); here we just replaced the last vector in (6.1) by its sum with the second one. So, this formula is different from the solution we got from the row reduction, but it is nevertheless correct.

The simplest way to check that (6.1) and (6.2) give us correct solutions, is to check that the first vector $(3, 1, 0, 2, 0)^T$ satisfies the equation $A\mathbf{x} = \mathbf{b}$, and that the other two (the ones with the parameters x_3 and x_5 or s and t in front of them) should satisfy the associated homogeneous equation $A\mathbf{x} = \mathbf{0}$.

If this checks out, we will be assured that any vector $\mathbf{x}$ defined by (6.1) or (6.2) is indeed a solution.

Note, that this method of checking the solution does not guarantee that (6.1) (or (6.2)) gives us all the solutions. For example, if we just somehow miss out the term with x_3, the above method of checking will still work fine.

So, how can we guarantee, that we did not miss any free variable, and there should not be extra term in (6.1)?

What comes to mind, is to count the pivots again. In this example, if one does row operations, the number of pivots is 3. So indeed, there should be 2 free variables, and it looks like we did not miss anything in (6.1).

To be able to *prove* this, we will need new notions of fundamental subspaces and of rank of a matrix. I should also mention that in this particular example, one does not have to perform all row operations to check that there are only 2 free variables, and that formulas (6.1) and (6.2) both give correct general solutions.

Exercises.

6.1. True or false

a) Any system of linear equations has at least one solution;

b) Any system of linear equations has at most one solution;

c) Any homogeneous system of linear equations has at least one solution;

d) Any system of n linear equations in n unknowns has at least one solution;

e) Any system of n linear equations in n unknowns has at most one solution;

f) If the homogeneous system corresponding to a given system of a linear equations has a solution, then the given system has a solution;

g) If the coefficient matrix of a homogeneous system of n linear equations in n unknowns is invertible, then the system has no non-zero solution;

h) The solution set of any system of m equations in n unknowns is a subspace in $\mathbb{R}^n$;

i) The solution set of any homogeneous system of m equations in n unknowns is a subspace in $\mathbb{R}^n$.

6.2. Find a 2×3 system (2 equations with 3 unknowns) such that its general solution has a form

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad s \in \mathbb{R}.$$

7. Fundamental subspaces of a matrix. Rank.

As we discussed above in Section 7 of Chapter 1, with any linear transformation $A : V \to W$ we can associate two subspaces, namely, its kernel, or null space

$$\operatorname{Ker} A = \operatorname{Null} A := \{\mathbf{v} \in V : A\mathbf{v} = \mathbf{0}\} \subset V,$$

and its range

$$\operatorname{Ran} A = \{\mathbf{w} \in W : \mathbf{w} = A\mathbf{v} \text{ for some } \mathbf{v} \in V\} \subset W.$$

In other words, the kernel $\operatorname{Ker} A$ is the solution set of the homogeneous equation $A\mathbf{x} = \mathbf{0}$, and the range $\operatorname{Ran} A$ is exactly the set of all right sides $\mathbf{b} \in W$ for which the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

If A is an $m \times n$ matrix, i.e. a mapping from $\mathbb{F}^n$ to $\mathbb{F}^m$, then it follows from the "column by coordinate" rule of the matrix multiplication that any vector $\mathbf{w} \in \operatorname{Ran} A$ can be represented as a linear combination of columns of A. This explains the name *column space* (notation $\operatorname{Col} A$), which is often used instead of $\operatorname{Ran} A$.

If A is a matrix, then in addition to $\operatorname{Ran} A$ and $\operatorname{Ker} A$ one can also consider the range and kernel for the transposed matrix A^T. Often the term *row space* is used for $\operatorname{Ran} A^T$ and the term *left null space* is used for $\operatorname{Ker} A^T$ (but usually no special notation is introduced).

The four subspaces $\operatorname{Ran} A$, $\operatorname{Ker} A$, $\operatorname{Ran} A^T$, $\operatorname{Ker} A^T$ are called the *fundamental subspaces* of the matrix A. In this section we will study important relations between the dimensions of the four fundamental subspaces.

We will need the following definition, which is one of the fundamental notions of Linear Algebra

Definition. Given a linear transformation (matrix) A its rank, $\operatorname{rank} A$, is the dimension of the range of A

$$\operatorname{rank} A := \dim \operatorname{Ran} A.$$

7.1. Computing fundamental subspaces and rank. To compute the fundamental subspaces and rank of a matrix, one needs to do echelon reduction. Namely, let A be the matrix, and A_{e} be its echelon form

1. The pivot columns of the *original* matrix A (i.e. the columns where after row operations we will have pivots in the echelon form) give us a basis (one of many possible) in $\operatorname{Ran} A$.
2. The pivot *rows* of the echelon from A_{e} give us a basis in the row space. Of course, it is possible just to transpose the matrix, and then do row operations. But if we already have the echelon form of A, say by computing $\operatorname{Ran} A$, then we get $\operatorname{Ran} A^T$ for free.
3. To find a basis in the null space $\operatorname{Ker} A$ one needs to solve the homogeneous equation $A\mathbf{x} = \mathbf{0}$: the details will be seen from the example below.

Example. Consider a matrix

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 \\ 2 & 2 & 1 & 1 & 1 \\ 3 & 3 & 3 & 3 & 2 \\ 1 & 1 & -1 & -1 & 0 \end{pmatrix}.$$

Performing row operations we get the echelon form

$$\begin{pmatrix} \boxed{1} & 1 & 2 & 2 & 1 \\ 0 & 0 & \boxed{-3} & -3 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

(the pivots are boxed here). So, the columns 1 and 3 of the *original matrix*, i.e. the columns

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \\ 3 \\ -1 \end{pmatrix}$$

give us a basis in $\operatorname{Ran} A$. We also get a basis for the row space $\operatorname{Ran} A^T$ for free: the first and second row of the *echelon form* of A, i.e. the vectors

$$\begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ -3 \\ -3 \\ -1 \end{pmatrix}$$

(we put the vectors vertically here. The question of whether to put vectors here vertically as columns, or horizontally as rows is is really a matter of convention. Our reason for putting them vertically is that although we call $\operatorname{Ran} A^T$ the *row space* we define it as a column space of A^T)

To compute the basis in the null space $\operatorname{Ker} A$ we need to solve the equation $A\mathbf{x} = \mathbf{0}$. Compute the *reduced* echelon form of A, which in this example is

$$\begin{pmatrix} \boxed{1} & 1 & 0 & 0 & 1/3 \\ 0 & 0 & \boxed{1} & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Note, that when solving the homogeneous equation $A\mathbf{x} = \mathbf{0}$, it is not necessary to write the whole augmented matrix, it is sufficient to work with the coefficient matrix. Indeed, in this case the last column of the augmented matrix is the column of zeroes, which does not change under row operations. So, we can just keep this column in mind, without actually writing it. Keeping this last zero column in mind, we can read the solution off the

reduced echelon form above:

$$\begin{cases} x_1 = -x_2 - \frac{1}{3}x_5, \\ x_2 \text{ is free}, \\ x_3 = -x_4 - \frac{1}{3}x_5 \\ x_4 \text{ is free}, \\ x_5 \text{ is free}, \end{cases}$$

or, in the vector form

$$\mathbf{x} = \begin{pmatrix} -x_2 - \frac{1}{3}x_5 \\ x_2 \\ -x_4 - \frac{1}{3}x_5 \\ x_4 \\ x_5 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -1/3 \\ 0 \\ -1/3 \\ 0 \\ 1 \end{pmatrix} \tag{7.1}$$

The vectors at each free variable, i.e. in our case the vectors

$$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1/3 \\ 0 \\ -1/3 \\ 0 \\ 1 \end{pmatrix}$$

form a basis in $\operatorname{Ker} A$.

Unfortunately, there is no shortcut for finding a basis in $\operatorname{Ker} A^T$, one must solve the equation $A^T\mathbf{x} = \mathbf{0}$. The knowledge of the echelon form of A does not help here.

7.2. Explanation of the computing bases in the fundamental subspaces. So, why do the above methods indeed give us bases in the fundamental subspaces?

7.2.1. *The null space* $\operatorname{Ker} A$. The case of the null space $\operatorname{Ker} A$ is probably the simplest one: since we solved the equation $A\mathbf{x} = \mathbf{0}$, i.e. found all the solutions, then any vector in $\operatorname{Ker} A$ is a linear combination of the vectors we obtained. Thus, the vectors we obtained form a spanning system in $\operatorname{Ker} A$. To see that the system is linearly independent, let us multiply each vector by the corresponding free variable and add everything, see (7.1). Then for each free variable x_k, the entry number k of the resulting vector is exactly x_k, see (7.1) again, so the only way this vector (the linear combination) can be $\mathbf{0}$ is when all free variables are 0.

7.2.2. *The column space* $\operatorname{Ran} A$. Let us now explain why the method for finding a basis in the column space $\operatorname{Ran} A$ works. First of all, notice that the pivot columns of the *reduced echelon* form A_{re} of A form a basis in $\operatorname{Ran} A_{\text{re}}$ (not in the column space of the original matrix, but of its reduced

echelon form!). Since row operations are just left multiplications by invertible matrices, they do not change linear independence. Therefore, the pivot columns of the *original* matrix A are linearly independent.

Let us now show that the pivot columns of A span the column space of A. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ be the pivot columns of A, and let $\mathbf{v}$ be an arbitrary column of A. We want to show that $\mathbf{v}$ can be represented as a linear combination of the pivot columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$,

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_r \mathbf{v}_r.$$

The reduced echelon form A_{re} is obtained from A by the left multiplication

$$A_{\mathrm{re}} = EA,$$

where E is a product of elementary matrices, so E is an invertible matrix. The vectors $E\mathbf{v}_1, E\mathbf{v}_2, \dots, E\mathbf{v}_r$ are the pivot columns of A_{re}, and the column $\mathbf{v}$ of A is transformed to the column $E\mathbf{v}$ of A_{re}. Since the pivot columns of A_{re} form a basis in $\operatorname{Ran} A_{\mathrm{re}}$, vector $E\mathbf{v}$ can be represented as a linear combination

$$E\mathbf{v} = \alpha_1 E\mathbf{v}_1 + \alpha_2 E\mathbf{v}_2 + \ldots + \alpha_r E\mathbf{v}_r.$$

Multiplying this equality by E^{-1} from the left we get the representation

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \ldots + \alpha_r \mathbf{v}_r,$$

so indeed the pivot columns of A span $\operatorname{Ran} A$.

7.2.3. *The row space* $\operatorname{Ran} A^T$. It is easy to see that the pivot rows of the *echelon form* A_{e} of A are linearly independent. Indeed, let $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r$ be the transposed (since we agreed always to put vectors vertically) pivot rows of A_{e}. Suppose

$$\alpha_1 \mathbf{w}_1 + \alpha_2 \mathbf{w}_2 + \ldots + \alpha_r \mathbf{w}_r = \mathbf{0}.$$

Consider the first non-zero entry of $\mathbf{w}_1$. Since for all other vectors $\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_r$ the corresponding entries equal 0 (by the definition of echelon form), we can conclude that $\alpha_1 = 0$. So we can just ignore the first term in the sum.

Consider now the first non-zero entry of $\mathbf{w}_2$. The corresponding entries of the vectors $\mathbf{w}_3, \dots, \mathbf{w}_r$ are 0, so $\alpha_2 = 0$. Repeating this procedure, we get that $\alpha_k = 0 \ \forall k = 1, 2, \dots, r$.

To see that vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_r$ span the row space, one can notice that *row operations do not change the row space*. This can be obtained directly from analyzing row operations, but we present here a more formal way to demonstrate this fact.

For a transformation A and a set X let us denote by $A(X)$ the set of all elements y which can represented as $y = A(x)$, $x \in X$,

$$A(X) := \{y = A(x) : x \in X\}.$$

If A is an $m \times n$ matrix, and A_{e} is its echelon form, A_{e} is obtained from A be left multiplication

$$A_{\mathrm{e}} = EA,$$

where E is an $m \times m$ invertible matrix (the product of the corresponding elementary matrices). Then

$$\operatorname{Ran} A_{\mathrm{e}}^T = \operatorname{Ran}(A^T E^T) = A^T(\operatorname{Ran} E^T) = A^T(\mathbb{R}^m) = \operatorname{Ran} A^T,$$

so indeed $\operatorname{Ran} A^T = \operatorname{Ran} A_{\mathrm{e}}^T$.

7.3. The Rank Theorem. Dimensions of fundamental subspaces. There are many applications in which one needs to find a basis in column space or in the null space of a matrix. For example, as it was shown above, solving a homogeneous equation $A\mathbf{x} = \mathbf{0}$ amounts to finding a basis in the null space $\operatorname{Ker} A$. Finding a basis in the column space means simply extracting a basis from a spanning set, by removing unnecessary vectors (columns).

However, the most important application of the above methods of computing bases of fundamental subspaces is the relations between their dimensions.

Theorem 7.1 (The Rank Theorem)**.** *For a matrix A*

$$\operatorname{rank} A = \operatorname{rank} A^T.$$

This theorem is often stated as follows:

The *column rank* of a matrix coincides with its *row rank.*

The proof of this theorem is trivial, since dimensions of both $\operatorname{Ran} A$ and $\operatorname{Ran} A^T$ are equal to the number of pivots in the echelon form of A.

The following theorem is gives us important relations between dimensions of the fundamental spaces. It is often also called the Rank Theorem

Theorem 7.2. *Let A be an $m \times n$ matrix, i.e. a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$. Then*

1. $\dim \operatorname{Ker} A + \operatorname{rank} A = n$ *(dimensionof the domain of A);*
2. $\dim \operatorname{Ker} A^T + \operatorname{rank} A = m$ *(dimension of the target space of A).*

Proof. The proof, modulo the above algorithms of finding bases in the fundamental subspaces, is almost trivial. The first statement is simply the fact that the number of free variables ($\dim \operatorname{Ker} A$) plus the number of basic variables (i.e. the number of pivots, i.e. $\operatorname{rank} A$) adds up to the number of columns (i.e. to n).

The second statement, if one takes into account that $\operatorname{rank} A = \operatorname{rank} A^T$ is simply the first statement applied to A^T. □

As an application of the above theorem, let us recall the example from Section 6. There we considered a system

$$\begin{pmatrix} 2 & 3 & 1 & 4 & -9 \\ 1 & 1 & 1 & 1 & -3 \\ 1 & 1 & 1 & 2 & -5 \\ 2 & 2 & 2 & 3 & -8 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 17 \\ 6 \\ 8 \\ 14 \end{pmatrix},$$

and we claimed that its general solution given by

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 2 \\ -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \qquad x_3, x_5 \in \mathbb{F},$$

or by

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{F}.$$

We checked in Section 6 that a vector $\mathbf{x}$ given by either formula is indeed a solution of the equation. But, how can we guarantee that any of the formulas describe *all* solutions?

First of all, we know that in either formula, the last 2 vectors (the ones multiplied by the parameters) belong to $\operatorname{Ker} A$. It is easy to see that in either case both vectors are linearly independent (two vectors are linearly dependent if and only if one is a multiple of the other).

Now, let us count dimensions: interchanging the first and the second rows and performing first round of row operations

$$\begin{matrix} \\ -2R_1 \\ -R_1 \\ -2R_1 \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 1 & -3 \\ 2 & 3 & 1 & 4 & -9 \\ 1 & 1 & 1 & 2 & -5 \\ 2 & 2 & 2 & 3 & -8 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 1 & -3 \\ 0 & 1 & -1 & 2 & -3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$$

we see that there are three pivots already, so $\operatorname{rank} A \geq 3$. (Actually, we already can see that the rank is 3, but it is enough just to have the estimate here). By Theorem 7.2, $\operatorname{rank} A + \dim \operatorname{Ker} A = 5$, hence $\dim \operatorname{Ker} A \leq 2$, and therefore there cannot be more than 2 linearly independent vectors in $\operatorname{Ker} A$. Therefore, last 2 vectors in either formula form a basis in $\operatorname{Ker} A$, so either formula give all solutions of the equation.

An important corollary of the rank theorem, is the following theorem connecting existence and uniqueness for linear equations.

Theorem 7.3. *Let A be an $m \times n$ matrix. Then the equation*

$$A\mathbf{x} = \mathbf{b}$$

has a solution for every $\mathbf{b} \in \mathbb{R}^m$ if and only if the dual equation

$$A^T\mathbf{x} = \mathbf{0}$$

has a unique (only the trivial) solution. (Note, that in the second equation we have A^T, not A).

Proof. The proof follows immediately from Theorem 7.2 by counting the dimensions. We leave the details as an exercise to the reader. □

There is a very nice geometric interpretation of the second rank theorem (Theorem 7.2). Namely, statement 1 of the theorem says, that if a transformation $A : \mathbb{F}^n \to \mathbb{F}^m$ has trivial kernel ($\operatorname{Ker} A = \{\mathbf{0}\}$), then the dimensions of the domain $\mathbb{F}^n$ and of the range $\operatorname{Ran} A$ coincide. If the kernel is non-trivial, then the transformation "kills" $\dim \operatorname{Ker} A$ dimensions, so $\dim \operatorname{Ran} A = n - \dim \operatorname{Ker} A$.

7.4. Completion of a linearly independent system to a basis. As Proposition 5.4 from Section 5 above asserts, any linearly independent system can be completed to a basis, i.e. given inearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ in a finite-dimensional vector space V, one can find vectors $\mathbf{v}_{r+1}, \mathbf{v}_{r+2} \dots, \mathbf{v}_n$ such that the system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a basis in V.

Theoretically, the proof of this proposition give us an algorithm of finding the vectors $\mathbf{v}_{r+1}, \mathbf{v}_{r+2} \dots, \mathbf{v}_n$, but this algorithm does not look too practical.

Ideas of this section give us a more practical way to perform the completion to a basis.

First of all, notice that if an $m \times n$ matrix is in an echelon form, then its non-zero rows (which are clearly linearly independent) can be easily completed to a basis in the whole space $\mathbb{F}^n$; one just needs to add some rows in appropriate places, so the resulting matrix is still in the echelon form and has pivots in every column.

Then, the non-zero rows of the new matrix form a basis, and we can order it any way we want, because property of being basis does not depend on the ordering.

Suppose now that we have linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$, $\mathbf{v}_k \in \mathbb{F}^n$. Consider the matrix A with rows $\mathbf{v}_1^T, \mathbf{v}_2^T, \dots, \mathbf{v}_r^T$ and perform row operations to get the echelon form A_{e}. As we discussed above, the rows of A_{e} can be easily completed to a basis in $\mathbb{R}^n$. And it turns out that the same

vectors that complete rows of A_{e} to a basis complete to a basis the original vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$.

To see that, let vectors $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ complete the rows of A_{e} to a basis in $\mathbb{F}^n$. Then, if we add to a matrix A_{e} rows $\mathbf{v}_{r+1}^T, \ldots, \mathbf{v}_n^T$, we get an invertible matrix. Let call this matrix $\widetilde{A}_{\mathrm{e}}$, and let $\widetilde{A}$ be the matrix obtained from A by adding rows $\mathbf{v}_{r+1}^T, \ldots, \mathbf{v}_n^T$. The matrix $\widetilde{A}_{\mathrm{e}}$ can be obtained from $\widetilde{A}$ by row operations, so

$$\widetilde{A}_{\mathrm{e}} = E\widetilde{A},$$

where E is the product of the corresponding elementary matrices. Then $\widetilde{A} = E^{-1}A_{\mathrm{e}}$ and $\widetilde{A}$ is invertible as a product of invertible matrices.

But that means that the rows of $\widetilde{A}$ form a basis in $\mathbb{F}^n$, which is exactly what we need.

Remark. The method of completion to a basis described above may be not the simplest one, but one of its principal advantages is that it works for vector spaces over an arbitrary field.

Exercises.

7.1. True or false:

a) The rank of a matrix is equal to the number of its non-zero columns;

b) The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0;

c) Elementary row operations preserve rank;

d) Elementary column operations do not necessarily preserve rank;

e) The rank of a matrix is equal to the maximum number of linearly independent columns in the matrix;

f) The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix;

g) The rank of an $n \times n$ matrix is at most n;

h) An $n \times n$ matrix having rank n is invertible.

7.2. A 54×37 matrix has rank 31. What are dimensions of all 4 fundamental subspaces?

7.3. Compute rank and find bases of all four fundamental subspaces for the matrices

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 & 3 & 1 & 1 \\ 1 & 4 & 0 & 1 & 2 \\ 0 & 2 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

7.4. Prove that if $A : X \to Y$ and V is a subspace of X then $\dim AV \le \operatorname{rank} A$. ($AV$ here means the subspace V transformed by the transformation A, i.e. any vector in AV can be represented as $A\mathbf{v}$, $\mathbf{v} \in V$). Deduce from here that $\operatorname{rank}(AB) \le \operatorname{rank} A$.

Remark: Here one can use the fact that if $V \subset W$ then $\dim V \le \dim W$. Do you understand why is it true?

7.5. Prove that if $A : X \to Y$ and V is a subspace of X then $\dim AV \le \dim V$. Deduce from here that $\operatorname{rank}(AB) \le \operatorname{rank} B$.

7.6. Prove that if the product AB of two $n \times n$ matrices is invertible, then both A and B are invertible. Even if you know about determinants, do not use them, we did not cover them yet. **Hint:** use previous 2 problems.

7.7. Prove that if $A\mathbf{x} = \mathbf{0}$ has unique solution, then the equation $A^T\mathbf{x} = \mathbf{b}$ has a solution for every right side $\mathbf{b}$.
Hint: count pivots

7.8. Write a matrix with the required property, or explain why no such matrix exists

a) Column space contains $(1,0,0)^T$, $(0,0,1)^T$, row space contains $(1,1)^T$, $(1,2)^T$;

b) Column space is spanned by $(1,1,1)^T$, nullspace is spanned by $(1,2,3)^T$;

c) Column space is $\mathbb{R}^4$, row space is $\mathbb{R}^3$.

Hint: Check first if the dimensions add up.

7.9. If A has the same four fundamental subspaces as B, does $A = B$?

7.10. Complete the rows of a matrix

$$\begin{pmatrix} e^3 & 3 & 4 & 0 & -\pi & 6 & -2 \\ 0 & 0 & 2 & -1 & \pi^e & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 & -3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

to a basis in $\mathbb{R}^7$.

7.11. For a matrix

$$\begin{pmatrix} 1 & 2 & -1 & 2 & 3 \\ 2 & 2 & 1 & 5 & 5 \\ 3 & 6 & -3 & 0 & 24 \\ -1 & -4 & 4 & -7 & 11 \end{pmatrix}$$

find bases in its column and row spaces.

7.12. For the matrix in the previous problem, complete the basis in the row space to a basis in $\mathbb{R}^5$

7.13. For the matrix

$$A = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}$$

compute $\operatorname{Ran} A$ and $\operatorname{Ker} A$. What can you say about relation between these subspaces?

7.14. Is it possible that for a real matrix A that $\operatorname{Ran} A = \operatorname{Ker} A^T$? Is it possible for a complex A?

7.15. Complete the vectors $(1,2,-1,2,3)^T$, $(2,2,1,5,5)^T$, $(-1,-4,4,7,-11)^T$ to a basis in $\mathbb{R}^5$.

8. Representation of a linear transformation in arbitrary bases. Change of coordinates formula.

The material we have learned about linear transformations and their matrices can be easily extended to transformations in abstract vector spaces with finite bases. In this section we will distinguish between a linear transformation T and its matrix, the reason being that we consider different bases, so a linear transformation can have different matrix representation.

8.1. Coordinate vector. Let V be a vector space with a basis $\mathcal{B} := \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$. Any vector $\mathbf{v} \in V$ admits a unique representation as a linear combination

$$\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2 + \dots + x_n\mathbf{b}_n = \sum_{k=1}^{n} x_k\mathbf{b}_k.$$

The numbers $x_1, x_2, \dots, x_n$ are called the *coordinates* of the vector $\mathbf{v}$ in the basis $\mathcal{B}$. It is convenient to join these coordinates into the so-called *coordinate vector* of $\mathbf{v}$ relative to the basis $\mathcal{B}$, which is the column vector

$$[\mathbf{v}]_{\mathcal{B}} := \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n.$$

Note that the mapping

$$\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$$

is an isomorphism between V and $\mathbb{F}^n$. It transforms the basis $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ to the standard basis $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ in $\mathbb{F}^n$.

8.2. Matrix of a linear transformation. Let $T : V \to W$ be a linear transformation, and let $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$, $\mathcal{B} := \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_m\}$ be bases in V and W respectively.

A matrix of the transformation T in (or with respect to) the bases $\mathcal{A}$ and $\mathcal{B}$ is an $m \times n$ matrix, denoted by $[T]_{\mathcal{B}\mathcal{A}}$, which relates the coordinate vectors $[T\mathbf{v}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{A}}$,

$$[T\mathbf{v}]_{\mathcal{B}} = [T]_{\mathcal{B}\mathcal{A}}[\mathbf{v}]_{\mathcal{A}};$$

notice the balance of symbols $\mathcal{A}$ and $\mathcal{B}$ here: this is the reason we put the first basis $\mathcal{A}$ into the second position.

The matrix $[T]_{\mathcal{B}\mathcal{A}}$ is easy to find: its kth column is just the coordinate vector $[T\mathbf{a}_k]_{\mathcal{B}}$ (compare this with finding the matrix of a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$).

As in the case of standard bases, composition of linear transformations is equivalent to multiplication of their matrices: one only has to be a bit more careful about bases. Namely, let $T_1 : X \to Y$ and $T_2 : Y \to Z$ be linear transformation, and let $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$ be bases in X, Y and Z respectively. Then for the composition $T = T_2 T_1$,

$$T : X \to Z, \qquad T\mathbf{x} := T_2(T_1(\mathbf{x}))$$

we have

$$[T]_{\mathcal{C}\mathcal{A}} = [T_2 T_1]_{\mathcal{C}\mathcal{A}} = [T_2]_{\mathcal{C}\mathcal{B}} [T_1]_{\mathcal{B}\mathcal{A}} \tag{8.1}$$

(notice again the balance of indices here).

The proof here goes exactly as in the case of $\mathbb{F}^n$ spaces with standard bases, so we do not repeat it here. Another possibility is to transfer everything to the spaces $\mathbb{F}^n$ via the coordinate isomorphisms $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$. Then one does not need any proof, everything follows from the results about matrix multiplication.

8.3. Change of coordinate matrix. Let us have two bases $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ in a vector space V. Consider the identity transformation $I = I_V$ and its matrix $[I]_{\mathcal{B}\mathcal{A}}$ in these bases. By the definition

$$[\mathbf{v}]_{\mathcal{B}} = [I]_{\mathcal{B}\mathcal{A}} [\mathbf{v}]_{\mathcal{A}}, \qquad \forall \mathbf{v} \in V,$$

i.e. for any vector $\mathbf{v} \in V$ the matrix $[I]_{\mathcal{B}\mathcal{A}}$ transforms its coordinates in the basis $\mathcal{A}$ into coordinates in the basis $\mathcal{B}$. The matrix $[I]_{\mathcal{B}\mathcal{A}}$ is often called the *change of coordinates* (from the basis $\mathcal{A}$ to the basis $\mathcal{B}$) matrix.

The matrix $[I]_{\mathcal{B}\mathcal{A}}$ is easy to compute: according to the general rule of finding the matrix of a linear transformation, its kth column is the coordinate representation $[\mathbf{a}_k]_{\mathcal{B}}$ of kth element of the basis $\mathcal{A}$

Note that

$$[I]_{\mathcal{A}\mathcal{B}} = ([I]_{\mathcal{B}\mathcal{A}})^{-1},$$

(follows immediately from the multiplication of matrices rule (8.1)), so any change of coordinate matrix is always invertible.

8.3.1. *An example: change of coordinates from the standard basis.* Let our space V be $\mathbb{F}^n$, and let us have a basis $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ there. We also have the standard basis $\mathcal{S} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ there. The change of coordinates matrix $[I]_{\mathcal{S}\mathcal{B}}$ is easy to compute:

$$[I]_{\mathcal{S}\mathcal{B}} = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n] =: B,$$

i.e. it is just the matrix B whose kth column is the vector (column) $\mathbf{v}_k$. And in the other direction

$$[I]_{\mathcal{B}\mathcal{S}} = ([I]_{\mathcal{S}\mathcal{B}})^{-1} = B^{-1}.$$

For example, consider a basis

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

in $\mathbb{F}^2$, and let $\mathcal{S}$ denote the standard basis there. Then

$$[I]_{\mathcal{SB}} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} =: B$$

and

$$[I]_{\mathcal{BS}} = [I]_{\mathcal{SB}}^{-1} = B^{-1} = \frac{1}{3} \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$$

(we know how to compute inverses, and it is also easy to check that the above matrix is indeed the inverse of B)

8.3.2. *An example: going through the standard basis.* In the space of polynomials of degree at most 1 we have bases

$$\mathcal{A} = \{1, 1+x\}, \quad \text{and} \quad \mathcal{B} = \{1+2x, 1-2x\},$$

and we want to find the change of coordinate matrix $[I]_{\mathcal{BA}}$.

Of course, we can always take vectors from the basis $\mathcal{A}$ and try to decompose them in the basis $\mathcal{B}$; it involves solving linear systems, and we know how to do that.

However, I think the following way is simpler. In $\mathbb{P}_1$ we also have the standard basis $\mathcal{S} = \{1, x\}$, and for this basis

$$[I]_{\mathcal{SA}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} =: A, \qquad [I]_{\mathcal{SB}} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} =: B,$$

and taking the inverses

$$[I]_{\mathcal{AS}} = A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad [I]_{\mathcal{BS}} = B^{-1} = \frac{1}{4} \begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix}.$$

Then

$$[I]_{\mathcal{BA}} = [I]_{\mathcal{BS}}[I]_{\mathcal{SA}} = B^{-1}A = \frac{1}{4} \begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and

$$[I]_{\mathcal{AB}} = [I]_{\mathcal{AS}}[I]_{\mathcal{SB}} = A^{-1}B = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}$$

Notice the balance of indices here.

8.4. Matrix of a transformation and change of coordinates. Let $T : V \to W$ be a linear transformation, and let $\mathcal{A}$, $\widetilde{\mathcal{A}}$ be two bases in V and let $\mathcal{B}$, $\widetilde{\mathcal{B}}$ be two bases in W. Suppose we know the matrix $[T]_{\mathcal{BA}}$, and we would like to find the matrix representation with respect to new bases $\widetilde{\mathcal{A}}$, $\widetilde{\mathcal{B}}$, i.e. the matrix $[T]_{\widetilde{\mathcal{B}}\widetilde{\mathcal{A}}}$. The rule is very simple:

to get the matrix in the "new" bases one has to surround the matrix in the "old" bases by change of coordinates matrices.

I did not mention here what change of coordinate matrix should go where, because we don't have any choice if we follow the balance of indices rule. Namely, matrix representation of a linear transformation changes according to the formula

Notice the balance of indices.

$$\boxed{[T]_{\widetilde{\mathcal{B}}\widetilde{\mathcal{A}}} = [I]_{\widetilde{\mathcal{B}}\mathcal{B}}[T]_{\mathcal{B}\mathcal{A}}[I]_{\mathcal{A}\widetilde{\mathcal{A}}}}$$

The proof can be done just by analyzing what each of the matrices does.

8.5. Case of one basis: similar matrices. Let V be a vector space and let $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ be a basis in V. Consider a linear transformation $T : V \to V$ and let $[T]_{\mathcal{A}\mathcal{A}}$ be its matrix in this basis (we use the same basis for "inputs" and "outputs")

The case when we use the same basis for "inputs" and "outputs" is very important (because in this case we can multiply a matrix by itself), so let us study this case a bit more carefully. Notice, that very often in this case the shorter notation $[T]_{\mathcal{A}}$ is used instead of $[T]_{\mathcal{A}\mathcal{A}}$. However, the two index notation $[T]_{\mathcal{A}\mathcal{A}}$ is better adapted to the balance of indices rule, so I recommend using it (or at least always keep it in mind) when doing change of coordinates.

$[T]_{\mathcal{A}}$ is often used instead of $[T]_{\mathcal{A}\mathcal{A}}$. It is shorter, but two index notation is better adapted to the balance of indices rule.

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be another basis in V. By the change of coordinate rule above

$$[T]_{\mathcal{B}\mathcal{B}} = [I]_{\mathcal{B}\mathcal{A}}[T]_{\mathcal{A}\mathcal{A}}[I]_{\mathcal{A}\mathcal{B}}$$

Recalling that

$$[I]_{\mathcal{B}\mathcal{A}} = [I]^{-1}_{\mathcal{A}\mathcal{B}}$$

and denoting $Q := [I]_{\mathcal{A}\mathcal{B}}$, we can rewrite the above formula as

$$[T]_{\mathcal{B}\mathcal{B}} = Q^{-1}[T]_{\mathcal{A}\mathcal{A}}Q.$$

This gives a motivation for the following definition

Definition 8.1. We say that a matrix A is similar to a matrix B if there exists an invertible matrix Q such that $A = Q^{-1}BQ$.

Since an invertible matrix must be square, it follows from counting dimensions, that similar matrices A and B have to be square and of the same size. If A is similar to B, i.e. if $A = Q^{-1}BQ$, then

$$B = QAQ^{-1} = (Q^{-1})^{-1}A(Q^{-1})$$

(since Q^{-1} is invertible), therefore B is similar to A. So, we can just say that A and B are *similar*.

The above reasoning shows, that it does not matter where to put Q and where Q^{-1}: one can use the formula $A = QBQ^{-1}$ in the definition of similarity.

The above discussion shows, that one can treat similar matrices as different matrix representation of the same linear operator (transformation).

Exercises.

8.1. True or false

a) Every change of coordinate matrix is square;

b) Every change of coordinate matrix is invertible;

c) The matrices A and B are called similar if $B = Q^T AQ$ for some matrix Q;

d) The matrices A and B are called similar if $B = Q^{-1}AQ$ for some matrix Q;

e) Similar matrices do not need to be square.

8.2. Consider the system of vectors

$$(1, 2, 1, 1)^T, \quad (0, 1, 3, 1)^T, \quad (0, 3, 2, 0)^T, \quad (0, 1, 0, 0)^T.$$

a) Prove that it is a basis in $\mathbb{F}^4$. Try to do minimal amount of computations.

b) Find the change of coordinate matrix that changes the coordinates in this basis to the standard coordinates in $\mathbb{F}^4$ (i.e. to the coordinates in the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_4$).

8.3. Find the change of coordinates matrix that changes the coordinates in the basis $1, 1 + t$ in $\mathbb{P}_1$ to the coordinates in the basis $1 - t, 2t$.

8.4. Let T be the linear operator in $\mathbb{F}^2$ defined (in the standard coordinates) by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x - 2y \end{pmatrix}$$

Find the matrix of T in the standard basis and in the basis

$$(1, 1)^T, \qquad (1, 2)^T.$$

8.5. Prove, that if A and B are similar matrices then $\operatorname{trace} A = \operatorname{trace} B$. **Hint:** recall how $\operatorname{trace}(XY)$ and $\operatorname{trace}(YX)$ are related.

8.6. Are the matrices

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} 0 & 2 \\ 4 & 2 \end{pmatrix}$$

similar? Justify.

INNER PRODUCT SPACES

Treil Chapter 5, Sections 1-5

Chapter 5

Inner product spaces

Theory of inner product spaces is developed only for real and complex spaces, so $\mathbb{F}$ in this Chapter is always $\mathbb{R}$ or $\mathbb{C}$; the results usually do not generalize to spaces over arbitrary fields.

Most of the results and calculations in this chapter hold (and the results have the same statements) in both real and complex cases. In rare situations when there is a difference between real and complex case, we state explicitly which case is considered: otherwise everything holds for both cases.

Finally, when the results and calculations hold for both complex and real cases, we use formulas for the complex case; in the real case they give correct, although sometimes a bit more complicated, formulas.

1. Inner product in $\mathbb{R}^n$ and $\mathbb{C}^n$. Inner product spaces.

1.1. Inner product and norm in $\mathbb{R}^n$. In dimensions 2 and 3, we defined the length of a vector $\mathbf{x}$ (i.e. the distance from its endpoint to the origin) by the Pythagorean rule, for example in $\mathbb{R}^3$ the length of the vector is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + x_3^2}.$$

It is natural to generalize this formula for all n, to define the *norm* of the vector $\mathbf{x} \in \mathbb{R}^n$ as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}.$$

The word *norm* is used as a fancy replacement for the word length.

The *dot product* in $\mathbb{R}^3$ was defined as $\mathbf{x} \cdot \mathbf{y} = x_1y_1 + x_2y_2 + x_3y_3$, where $\mathbf{x} = (x_1, x_2, x_3)^T$ and $\mathbf{y} = (y_1, y_2, y_3)^T$.

While the notation $\mathbf{x} \cdot \mathbf{y}$ and term "dot product" is often used for the inner product, for reasons which will be clear later, we prefer the notation $(\mathbf{x}, \mathbf{y})$

Similarly, in $\mathbb{R}^n$ one can define the *inner product* $(\mathbf{x}, \mathbf{y})$ of two vectors $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$, $\mathbf{y} = (y_1, y_2, \dots, y_n)^T$ by

$$(\mathbf{x}, \mathbf{y}) := x_1y_1 + x_2y_2 + \dots + x_ny_n = \mathbf{y}^T\mathbf{x},$$

so $\|\mathbf{x}\| = \sqrt{(\mathbf{x}, \mathbf{x})}$.

Note, that $\mathbf{y}^T\mathbf{x} = \mathbf{x}^T\mathbf{y}$, and we use the notation $\mathbf{y}^T\mathbf{x}$ only to be consistent.

1.2. Inner product and norm in $\mathbb{C}^n$. Let us now define norm and inner product for $\mathbb{C}^n$. As we have seen before, the complex space $\mathbb{C}^n$ is the most natural space from the point of view of spectral theory: even if one starts from a matrix with real coefficients (or operator on a real vectors space), the eigenvalues can be complex, and one needs to work in a complex space.

For a complex number $z = x + iy$, we have $|z|^2 = x^2 + y^2 = z\overline{z}$. If $\mathbf{z} \in \mathbb{C}^n$ is given by

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ \vdots \\ x_n + iy_n \end{pmatrix},$$

it is natural to define its norm $\|\mathbf{z}\|$ by

$$\|\mathbf{z}\|^2 = \sum_{k=1}^{n} (x_k^2 + y_k^2) = \sum_{k=1}^{n} |z_k|^2.$$

Let us try to define an inner product on $\mathbb{C}^n$ such that $\|\mathbf{z}\|^2 = (\mathbf{z}, \mathbf{z})$. One of the choices is to define $(\mathbf{z}, \mathbf{w})$ by

$$(\mathbf{z}, \mathbf{w}) = z_1\overline{w}_1 + z_2\overline{w}_2 + \dots + z_n\overline{w}_n = \sum_{k=1}^{n} z_k\overline{w}_k,$$

and that will be our definition of the *standard* inner product in $\mathbb{C}^n$.

To simplify the notation, let us introduce a new notion. For a matrix A let us define its *Hermitian adjoint*, or simply *adjoint* A^* by $A^* = \overline{A}^T$, meaning that we take the transpose of the matrix, and then take the complex conjugate of each entry. Note, that for a real matrix A, $A^* = A^T$.

Using the notion of A^*, one can write the standard inner product in $\mathbb{C}^n$ as

$$(\mathbf{z}, \mathbf{w}) = \mathbf{w}^*\mathbf{z}.$$

Remark. It is easy to see that one can define a different inner product in $\mathbb{C}^n$ such that $\|\mathbf{z}\|^2 = (\mathbf{z}, \mathbf{z})$, namely the inner product given by

$$(\mathbf{z}, \mathbf{w})_1 = \overline{z}_1 w_1 + \overline{z}_2 w_2 + \ldots + \overline{z}_n w_n = \mathbf{z}^* \mathbf{w}.$$

We did not specify what properties we want the inner product to satisfy, but $\mathbf{z}^*\mathbf{w}$ and $\mathbf{w}^*\mathbf{z}$ are the only reasonable choices giving $\|\mathbf{z}\|^2 = (\mathbf{z}, \mathbf{z})$.

Note, that the above two choices of the inner product are essentially equivalent: the only difference between them is notational, because $(\mathbf{z}, \mathbf{w})_1 = (\mathbf{w}, \mathbf{z})$.

While the second choice of the inner product looks more natural, the first one, $(\mathbf{z}, \mathbf{w}) = \mathbf{w}^*\mathbf{z}$ is more widely used, so we will use it as well.

1.3. Inner product spaces. The inner product we defined for $\mathbb{R}^n$ and $\mathbb{C}^n$ satisfies the following properties:

1. (Conjugate) symmetry: $(\mathbf{x}, \mathbf{y}) = \overline{(\mathbf{y}, \mathbf{x})}$; note, that for a real space, this property is just symmetry, $(\mathbf{x}, \mathbf{y}) = (\mathbf{y}, \mathbf{x})$;
2. Linearity: $(\alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z}) = \alpha(\mathbf{x}, \mathbf{z}) + \beta(\mathbf{y}, \mathbf{z})$ for all vector $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and all scalars α, β;
3. Non-negativity: $(\mathbf{x}, \mathbf{x}) \geq 0$ $\forall \mathbf{x}$;
4. Non-degeneracy: $(\mathbf{x}, \mathbf{x}) = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Let V be a (complex or real) vector space. An *inner product* on V is a function, that assign to each pair of vectors $\mathbf{x}$, $\mathbf{y}$ a scalar, denoted by $(\mathbf{x}, \mathbf{y})$ such that the above properties 1–4 are satisfied.

Note that for a real space V we assume that $(\mathbf{x}, \mathbf{y})$ is always real, and for a complex space the inner product $(\mathbf{x}, \mathbf{y})$ can be complex.

A space V together with an inner product on it is called *an inner product space*. Given an inner product space, one defines the norm on it by

$$\|\mathbf{x}\| = \sqrt{(\mathbf{x}, \mathbf{x})}.$$

1.3.1. *Examples.*

Example 1.1. Let V be $\mathbb{R}^n$ or $\mathbb{C}^n$. We already have an inner product $(\mathbf{x}, \mathbf{y}) = \mathbf{y}^*\mathbf{x} = \sum_{k=1}^n x_k \overline{y}_k$ defined above.

This inner product is called the *standard* inner product in $\mathbb{R}^n$ or $\mathbb{C}^n$

We will use symbol $\mathbb{F}$ to denote both $\mathbb{C}$ and $\mathbb{R}$. When we have some statement about the space $\mathbb{F}^n$, it means the statement is true for both $\mathbb{R}^n$ and $\mathbb{C}^n$.

Example 1.2. Let V be the space $\mathbb{P}_n$ of polynomials of degree at most n. Define the inner product by

$$(f, g) = \int_{-1}^{1} f(t)\overline{g(t)}dt.$$

It is easy to check, that the above properties 1–4 are satisfied.

This definition works both for complex and real cases. In the real case we only allow polynomials with real coefficients, and we do not need the complex conjugate here.

Let us recall, that for a square matrix A, its *trace* is defined as the sum of the diagonal entries,

$$\operatorname{trace} A := \sum_{k=1}^{n} a_{k,k}.$$

Example 1.3. For the space $M_{m\times n}$ of $m \times n$ matrices let us define the so-called Frobenius inner product by

$$(A, B) = \operatorname{trace}(B^* A).$$

Again, it is easy to check that the properties 1–4 are satisfied, i.e. that we indeed defined an inner product.

Note, that

$$\operatorname{trace}(B^* A) = \sum_{j,k} A_{j,k}\overline{B}_{j,k},$$

so this inner product coincides with the standard inner product in $\mathbb{C}^{mn}$.

1.4. Properties of inner product. The statements we get in this section are true for any abstract inner product space, not only for $\mathbb{F}^n$. To prove them we use only properties 1–4 of the inner product.

First of all let us notice, that properties 1 and 2 imply that

$2'$. $(\mathbf{x}, \alpha\mathbf{y} + \beta\mathbf{z}) = \overline{\alpha}(\mathbf{x}, \mathbf{y}) + \overline{\beta}(\mathbf{x}, \mathbf{z})$.

Indeed,

$$(\mathbf{x}, \alpha\mathbf{y} + \beta\mathbf{z}) = \overline{(\alpha\mathbf{y} + \beta\mathbf{z}, \mathbf{x})} = \overline{\alpha(\mathbf{y}, \mathbf{x}) + \beta(\mathbf{z}, \mathbf{x})} =$$
$$= \overline{\alpha}\overline{(\mathbf{y}, \mathbf{x})} + \overline{\beta}\,\overline{(\mathbf{z}, \mathbf{x})} = \overline{\alpha}(\mathbf{x}, \mathbf{y}) + \overline{\beta}(\mathbf{x}, \mathbf{z})$$

Note also that property 2 implies that for all vectors $\mathbf{x}$

$$(\mathbf{0}, \mathbf{x}) = (\mathbf{x}, \mathbf{0}) = 0.$$

Lemma 1.4. *Let $\mathbf{x}$ be a vector in an inner product space V. Then $\mathbf{x} = \mathbf{0}$ if and only if*

$$(\mathbf{x}, \mathbf{y}) = 0 \qquad \forall \mathbf{y} \in V. \tag{1.1}$$

Proof. Since $(\mathbf{0}, \mathbf{y}) = 0$ we only need to show that (1.1) implies $\mathbf{x} = \mathbf{0}$. Putting $\mathbf{y} = \mathbf{x}$ in (1.1) we get $(\mathbf{x}, \mathbf{x}) = 0$, so $\mathbf{x} = \mathbf{0}$. □

Applying the above lemma to the difference $\mathbf{x} - \mathbf{y}$ we get the following

Corollary 1.5. *Let $\mathbf{x}, \mathbf{y}$ be vectors in an inner product space V. The equality $\mathbf{x} = \mathbf{y}$ holds if and only if*

$$(\mathbf{x}, \mathbf{z}) = (\mathbf{y}, \mathbf{z}) \qquad \forall \mathbf{z} \in V.$$

The following corollary is very simple, but will be used a lot

Corollary 1.6. *Suppose two operators $A, B : X \to Y$ satisfy*

$$(A\mathbf{x}, \mathbf{y}) = (B\mathbf{x}, \mathbf{y}) \qquad \forall \mathbf{x} \in X,\ \forall \mathbf{y} \in Y.$$

Then $A = B$.

Proof. By the previous corollary (fix $\mathbf{x}$ and take all possible $\mathbf{y}$'s) we get $A\mathbf{x} = B\mathbf{x}$. Since this is true for all $\mathbf{x} \in X$, the transformations A and B coincide. □

The following property relates the norm and the inner product.

Theorem 1.7 (Cauchy–Schwarz inequality)**.**

$$|(\mathbf{x}, \mathbf{y})| \le \|\mathbf{x}\| \cdot \|\mathbf{y}\|.$$

Proof. The proof we are going to present, is not the shortest one, but it shows where the main ideas came from.

Let us consider the real case first. If $\mathbf{y} = \mathbf{0}$, the statement is trivial, so we can assume that $\mathbf{y} \neq \mathbf{0}$. By the properties of an inner product, for all scalar t

$$0 \le \|\mathbf{x} - t\mathbf{y}\|^2 = (\mathbf{x} - t\mathbf{y}, \mathbf{x} - t\mathbf{y}) = \|\mathbf{x}\|^2 - 2t(\mathbf{x}, \mathbf{y}) + t^2\|\mathbf{y}\|^2.$$

In particular, this inequality should hold for $t = \frac{(\mathbf{x},\mathbf{y})}{\|\mathbf{y}\|^2}$ [1], and for this point the inequality becomes

$$0 \le \|\mathbf{x}\|^2 - 2\frac{(\mathbf{x}, \mathbf{y})^2}{\|\mathbf{y}\|^2} + \frac{(\mathbf{x}, \mathbf{y})^2}{\|\mathbf{y}\|^2} = \|\mathbf{x}\|^2 - \frac{(\mathbf{x}, \mathbf{y})^2}{\|\mathbf{y}\|^2},$$

which is exactly the inequality we need to prove.

There are several possible ways to treat the complex case. One is to replace $\mathbf{x}$ by $\alpha\mathbf{x}$, where α is a complex constant, $|\alpha| = 1$ such that $(\alpha\mathbf{x}, \mathbf{y})$ is real, and then repeat the proof for the real case.

The other possibility is again to consider

$$\begin{aligned} 0 \le \|\mathbf{x} - t\mathbf{y}\|^2 &= (\mathbf{x} - t\mathbf{y}, \mathbf{x} - t\mathbf{y}) = (\mathbf{x}, \mathbf{x} - t\mathbf{y}) - t(\mathbf{y}, \mathbf{x} - t\mathbf{y}) \\ &= \|\mathbf{x}\|^2 - t(\mathbf{y}, \mathbf{x}) - \bar{t}(\mathbf{x}, \mathbf{y}) + |t|^2\|\mathbf{y}\|^2. \end{aligned}$$

[1] That is the point where the above quadratic polynomial has a minimum: it can be computed, for example by taking the derivative in t and equating it to 0

Substituting $t = \frac{(\mathbf{x},\mathbf{y})}{\|\mathbf{y}\|^2} = \frac{\overline{(\mathbf{y},\mathbf{x})}}{\|\mathbf{y}\|^2}$ into this inequality, we get

$$0 \leq \|\mathbf{x}\|^2 - \frac{|(\mathbf{x},\mathbf{y})|^2}{\|\mathbf{y}\|^2}$$

which is the inequality we need.

Note, that the above paragraph is in fact a complete formal proof of the theorem. The reasoning before that was only to explain why do we need to pick this particular value of t. □

An immediate Corollary of the Cauchy–Schwarz Inequality is the following lemma.

Lemma 1.8 (Triangle inequality)**.** *For any vectors* $\mathbf{x}$*,* $\mathbf{y}$ *in an inner product space*

$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Proof.

$$\begin{aligned}\|\mathbf{x}+\mathbf{y}\|^2 = (\mathbf{x}+\mathbf{y},\mathbf{x}+\mathbf{y}) &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + (\mathbf{x},\mathbf{y}) + (\mathbf{y},\mathbf{x}) \\ &\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2|(\mathbf{x},\mathbf{y})| \\ &\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2\|\mathbf{x}\|\cdot\|\mathbf{y}\| = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2.\end{aligned}$$

□

The following *polarization identities* allow one to reconstruct the inner product from the norm:

Lemma 1.9 (Polarization identities)**.** *For* $\mathbf{x}, \mathbf{y} \in V$

$$(\mathbf{x},\mathbf{y}) = \frac{1}{4}\left(\|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2\right)$$

if V *is a real inner product space, and*

$$(\mathbf{x},\mathbf{y}) = \frac{1}{4}\sum_{\alpha=\pm1,\pm i} \alpha\|\mathbf{x}+\alpha\mathbf{y}\|^2$$

if V *is a complex inner product space.*

The lemma is proved by direct computation. We leave the proof as an exercise for the reader.

Another important property of the norm in an inner product space can be also checked by direct calculation.

Lemma 1.10 (Parallelogram Identity)**.** *For any vectors* $\mathbf{u}, \mathbf{v}$

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2).$$

In 2-dimensional space this lemma relates sides of a parallelogram with its diagonals, which explains the name. It is a well-known fact from planar geometry.

1.5. Norm. Normed spaces. We have proved before that the norm $\|\mathbf{v}\|$ satisfies the following properties:

1. Homogeneity: $\|\alpha\mathbf{v}\| = |\alpha| \cdot \|\mathbf{v}\|$ for all vectors $\mathbf{v}$ and all scalars α.
2. Triangle inequality: $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.
3. Non-negativity: $\|\mathbf{v}\| \ge 0$ for all vectors $\mathbf{v}$.
4. Non-degeneracy: $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Suppose in a vector space V we assigned to each vector $\mathbf{v}$ a number $\|\mathbf{v}\|$ such that above properties 1–4 are satisfied. Then we say that the function $\mathbf{v} \mapsto \|\mathbf{v}\|$ is a norm. A vector space V equipped with a norm is called a *normed space*.

Any inner product space is a normed space, because the norm $\|\mathbf{v}\| = \sqrt{(\mathbf{v}, \mathbf{v})}$ satisfies the above properties 1–4. However, there are many other normed spaces. For example, given p, $1 \le p < \infty$ one can define the norm $\|\cdot\|_p$ on $\mathbb{R}^n$ or $\mathbb{C}^n$ by

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + \ldots + |x_n|^p)^{1/p} = \left(\sum_{k=1}^{n} |x_k|^p\right)^{1/p}.$$

One can also define the norm $\|\cdot\|_\infty$ ($p = \infty$) by

$$\|\mathbf{x}\|_\infty = \max\{|x_k| : k = 1, 2, \ldots, n\}.$$

The norm $\|\cdot\|_p$ for $p = 2$ coincides with the regular norm obtained from the inner product.

To check that $\|\cdot\|_p$ is indeed a norm one has to check that it satisfies all the above properties 1–4. Properties 1, 3 and 4 are very easy to check, we leave it as an exercise for the reader. The triangle inequality (property 2) is easy to check for $p = 1$ and $p = \infty$ (and we proved it for $p = 2$).

For all other p the triangle inequality is true, but the proof is not so simple, and we will not present it here. The triangle inequality for $\|\cdot\|_p$ even has special name: its called *Minkowski inequality*, after the German mathematician H. Minkowski.

Note, that the norm $\|\cdot\|_p$ for $p \ne 2$ cannot be obtained from an inner product. It is easy to see that this norm is not obtained from the *standard* inner product in $\mathbb{R}^n$ ($\mathbb{C}^n$). But we claim more! We claim that it is *impossible to introduce* an inner product which gives rise to the norm $\|\cdot\|_p$, $p \ne 2$.

This statement is actually quite easy to prove. By Lemma 1.10 any norm obtained from an inner product must satisfy the Parallelogram Identity. It

is easy to see that the Parallelogram Identity fails for the norm $\|\cdot\|_p$, $p \neq 2$, and one can easily find a counterexample in $\mathbb{R}^2$, which then gives rise to a counterexample in all other spaces.

In fact, the Parallelogram Identity, as the theorem below asserts, completely characterizes norms obtained from an inner product.

Theorem 1.11. *A norm in a normed space is obtained from some inner product if and only if it satisfies the Parallelogram Identity*

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2) \qquad \forall \mathbf{u}, \mathbf{v} \in V.$$

Lemma 1.10 asserts that a norm obtained from an inner product satisfies the Parallelogram Identity.

The converse implication is more complicated. If we are given a norm, and this norm came from an inner product, then we do not have any choice; this inner product must be given by the polarization identities, see Lemma 1.9. But, we need to show that $(\mathbf{x}, \mathbf{y})$ which we got from the polarization identities is indeed an inner product, i.e. that it satisfies all the properties.

It is indeed possible to verify that if the norm satisfies the parallelogram identity then the inner product $(\mathbf{x}, \mathbf{y})$ obtained from the polarization identities is indeed an inner product (i.e. satisfies all the properties of an inner product). However, the proof is a bit too involved, so we do not present it here.

Exercises.

1.1. Compute

$$(3+2i)(5-3i), \qquad \frac{2-3i}{1-2i}, \qquad \operatorname{Re}\left(\frac{2-3i}{1-2i}\right), \qquad (1+2i)^3, \qquad \operatorname{Im}((1+2i)^3).$$

1.2. For vectors $\mathbf{x} = (1, 2i, 1+i)^T$ and $\mathbf{y} = (i, 2-i, 3)^T$ compute

a) $(\mathbf{x}, \mathbf{y})$, $\|\mathbf{x}\|^2$, $\|\mathbf{y}\|^2$, $\|\mathbf{y}\|$;

b) $(3\mathbf{x}, 2i\mathbf{y})$, $(2\mathbf{x}, i\mathbf{x} + 2\mathbf{y})$;

c) $\|\mathbf{x} + 2\mathbf{y}\|$.

Remark: After you have done part a), you can do parts b) and c) without actually computing all vectors involved, just by using the properties of inner product.

1.3. Let $\|\mathbf{u}\| = 2$, $\|\mathbf{v}\| = 3$, $(\mathbf{u}, \mathbf{v}) = 2 + i$. Compute

$$\|\mathbf{u}+\mathbf{v}\|^2, \qquad \|\mathbf{u}-\mathbf{v}\|^2, \qquad (\mathbf{u}+\mathbf{v}, \mathbf{u}-i\mathbf{v}), \qquad (\mathbf{u}+3i\mathbf{v}, 4i\mathbf{u}).$$

1.4. Prove that for vectors in a inner product space

$$\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \pm 2\operatorname{Re}(\mathbf{x}, \mathbf{y})$$

Recall that $\operatorname{Re} z = \frac{1}{2}(z + \overline{z})$

1.5. Explain why each of the following *is not* an inner product on a given vector space:

a) $(\mathbf{x}, \mathbf{y}) = x_1 y_1 - x_2 y_2$ on $\mathbb{R}^2$;

b) $(A, B) = \operatorname{trace}(A + B)$ on the space of real 2×2 matrices'

c) $(f, g) = \int_0^1 f'(t)\overline{g(t)}dt$ on the space of polynomials; $f'(t)$ denotes derivative.

1.6 (Equality in Cauchy–Schwarz inequality)**.** Prove that

$$|(\mathbf{x}, \mathbf{y})| = \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

if and only if one of the vectors is a multiple of the other. **Hint:** Analyze the proof of the Cauchy–Schwarz inequality.

1.7. Prove the parallelogram identity for an inner product space V,

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2).$$

1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ be a spanning set (in particular, a basis) in an inner product space V. Prove that

a) If $(\mathbf{x}, \mathbf{v}) = 0$ for all $\mathbf{v} \in V$, then $\mathbf{x} = \mathbf{0}$;

b) If $(\mathbf{x}, \mathbf{v}_k) = 0$ $\forall k$, then $\mathbf{x} = \mathbf{0}$;

c) If $(\mathbf{x}, \mathbf{v}_k) = (\mathbf{y}, \mathbf{v}_k)$ $\forall k$, then $\mathbf{x} = \mathbf{y}$.

1.9. Consider the space $\mathbb{R}^2$ with the norm $\|\cdot\|_p$, introduced in Section 1.5. For $p = 1, 2, \infty$, draw the "unit ball" B_p in the norm $\|\cdot\|_p$

$$B_p := \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_p \le 1\}.$$

Can you guess what the balls B_p for other p look like?

2. Orthogonality. Orthogonal and orthonormal bases.

Definition 2.1. Two vectors $\mathbf{u}$ and $\mathbf{v}$ are called *orthogonal* (also *perpendicular*) if $(\mathbf{u}, \mathbf{v}) = 0$. We will write $\mathbf{u} \perp \mathbf{v}$ to say that the vectors are orthogonal.

Note, that for orthogonal vectors $\mathbf{u}$ and $\mathbf{v}$ we have the following, so-called Pythagorean identity:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \qquad \text{if } \mathbf{u} \perp \mathbf{v}.$$

The proof is straightforward computation,

$$\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) = (\mathbf{u}, \mathbf{u}) + (\mathbf{v}, \mathbf{v}) + (\mathbf{u}, \mathbf{v}) + (\mathbf{v}, \mathbf{u}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

($(\mathbf{u}, \mathbf{v}) = (\mathbf{v}, \mathbf{u}) = 0$ because of orthogonality).

Definition 2.2. We say that a vector $\mathbf{v}$ is orthogonal to a subspace E if $\mathbf{v}$ is orthogonal to all vectors $\mathbf{w}$ in E.

We say that subspaces E and F are orthogonal if all vectors in E are orthogonal to F, i.e. all vectors in E are orthogonal to all vectors in F

The following lemma shows how to check that a vector is orthogonal to a subspace.

Lemma 2.3. *Let E be spanned by vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$. Then $\mathbf{v} \perp E$ if and only if*

$$\mathbf{v} \perp \mathbf{v}_k, \qquad \forall k = 1, 2, \dots, r.$$

Proof. By the definition, if $\mathbf{v} \perp E$ then $\mathbf{v}$ is orthogonal to all vectors in E. In particular, $\mathbf{v} \perp \mathbf{v}_k$, $k = 1, 2, \dots, r$.

On the other hand, let $\mathbf{v} \perp \mathbf{v}_k$, $k = 1, 2, \dots, r$. Since the vectors $\mathbf{v}_k$ span E, any vector $\mathbf{w} \in E$ can be represented as a linear combination $\sum_{k=1}^{r} \alpha_k \mathbf{v}_k$. Then

$$(\mathbf{v}, \mathbf{w}) = \sum_{k=1}^{r} \alpha_k (\mathbf{v}, \mathbf{v}_k) = 0,$$

so $\mathbf{v} \perp \mathbf{w}$. □

Definition 2.4. A system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is called *orthogonal* if any two vectors are orthogonal to each other (i.e. if $(\mathbf{v}_j, \mathbf{v}_k) = 0$ for $j \neq k$).

If, in addition $\|\mathbf{v}_k\| = 1$ for all k, we call the system *orthonormal*.

Lemma 2.5 (Generalized Pythagorean identity)**.** *Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be an orthogonal system. Then*

$$\left\| \sum_{k=1}^{n} \alpha_k \mathbf{v}_k \right\|^2 = \sum_{k=1}^{n} |\alpha_k|^2 \|\mathbf{v}_k\|^2$$

This formula looks particularly simple for orthonormal systems, where $\|\mathbf{v}_k\| = 1$.

Proof of the Lemma.

$$\left\| \sum_{k=1}^{n} \alpha_k \mathbf{v}_k \right\|^2 = \Big(\sum_{k=1}^{n} \alpha_k \mathbf{v}_k, \sum_{j=1}^{n} \alpha_j \mathbf{v}_j \Big) = \sum_{k=1}^{n} \sum_{j=1}^{n} \alpha_k \overline{\alpha}_j (\mathbf{v}_k, \mathbf{v}_j).$$

Because of orthogonality $(\mathbf{v}_k, \mathbf{v}_j) = 0$ if $j \neq k$. Therefore we only need to sum the terms with $j = k$, which gives exactly

$$\sum_{k=1}^{n} |\alpha_k|^2 (\mathbf{v}_k, \mathbf{v}_k) = \sum_{k=1}^{n} |\alpha_k|^2 \|\mathbf{v}_k\|^2.$$

□

Corollary 2.6. *Any orthogonal system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ of non-zero vectors is linearly independent.*

Proof. Suppose for some $\alpha_1, \alpha_2, \dots, \alpha_n$ we have $\sum_{k=1}^{n} \alpha_k \mathbf{v}_k = \mathbf{0}$. Then by the Generalized Pythagorean identity (Lemma 2.5)

$$0 = \|\mathbf{0}\|^2 = \sum_{k=1}^{n} |\alpha_k|^2 \|\mathbf{v}_k\|^2.$$

Since $\|\mathbf{v}_k\| \neq \mathbf{0}$ ($\mathbf{v}_k \neq \mathbf{0}$) we conclude that

$$\alpha_k = 0 \qquad \forall k,$$

so only the trivial linear combination gives $\mathbf{0}$. □

Remark. In what follows we will usually mean by an orthogonal system an orthogonal system of non-zero vectors. Since the zero vector $\mathbf{0}$ is orthogonal to everything, it always can be added to any orthogonal system, but it is really not interesting to consider orthogonal systems with zero vectors.

2.1. Orthogonal and orthonormal bases.

Definition 2.7. An orthogonal (orthonormal) system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ which is also a basis is called an orthogonal (orthonormal) basis.

It is clear that in $\dim V = n$ then any orthogonal system of n non-zero vectors is an orthogonal basis.

As we studied before, to find coordinates of a vector in a basis one needs to solve a linear system. However, for an orthogonal basis finding coordinates of a vector is much easier. Namely, suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is an orthogonal basis, and let

$$\mathbf{x} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots + \alpha_n\mathbf{v}_n = \sum_{j=1}^{n} \alpha_j\mathbf{v}_j.$$

Taking inner product of both sides of the equation with $\mathbf{v}_1$ we get

$$(\mathbf{x}, \mathbf{v}_1) = \sum_{j=1}^{n} \alpha_j(\mathbf{v}_j, \mathbf{v}_1) = \alpha_1(\mathbf{v}_1, \mathbf{v}_1) = \alpha_1\|\mathbf{v}_1\|^2$$

(all inner products $(\mathbf{v}_j, \mathbf{v}_1) = 0$ if $j \neq 1$), so

$$\alpha_1 = \frac{(\mathbf{x}, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}.$$

Similarly, multiplying both sides by $\mathbf{v}_k$ we get

$$(\mathbf{x}, \mathbf{v}_k) = \sum_{j=1}^{n} \alpha_j(\mathbf{v}_j, \mathbf{v}_k) = \alpha_k(\mathbf{v}_k, \mathbf{v}_k) = \alpha_k\|\mathbf{v}_k\|^2$$

so

$$\alpha_k = \frac{(\mathbf{x}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}. \tag{2.1}$$

Therefore,

to find coordinates of a vector in an orthogonal basis one does not need to solve a linear system, the coordinates are determined by the formula (2.1).

This formula is especially simple for orthonormal bases, when $\|\mathbf{v}_k\| = 1$.

Namely, if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is an orthonormal basis, any vector $\mathbf{v}$ can be represented as

$$\mathbf{v} = \sum_{k=1}^{n} (\mathbf{v}, \mathbf{v}_k)\mathbf{v}_k. \tag{2.2}$$

This formula is sometimes called (a baby version of) the *abstract orthogonal Fourier decomposition*. The classical (non-abstract) Fourier decomposition deals with a concrete orthonormal system (sines and cosines or complex exponentials). We call this formula a *baby version* because the real Fourier decomposition deals with infinite orthonormal systems.

This is a very important remark allowing one to translate any statement about the standard inner product space $\mathbb{F}^n$ to an inner product space with an orthonormal basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Remark 2.8. The importance of orthonormal bases is that if we fix an orthonormal basis in an inner product space V, we can work with coordinates in this basis the same way we work with vectors in $\mathbb{F}^n$. Namely, as it was discussed in the very beginning of the book, see Remark 2.4 in Chapter 1, if we have a vector space V (over a field $\mathbb{F}$) with a basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then we can perform the standard vector operations (vector addition and multiplication by a scalar) by working with the columns of coordinates in the basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in absolutely the same way we work with vectors in the standard coordinate space $\mathbb{F}^n$.

Exercise 2.3 below shows that if we have an *orthonormal* basis in an inner product space V, we can compute the inner product of 2 vectors in V by taking columns of their coordinates in this orthonormal basis and computing the standard inner product (in $\mathbb{C}^n$ or $\mathbb{R}^n$) of these columns.

As it will be shown below in Section 3 any finite-dimensional inner product space has an orthonormal basis. Thus, the standard inner product spaces $\mathbb{C}^n$ (or $\mathbb{R}^n$ in the case of real spaces) are essentially the only examples of a finite-dimensional inner product spaces.

Exercises.

2.1. Find all vectors in $\mathbb{R}^4$ orthogonal to vectors $(1,1,1,1)^T$ and $(1,2,3,4)^T$.

2.2. Let A be a real $m \times n$ matrix. Describe $(\operatorname{Ran} A^T)^\perp$, $(\operatorname{Ran} A)^\perp$

2.3. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be an orthonormal basis in V.

a) Prove that for any $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{n} \alpha_k \overline{\beta}_k.$$

b) Deduce from this the *Parseval's identity*

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{n} (\mathbf{x}, \mathbf{v}_k)\overline{(\mathbf{y}, \mathbf{v}_k)}$$

c) Assume now that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is only an orthogonal basis, not an orthonormal one. Can you write down Parseval's identity in this case?

This problem shows that if we have an orthonormal basis, we can use the coordinates in this basis absolutely the same way we use the standard coordinates in $\mathbb{C}^n$ (or $\mathbb{R}^n$).

The problem below shows that we can define an inner product by declaring a basis to be an orthonormal one.

2.4. Let V be a vector space and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in V. For $\mathbf{x} = \sum_{k=1}^n \alpha_k \mathbf{v}_k$, $\mathbf{y} = \sum_{k=1}^n \beta_k \mathbf{v}_k$ define $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{k=1}^n \alpha_k \bar{\beta}_k$.

Prove that $\langle \mathbf{x}, \mathbf{y} \rangle$ defines an inner product in V.

2.5. Let A be a real $m \times n$ matrix. Describe the set of all vectors in $\mathbb{F}^m$ orthogonal to to $\operatorname{Ran} A$.

3. Orthogonal projection and Gram-Schmidt orthogonalization

Recalling the definition of orthogonal projection from the classical planar (2-dimensional) geometry, one can introduce the following definition. Let E be a subspace of an inner product space V.

Definition 3.1. For a vector $\mathbf{v}$ its orthogonal projection $P_E\mathbf{v}$ onto the subspace E is a vector $\mathbf{w}$ such that

1. $\mathbf{w} \in E$;
2. $\mathbf{v} - \mathbf{w} \perp E$.

We will use notation $\mathbf{w} = P_E\mathbf{v}$ for the orthogonal projection.

After introducing an object, it is natural to ask:

1. Does the object exist?
2. Is the object unique?
3. How does one find it?

We will show first that the projection is unique. Then we present a method of finding the projection, proving its existence.

The following theorem shows why the orthogonal projection is important and also proves that it is unique.

Theorem 3.2. *The orthogonal projection $\mathbf{w} = P_E\mathbf{v}$ minimizes the distance from $\mathbf{v}$ to E, i.e. for all $\mathbf{x} \in E$*

$$\|\mathbf{v} - \mathbf{w}\| \le \|\mathbf{v} - \mathbf{x}\|.$$

Moreover, if for some $\mathbf{x} \in E$

$$\|\mathbf{v} - \mathbf{w}\| = \|\mathbf{v} - \mathbf{x}\|,$$

then $\mathbf{x} = \mathbf{w}$.

Proof. Let $\mathbf{y} = \mathbf{w} - \mathbf{x}$. Then

$$\mathbf{v} - \mathbf{x} = \mathbf{v} - \mathbf{w} + \mathbf{w} - \mathbf{x} = \mathbf{v} - \mathbf{w} + \mathbf{y}.$$

Since $\mathbf{v} - \mathbf{w} \perp E$ we have $\mathbf{y} \perp \mathbf{v} - \mathbf{w}$ and so by Pythagorean Theorem

$$\|\mathbf{v} - \mathbf{x}\|^2 = \|\mathbf{v} - \mathbf{w}\|^2 + \|\mathbf{y}\|^2 \geq \|\mathbf{v} - \mathbf{w}\|^2.$$

Note that equality happens only if $\mathbf{y} = \mathbf{0}$ i.e. if $\mathbf{x} = \mathbf{w}$. □

The following proposition shows how to find an orthogonal projection if we know an orthogonal basis in E.

Proposition 3.3. *Let* $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ *be an orthogonal basis in* E. *Then the orthogonal projection* $P_E\mathbf{v}$ *of a vector* $\mathbf{v}$ *is given by the formula*

$$P_{_E}\mathbf{v} = \sum_{k=1}^{r} \alpha_k \mathbf{v}_k, \qquad \textit{where} \qquad \alpha_k = \frac{(\mathbf{v}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}\,.$$

In other words

$$P_{_E}\mathbf{v} = \sum_{k=1}^{r} \frac{(\mathbf{v}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}\, \mathbf{v}_k. \tag{3.1}$$

Note that the formula for α_k coincides with (2.1), i.e. this formula applied to an orthogonal system (not a basis) gives us a projection onto its span.

Remark 3.4. It is easy to see now from formula (3.1) that the orthogonal projection $P_{_E}$ is a linear transformation.

One can also see linearity of $P_{_E}$ directly, from the definition and uniqueness of the orthogonal projection. Indeed, it is easy to check that for any $\mathbf{x}$ and $\mathbf{y}$ the vector $\alpha\mathbf{x} + \beta\mathbf{y} - (\alpha P_{_E}\mathbf{x} - \beta P_{_E}\mathbf{y})$ is orthogonal to any vector in E, so by the definition $P_{_E}(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha P_{_E}\mathbf{x} + \beta P_{_E}\mathbf{y}$.

Remark 3.5. Recalling the definition of inner product in $\mathbb{C}^n$ and $\mathbb{R}^n$ one can get from the above formula (3.1) the matrix of the orthogonal projection $P_{_E}$ onto E in $\mathbb{C}^n$ ($\mathbb{R}^n$) is given by

$$P_{_E} = \sum_{k=1}^{r} \frac{1}{\|\mathbf{v}_k\|^2}\ \mathbf{v}_k\mathbf{v}_k^* \tag{3.2}$$

where columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ form an orthogonal basis in E.

Proof of Proposition 3.3. Let

$$\mathbf{w} := \sum_{k=1}^{r} \alpha_k \mathbf{v}_k, \qquad \text{where} \qquad \alpha_k = \frac{(\mathbf{v}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}\,.$$

We want to show that $\mathbf{v} - \mathbf{w} \perp E$. By Lemma 2.3 it is sufficient to show that $\mathbf{v} - \mathbf{w} \perp \mathbf{v}_k$, $k = 1, 2, \dots, n$. Computing the inner product we get for $k = 1, 2, \dots, r$

$$\begin{aligned}(\mathbf{v} - \mathbf{w}, \mathbf{v}_k) &= (\mathbf{v}, \mathbf{v}_k) - (\mathbf{w}, \mathbf{v}_k) = (\mathbf{v}, \mathbf{v}_k) - \sum_{j=1}^{r} \alpha_j (\mathbf{v}_j, \mathbf{v}_k) \\ &= (\mathbf{v}, \mathbf{v}_k) - \alpha_k (\mathbf{v}_k, \mathbf{v}_k) = (\mathbf{v}, \mathbf{v}_k) - \frac{(\mathbf{v}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2} \|\mathbf{v}_k\|^2 = 0.\end{aligned}$$

□

So, if we know an orthogonal basis in E we can find the orthogonal projection onto E. In particular, since any system consisting of one vector is an orthogonal system, we know how to perform orthogonal projection onto one-dimensional spaces.

But how do we find an orthogonal projection if we are only given a basis in E? Fortunately, there exists a simple algorithm allowing one to get an orthogonal basis from a basis.

3.1. Gram-Schmidt orthogonalization algorithm. Suppose we have a linearly independent system $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. The Gram-Schmidt method constructs from this system an orthogonal system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that

$$\operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}.$$

Moreover, for all $r \le n$ we get

$$\operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r\} = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$$

Now let us describe the algorithm.

Step 1. Put $\mathbf{v}_1 := \mathbf{x}_1$. Denote by $E_1 := \operatorname{span}\{\mathbf{x}_1\} = \operatorname{span}\{\mathbf{v}_1\}$.

Step 2. Define $\mathbf{v}_2$ by

$$\mathbf{v}_2 = \mathbf{x}_2 - P_{E_1}\mathbf{x}_2 = \mathbf{x}_2 - \frac{(\mathbf{x}_2, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}\mathbf{v}_1.$$

Define $E_2 = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Note that $\operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2\} = E_2$.

Step 3. Define $\mathbf{v}_3$ by

$$\mathbf{v}_3 := \mathbf{x}_3 - P_{E_2}\mathbf{x}_3 = \mathbf{x}_3 - \frac{(\mathbf{x}_3, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{(\mathbf{x}_3, \mathbf{v}_2)}{\|\mathbf{v}_2\|^2}\mathbf{v}_2$$

Put $E_3 := \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Note that $\operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\} = E_3$. Note also that $\mathbf{x}_3 \notin E_2$ so $\mathbf{v}_3 \neq \mathbf{0}$.

...

Step $r+1$**.** Suppose that we already made r steps of the process, constructing an orthogonal system (consisting of non-zero vectors) $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ such that $E_r := \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\} = \operatorname{span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r\}$. Define

$$\mathbf{v}_{r+1} := \mathbf{x}_{r+1} - P_{E_r}\mathbf{x}_{r+1} = \mathbf{x}_{r+1} - \sum_{k=1}^{r} \frac{(\mathbf{x}_{r+1}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}\mathbf{v}_k$$

Note,that $\mathbf{x}_{r+1} \notin E_r$ so $\mathbf{v}_{r+1} \neq \mathbf{0}$.

...

Continuing this algorithm we get an orthogonal system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.

3.2. An example. Suppose we are given vectors

$$\mathbf{x}_1 = (1,1,1)^T, \qquad \mathbf{x}_2 = (0,1,2)^T, \qquad \mathbf{x}_3 = (1,0,2)^T,$$

and we want to orthogonalize it by Gram-Schmidt. On the first step define

$$\mathbf{v}_1 = \mathbf{x}_1 = (1,1,1)^T.$$

On the second step we get

$$\mathbf{v}_2 = \mathbf{x}_2 - P_{E_1}\mathbf{x}_2 = \mathbf{x}_2 - \frac{(\mathbf{x}_2, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}\mathbf{v}_1.$$

Computing

$$(\mathbf{x}_2, \mathbf{v}_1) = \left(\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = 3, \quad \|\mathbf{v}_1\|^2 = 3,$$

we get

$$\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - \frac{3}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Finally, define

$$\mathbf{v}_3 = \mathbf{x}_3 - P_{E_2}\mathbf{x}_3 = \mathbf{x}_3 - \frac{(\mathbf{x}_3, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{(\mathbf{x}_3, \mathbf{v}_2)}{\|\mathbf{v}_2\|^2}\mathbf{v}_2.$$

Computing

$$\left(\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = 3, \quad \left(\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}\right) = 1, \ \|\mathbf{v}_1\|^2 = 3, \ \|\mathbf{v}_2\|^2 = 2$$

($\|\mathbf{v}_1\|^2$ was already computed before) we get

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{3}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -1 \\ \frac{1}{2} \end{pmatrix}$$

Remark. Since the multiplication by a scalar does not change the orthogonality, one can multiply vectors $\mathbf{v}_k$ obtained by Gram-Schmidt by any non-zero numbers.

In particular, in many theoretical constructions one *normalizes* vectors $\mathbf{v}_k$ by dividing them by their respective norms $\|\mathbf{v}_k\|$. Then the resulting system will be orthonormal, and the formulas will look simpler.

On the other hand, when performing the computations one may want to avoid fractional entries by multiplying a vector by the least common denominator of its entries. Thus one may want to replace the vector $\mathbf{v}_3$ from the above example by $(1, -2, 1)^T$.

3.3. Orthogonal complement. Decomposition $V = E \oplus E^\perp$.

Definition. For a subspace E its *orthogonal complement* $E^\perp$ is the set of all vectors orthogonal to E,

$$E^\perp := \{\mathbf{x} : \mathbf{x} \perp E\}.$$

If $\mathbf{x}, \mathbf{y} \perp E$ then for any linear combination $\alpha\mathbf{x} + \beta\mathbf{y} \perp E$ (can you see why?). Therefore $E^\perp$ is a subspace.

By the definition of orthogonal projection any vector in an inner product space V admits a unique representation

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2, \qquad \mathbf{v}_1 \in E,\ \mathbf{v}_2 \perp E \ (\text{eqv. } \mathbf{v}_2 \in E^\perp)$$

(where clearly $\mathbf{v}_1 = P_{_E}\mathbf{v}$).

This statement is often symbolically written as $V = E \oplus E^\perp$, which mean exactly that any vector admits the unique decomposition above.

The following proposition gives an important property of the orthogonal complement.

Proposition 3.6. *For a subspace E*

$$(E^\perp)^\perp = E.$$

The proof is left as an exercise, see Problem 3.12 below.

Exercises.

3.1. Apply Gram–Schmidt orthogonalization to the system of vectors $(1, 2, -2)^T$, $(1, -1, 4)^T$, $(2, 1, 1)^T$.

3.2. Apply Gram–Schmidt orthogonalization to the system of vectors $(1, 2, 3)^T$, $(1, 3, 1)^T$. Write the matrix of the orthogonal projection onto 2-dimensional subspace spanned by these vectors.

3.3. Complete an orthogonal system obtained in the previous problem to an orthogonal basis in $\mathbb{R}^3$, i.e. add to the system some vectors (how many?) to get an orthogonal basis.

Can you describe how to complete an orthogonal system to an orthogonal basis in general situation of $\mathbb{R}^n$ or $\mathbb{C}^n$?

3.4. Find the distance from a vector $(2,3,1)^T$ to the subspace spanned by the vectors $(1,2,3)^T$, $(1,3,1)^T$. Note, that I am only asking to find the distance to the subspace, not the orthogonal projection.

3.5. Find the orthogonal projection of a vector $(1,1,1,1)^T$ onto the subspace spanned by the vectors $\mathbf{v}_1 = (1,3,1,1)^T$ and $\mathbf{v}_2 = (2,-1,1,0)^T$ (note that $\mathbf{v}_1 \perp \mathbf{v}_2$).

3.6. Find the distance from a vector $(1,2,3,4)$ to the subspace spanned by the vectors $\mathbf{v}_1 = (1,-1,1,0)^T$ and $\mathbf{v}_2 = (1,2,1,1)^T$ (note that $\mathbf{v}_1 \perp \mathbf{v}_2$). Can you find the distance without actually computing the projection? That would simplify the calculations.

3.7. True or false: if E is a subspace of V, then $\dim E + \dim(E^\perp) = \dim V$? Justify.

3.8. Let P be the orthogonal projection onto a subspace E of an inner product space V, $\dim V = n$, $\dim E = r$. Find the eigenvalues and the eigenvectors (eigenspaces). Find the algebraic and geometric multiplicities of each eigenvalue.

3.9. (Using eigenvalues to compute determinants).

a) Find the matrix of the orthogonal projection onto the one-dimensional subspace in $\mathbb{R}^n$ spanned by the vector $(1,1,\dots,1)^T$;

b) Let A be the $n \times n$ matrix with all entries equal 1. Compute its eigenvalues and their multiplicities (use the previous problem);

c) Compute eigenvalues (and multiplicities) of the matrix $A - I$, i.e. of the matrix with zeroes on the main diagonal and ones everywhere else;

d) Compute $\det(A - I)$.

3.10 (Legendre's polynomials:)**.** Let an inner product on the space of polynomials be defined by $(f,g) = \int_{-1}^{1} f(t)\overline{g(t)}dt$. Apply Gram-Schmidt orthogonalization to the system $1, t, t^2, t^3$.

Legendre's polynomials are particular case of the so-called orthogonal polynomials, which play an important role in many branches of mathematics.

3.11. Let $P = P_E$ be the matrix of an orthogonal projection onto a subspace E. Show that

a) The matrix P is *self-adjoint*, meaning that $P^* = P$.

b) $P^2 = P$.

Remark: The above 2 properties completely characterize orthogonal projection, i.e. any matrix P satisfying these properties is the matrix of some orthogonal projection. We will discuss this some time later.

3.12. Show that for a subspace E we have $(E^\perp)^\perp = E$. **Hint:** It is easy to see that E is orthogonal to $E^\perp$ (why?). To show that *any* vector $\mathbf{x}$ orthogonal to $E^\perp$ belongs to E use the decomposition $V = E \oplus E^\perp$ from Section 3.3 above.

3.13. Suppose P is the orthogonal projection onto a subspace E, and Q is the orthogonal projection onto the orthogonal complement $E^\perp$.

a) What are $P + Q$ and PQ?

b) Show that $P - Q$ is its own inverse.

4. Least square solution. Formula for the orthogonal projection

As it was discussed before in Chapter 2, the equation

$$A\mathbf{x} = \mathbf{b}$$

has a solution if and only if $\mathbf{b} \in \operatorname{Ran} A$. But what do we do to solve an equation that does not have a solution?

This seems to be a silly question, because if there is no solution, then there is no solution. But, situations when we want to solve an equation that does not have a solution can appear naturally, for example, if we obtained the equation from an experiment. If we do not have any errors, the right side $\mathbf{b}$ belongs to the column space $\operatorname{Ran} A$, and equation is consistent. But, in real life it is impossible to avoid errors in measurements, so it is possible that an equation that in theory should be consistent, does not have a solution. So, what can one do in this situation?

4.1. Least square solution. The simplest idea is to write down the error

$$\|A\mathbf{x} - \mathbf{b}\|$$

and try to find $\mathbf{x}$ minimizing it. If we can find $\mathbf{x}$ such that the error is 0, the system is consistent and we have exact solution. Otherwise, we get the so-called *least square* solution.

The term *least square* arises from the fact that minimizing $\|A\mathbf{x} - \mathbf{b}\|$ is equivalent to minimizing

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{k=1}^{m} |(A\mathbf{x})_k - b_k|^2 = \sum_{k=1}^{m} \Big| \sum_{j=1}^{n} A_{k,j} x_j - b_k \Big|^2$$

i.e. to minimizing the sum of squares of linear functions.

There are several ways to find the least square solution. If we are in $\mathbb{R}^n$, and everything is real, we can forget about absolute values. Then we can just take partial derivatives with respect to x_j and find the where all of them are 0, which gives us the minimum.

4.1.1. *Geometric approach.* However, there is a simpler way of finding the minimum. Namely, if we take all possible vectors $\mathbf{x}$, then $A\mathbf{x}$ gives us all possible vectors in $\operatorname{Ran} A$, so minimum of $\|A\mathbf{x} - \mathbf{b}\|$ is exactly the distance from $\mathbf{b}$ to $\operatorname{Ran} A$. Therefore the value of $\|A\mathbf{x} - \mathbf{b}\|$ is minimal if and only if $A\mathbf{x} = P_{\operatorname{Ran} A}\mathbf{b}$, where $P_{\operatorname{Ran} A}$ stands for the orthogonal projection onto the column space $\operatorname{Ran} A$.

So, to find the least square solution we simply need to solve the equation

$$A\mathbf{x} = P_{\operatorname{Ran} A}\mathbf{b}.$$

If we know an orthogonal basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in $\operatorname{Ran} A$, we can find vector $P_{\operatorname{Ran} A}\mathbf{b}$ by the formula

$$P_{\operatorname{Ran} A}\mathbf{b} = \sum_{k=1}^{n} \frac{(\mathbf{b}, \mathbf{v}_k)}{\|\mathbf{v}_k\|^2}\mathbf{v}_k.$$

If we only know a basis in $\operatorname{Ran} A$, we need to use the Gram–Schmidt orthogonalization to obtain an orthogonal basis from it.

So, theoretically, the problem is solved, but the solution is not very simple: it involves Gram–Schmidt orthogonalization, which can be computationally intensive. Fortunately, there exists a simpler solution.

4.1.2. *Normal equation.* Namely, $A\mathbf{x}$ is the orthogonal projection $P_{\operatorname{Ran} A}\mathbf{b}$ if and only if $\mathbf{b} - A\mathbf{x} \perp \operatorname{Ran} A$ ($A\mathbf{x} \in \operatorname{Ran} A$ for all $\mathbf{x}$).

If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are columns of A, then the condition $A\mathbf{x} \perp \operatorname{Ran} A$ can be rewritten as

$$\mathbf{b} - A\mathbf{x} \perp \mathbf{a}_k, \qquad \forall k = 1, 2, \dots, n.$$

That means

$$0 = (\mathbf{b} - A\mathbf{x}, \mathbf{a}_k) = \mathbf{a}_k^*(\mathbf{b} - A\mathbf{x}) \qquad \forall k = 1, 2, \dots, n.$$

Joining rows $\mathbf{a}_k^*$ together we get that these equations are equivalent to

$$A^*(\mathbf{b} - A\mathbf{x}) = \mathbf{0},$$

which in turn is equivalent to the so-called *normal equation*

$$A^*A\mathbf{x} = A^*\mathbf{b}.$$

A solution of this equation gives us the least square solution of $A\mathbf{x} = \mathbf{b}$.

Note, that the least square solution is unique if and only if A^*A is invertible.

4.2. Formula for the orthogonal projection. As we already discussed above, if $\mathbf{x}$ is a solution of the *normal equation* $A^*A\mathbf{x} = A^*\mathbf{b}$ (i.e. a least square solution of $A\mathbf{x} = \mathbf{b}$), then $A\mathbf{x} = P_{\operatorname{Ran} A}\mathbf{b}$. So, to find the orthogonal projection of $\mathbf{b}$ onto the column space $\operatorname{Ran} A$ we need to solve the normal equation $A^*A\mathbf{x} = A^*\mathbf{b}$, and then multiply the solution by A.

If the operator A^*A is invertible, the solution of the normal equation $A^*A\mathbf{x} = A^*\mathbf{b}$ is given by $\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b}$, so the orthogonal projection $P_{\operatorname{Ran} A}\mathbf{b}$ can be computed as

$$P_{\operatorname{Ran} A}\mathbf{b} = A(A^*A)^{-1}A^*\mathbf{b}.$$

Since this is true for all $\mathbf{b}$,

$$P_{\operatorname{Ran} A} = A(A^*A)^{-1}A^*$$

is the formula for the matrix of the orthogonal projection onto $\operatorname{Ran} A$.

The following theorem implies that for an $m \times n$ matrix A the matrix A^*A is invertible if and only if $\operatorname{rank} A = n$.

Theorem 4.1. *For an $m \times n$ matrix A*

$$\operatorname{Ker} A = \operatorname{Ker}(A^*A).$$

Indeed, according to the rank theorem $\operatorname{Ker} A = \{\mathbf{0}\}$ if and only if rank A is n. Therefore $\operatorname{Ker}(A^*A) = \{\mathbf{0}\}$ if and only if $\operatorname{rank} A = n$. Since the matrix A^*A is square, it is invertible if and only if $\operatorname{rank} A = n$.

We leave the proof of the theorem as an exercise. To prove the equality $\operatorname{Ker} A = \operatorname{Ker}(A^*A)$ one needs to prove two inclusions $\operatorname{Ker}(A^*A) \subset \operatorname{Ker} A$ and $\operatorname{Ker} A \subset \operatorname{Ker}(A^*A)$. One of the inclusions is trivial, for the other one use the fact that

$$\|A\mathbf{x}\|^2 = (A\mathbf{x}, A\mathbf{x}) = (A^*A\mathbf{x}, \mathbf{x}).$$

4.3. An example: line fitting. Let us introduce a few examples where the least square solution appears naturally. Suppose that we know that two quantities x and y are related by the law $y = a + bx$. The coefficients a and b are unknown, and we would like to find them from experimental data.

Suppose we run the experiment n times, and we get n pairs (x_k, y_k), $k = 1, 2, \dots, n$. Ideally, all the points (x_k, y_k) should be on a straight line, but because of errors in measurements, it usually does not happen: the point are usually close to some line, but not exactly on it. That is where the least square solution helps!

Ideally, the coefficients a and b should satisfy the equations

$$a + bx_k = y_k, \qquad k = 1, 2, \dots, n$$

(note that here, x_k and y_k are some fixed numbers, and the unknowns are a and b). If it is possible to find such a and b we are lucky. If not, the standard thing to do is to minimize the total quadratic error

$$\sum_{k=1}^{n} |a + bx_k - y_k|^2.$$

But, minimizing this error is exactly finding the least square solution of the system

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

(recall that x_k y_k are some given numbers, and the unknowns are a and b).

4.3.1. *An example.* Suppose our data (x_k, y_k) consist of pairs

$$(-2,4),\ (-1,2),\ (0,1),\ (2,1),\ (3,1).$$

Then we need to find the least square solution of

$$\begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{pmatrix} 4 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Then

$$A^*A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 2 & 18 \end{pmatrix}$$

and

$$A^*\mathbf{b} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ -5 \end{pmatrix}$$

so the normal equation $A^*A\mathbf{x} = A^*\mathbf{b}$ is rewritten as

$$\begin{pmatrix} 5 & 2 \\ 2 & 18 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 9 \\ -5 \end{pmatrix}.$$

The solution of this equation is

$$a = 2, b = -1/2,$$

so the best fitting straight line is

$$y = 2 - 1/2x.$$

4.4. Other examples: curves and planes. The least square method is not limited to the line fitting. It can also be applied to more general curves, as well as to surfaces in higher dimensions.

The only constraint here is that the parameters we want to find be involved linearly. The general algorithm is as follows:

1. Find the equations that your data should satisfy if there is exact fit;
2. Write these equations as a linear system, where unknowns are the parameters you want to find. Note, that the system need not to be consistent (and usually is not);
3. Find the least square solution of the system.

4.4.1. *An example: curve fitting.* For example, suppose we know that the relation between x and y is given by the quadratic law $y = a + bx + cx^2$, so we want to fit a parabola $y = a + bx + cx^2$ to the data. Then our unknowns a, b, c should satisfy the equations

$$a + bx_k + cx_k^2 = y_k, \qquad k = 1, 2, \ldots, n$$

or, in matrix form

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

For example, for the data from the previous example we need to find the least square solution of

$$\begin{pmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$A^*A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 & 3 \\ 4 & 1 & 0 & 4 & 9 \end{pmatrix} \begin{pmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 18 \\ 2 & 18 & 26 \\ 18 & 26 & 114 \end{pmatrix}$$

and

$$A^*\mathbf{b} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 & 3 \\ 4 & 1 & 0 & 4 & 9 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ -5 \\ 31 \end{pmatrix}.$$

Therefore the normal equation $A^*A\mathbf{x} = A^*\mathbf{b}$ is

$$\begin{pmatrix} 5 & 2 & 18 \\ 2 & 18 & 26 \\ 18 & 26 & 114 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 9 \\ -5 \\ 31 \end{pmatrix}$$

which has the unique solution

$$a = 86/77, \qquad b = -62/77, \qquad c = 43/154.$$

Therefore,

$$y = 86/77 - 62x/77 + 43x^2/154$$

is the best fitting parabola.

4.4.2. *Plane fitting.* As another example, let us fit a plane $z = a + bx + cy$ to the data

$$(x_k, y_k, z_k) \in \mathbb{R}^3, \qquad k = 1, 2, \dots n.$$

The equations we should have in the case of exact fit are

$$a + bx_k + cy_k = z_k, \qquad k = 1, 2, \dots, n,$$

or, in the matrix form

$$\begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & y_n \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}.$$

So, to find the best fitting plane, we need to find the best square solution of this system (the unknowns are a, b, c).

Exercises.

4.1. Find the least square solution of the system

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

4.2. Find the matrix of the orthogonal projection P onto the column space of

$$\begin{pmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{pmatrix}.$$

Use two methods: Gram–Schmidt orthogonalization and formula for the projection.

Compare the results.

4.3. Find the best straight line fit (least square solution) to the points $(-2, 4)$, $(-1, 3)$, $(0, 1)$, $(2, 0)$.

4.4. Fit a plane $z = a + bx + cy$ to four points $(1, 1, 3)$, $(0, 3, 6)$, $(2, 1, 5)$, $(0, 0, 0)$.

To do that

a) Find 4 equations with 3 unknowns a, b, c such that the plane pass through all 4 points (this system does not have to have a solution);

b) Find the least square solution of the system.

4.5. *Minimal norm solution.* let an equation $A\mathbf{x} = \mathbf{b}$ has a solution, and let A has non-trivial kernel (so the solution is not unique). Prove that

a) There exist a unique solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{b}$ minimizing the norm $\|\mathbf{x}\|$, i.e. that there exists unique $\mathbf{x}_0$ such that $A\mathbf{x}_0 = \mathbf{b}$ and $\|\mathbf{x}_0\| \le \|\mathbf{x}\|$ for any $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{b}$.

b) $\mathbf{x}_0 = P_{(\operatorname{Ker} A)^\perp} \mathbf{x}$ for any $\mathbf{x}$ satisfying $A\mathbf{x} = \mathbf{b}$.

4.6. *Minimal norm least square solution.* Applying previous problem to the equation $A\mathbf{x} = P_{\operatorname{Ran} A}\mathbf{b}$ show that

a) There exists a unique least square solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{b}$ minimizing the norm $\|\mathbf{x}\|$.

b) $\mathbf{x}_0 = P_{(\operatorname{Ker} A)^\perp}\mathbf{x}$ for any least square solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$.

5. Adjoint of a linear transformation. Fundamental subspaces revisited.

5.1. Adjoint matrices and adjoint operators. Let as recall that for an $m \times n$ matrix A its *Hermitian adjoint* (or simply *adjoint*) A^* is defined by $A^* := \overline{A^T}$. In other words, the matrix A^* is obtained from the transposed matrix A^T by taking complex conjugate of each entry.

The following identity is the main property of adjoint matrix:

$$\boxed{(A\mathbf{x}, \mathbf{y}) = (\mathbf{x}, A^*\mathbf{y}) \qquad \forall \mathbf{x} \in \mathbb{C}^n,\ \forall \mathbf{y} \in \mathbb{C}^m.}$$

Before proving this identity, let us introduce some useful formulas. Let us recall that for transposed matrices we have the identity $(AB)^T = B^T A^T$. Since for complex numbers z and w we have $\overline{zw} = \overline{z}\,\overline{w}$, the identity

$$(AB)^* = B^* A^*$$

holds for the adjoint.

Also, since $(A^T)^T = A$ and $\overline{\overline{z}} = z$,

$$(A^*)^* = A.$$

Now, we are ready to prove the main identity:

$$(A\mathbf{x}, \mathbf{y}) = \mathbf{y}^* A\mathbf{x} = (A^*\mathbf{y})^*\mathbf{x} = (\mathbf{x}, A^*\mathbf{y});$$

the first and the last equalities here follow from the definition of inner product in $\mathbb{F}^n$, and the middle one follows from the fact that

$$(A^*\mathbf{x})^* = \mathbf{x}^*(A^*)^* = \mathbf{x}^* A.$$

5.1.1. *Uniqueness of the adjoint.* The above main identity $(A\mathbf{x}, \mathbf{y}) = (\mathbf{x}, A^*\mathbf{y})$ is often used as the definition of the adjoint operator. Let us first notice that the adjoint operator is unique: if a matrix B satisfies

$$(A\mathbf{x}, \mathbf{y}) = (\mathbf{x}, B\mathbf{y}) \qquad \forall \mathbf{x}, \mathbf{y},$$

then $B = A^*$. Indeed, by the definition of A^* for a given $\mathbf{y}$ we have

$$(\mathbf{x}, A^*\mathbf{y}) = (\mathbf{x}, B\mathbf{y}) \qquad \forall \mathbf{x},$$

and therefore by Corollary 1.5 $A^*\mathbf{y} = B\mathbf{y}$. Since it is true for all $\mathbf{y}$, the linear transformations, and therefore the matrices A^* and B coincide.

5.1.2. *Adjoint transformation in abstract setting.* The above main identity $(A\mathbf{x}, \mathbf{y}) = (\mathbf{x}, A^*\mathbf{y})$ can be used to define the adjoint operator in abstract setting, where $A : V \to W$ is an operator acting from one inner product space to another. Namely, we define $A^* : W \to V$ to be the operator satisfying

$$(A\mathbf{x}, \mathbf{y}) = (\mathbf{x}, A^*\mathbf{y}) \qquad \forall \mathbf{x} \in V,\ \forall \mathbf{y} \in W.$$

Why does such an operator exists? We can simply construct it: consider orthonormal bases $\mathcal{A} = \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in V and $\mathcal{B} = \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m$ in W. If $[A]_{\mathcal{BA}}$ is the matrix of A with respect to these bases, we define the operator A^* by defining its matrix $[A^*]_{\mathcal{AB}}$ as

$$[A^*]_{\mathcal{AB}} = ([A]_{\mathcal{BA}})^*.$$

We leave the proof that this indeed gives the adjoint operator as an exercise for the reader.

Note, that the reasoning in the above Sect. 5.1.1 implies that the adjoint operator is unique.

5.1.3. *Useful formulas.* Below we present the properties of the adjoint operators (matrices) we will use a lot. We leave the proofs as an exercise for the reader.

1. $(A + B)^* = A^* + B^*$;
2. $(\alpha A)^* = \overline{\alpha} A^*$;
3. $(AB)^* = B^* A^*$;
4. $(A^*)^* = A$;
5. $(\mathbf{y}, A\mathbf{x}) = (A^*\mathbf{y}, \mathbf{x})$.

5.2. Relation between fundamental subspaces.

Theorem 5.1. *Let $A : V \to W$ be an operator acting from one inner product space to another. Then*

1. $\operatorname{Ker} A^* = (\operatorname{Ran} A)^\perp$;
2. $\operatorname{Ker} A = (\operatorname{Ran} A^*)^\perp$;
3. $\operatorname{Ran} A = (\operatorname{Ker} A^*)^\perp$;
4. $\operatorname{Ran} A^* = (\operatorname{Ker} A)^\perp$.

Remark. Earlier in Section 7 of Chapter 2 the fundamental subspaces were defined (as it is often done in the literature) using A^T instead of A^*. Of course, there is no difference for real matrices, so in the real case the above theorem gives the geometric description of the fundamentals subspaces defined there.

Geometric interpretation of the fundamental subspaces defined using A^T is presented in Chapter 8 below, see Section 3 there (Theorem 3.7). The

formulas in this theorem are essentially the same as in Theorem 5.1 here, only the interpretation is a bit different.

Proof of Theorem 5.1. First of all, let us notice, that since for a subspace E we have $(E^\perp)^\perp = E$, the statements 1 and 3 are equivalent. Similarly, for the same reason, the statements 2 and 4 are equivalent as well. Finally, statement 2 is exactly statement 1 applied to the operator A^* (here we use the fact that $(A^*)^* = A$).

So, to prove the theorem we only need to prove statement 1.

We will present 2 proofs of this statement: a "matrix" proof, and an "invariant", or "coordinate-free" one.

In the "matrix" proof, we assume that A is an $m \times n$ matrix, i.e. that $A : \mathbb{F}^n \to \mathbb{F}^m$. The general case can be always reduced to this one by picking orthonormal bases in V and W, and considering the matrix of A in this bases.

Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ be the columns of A. Note, that $\mathbf{x} \in (\operatorname{Ran} A)^\perp$ if and only if $\mathbf{x} \perp \mathbf{a}_k$ (i.e. $(\mathbf{x}, \mathbf{a}_k) = 0$) $\forall k = 1, 2, \dots, n$.

By the definition of the inner product in $\mathbb{F}^n$, that means

$$0 = (\mathbf{x}, \mathbf{a}_k) = \mathbf{a}_k^* \mathbf{x} \qquad \forall k = 1, 2, \dots, n.$$

Since $\mathbf{a}_k^*$ is the row number k of A^*, the above n equalities are equivalent to the equation

$$A^*\mathbf{x} = \mathbf{0}.$$

So, we proved that $\mathbf{x} \in (\operatorname{Ran} A)^\perp$ if and only if $A^*\mathbf{x} = \mathbf{0}$, and that is exactly the statement 1.

Now, let us present the "coordinate-free" proof. The inclusion $\mathbf{x} \in (\operatorname{Ran} A)^\perp$ means that $\mathbf{x}$ is orthogonal to all vectors of the form $A\mathbf{y}$, i.e. that

$$(\mathbf{x}, A\mathbf{y}) = 0 \qquad \forall \mathbf{y}.$$

Since $(\mathbf{x}, A\mathbf{y}) = (A^*\mathbf{x}, \mathbf{y})$, this identity is equivalent to

$$(A^*\mathbf{x}, \mathbf{y}) = 0 \qquad \forall \mathbf{y},$$

and by Lemma 1.4 this happens if and only if $A^*\mathbf{x} = \mathbf{0}$. So we proved that $x \in (\operatorname{Ran} A)^\perp$ if and only if $A^*\mathbf{x} = \mathbf{0}$, which is exactly the statement 1 of the theorem. □

5.3. The "essential" part of a linear transformation. The above theorem makes the structure of the operator A and the geometry of fundamental subspaces much more transparent. It follows from this theorem that the operator A can be represented as a composition of orthogonal projection onto $\operatorname{Ran} A^*$ and an isomorphism from $\operatorname{Ran} A^*$ to $\operatorname{Ran} A$.

Indeed, let $\widetilde{A} : \operatorname{Ran} A^* \to \operatorname{Ran} A$ be the restriction of A to the domain $\operatorname{Ran} A^*$ and the target space $\operatorname{Ran} A$,

$$\widetilde{A}\mathbf{x} = A\mathbf{x}, \qquad \forall \mathbf{x} \in \operatorname{Ran} A^*.$$

Since $\operatorname{Ker} A = (\operatorname{Ran} A^*)^\perp$, we have

$$A\mathbf{x} = AP_{\operatorname{Ran} A^*}\mathbf{x} = \widetilde{A}P_{\operatorname{Ran} A^*}\mathbf{x} \qquad \forall \mathbf{x} \in X;$$

the fact that $\mathbf{x} - P_{\operatorname{Ran} A^*}\mathbf{x} \in (\operatorname{Ran} A^*)^\perp = \operatorname{Ker} A$ is used here. Therefore we can write

$$A\mathbf{x} = \widetilde{A}P_{\operatorname{Ran} A^*}\mathbf{x} \qquad \forall \mathbf{x} \in X, \tag{5.1}$$

or, equivalently, $A = \widetilde{A}P_{\operatorname{Ran} A^*}$.

Note also that $\widetilde{A} : \operatorname{Ran} A^* \to \operatorname{Ran} A$ is an invertible transformation. First we notice that $\operatorname{Ker} \widetilde{A} = \{\mathbf{0}\}$: if $\mathbf{x} \in \operatorname{Ran} A^*$ is such that $\widetilde{A}\mathbf{x} = A\mathbf{x} = \mathbf{0}$, then $\mathbf{x} \in \operatorname{Ker} A = (\operatorname{Ran} A^*)^\perp$, so $\mathbf{x} \in \operatorname{Ran} A^* \cap (\operatorname{Ran} A^*)^\perp$, thus $\mathbf{x} = \mathbf{0}$. Then to see that $\widetilde{A}$ is invertible, it is sufficient to she that $\widetilde{A}$ is onto (surjective). But this immediately follows from (5.1):

$$\operatorname{Ran} \widetilde{A} = \widetilde{A} \operatorname{Ran} A^* = AP_{\operatorname{Ran} A^*}X = AX = \operatorname{Ran} A.$$

The isomorphism $\widetilde{A}$ is sometimes called the "essential part" of the operator A (a non-standard terminology).

The fact the "essential part" $\widetilde{A} : \operatorname{Ran} A^* \to \operatorname{Ran} A$ of A is an isomorphism implies the following "complex" rank theorem: $\operatorname{rank} A = \operatorname{rank} A^*$. But, of course, this theorem also follows from an elementary observation that complex conjugation does not change rank of a matrix, $\operatorname{rank} A = \operatorname{rank} \overline{A}$.

Exercises.

5.1. Show that for a square matrix A the equality $\det(A^*) = \overline{\det(A)}$ holds.

5.2. Find matrices of orthogonal projections onto all 4 fundamental subspaces of the matrix

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 2 & 4 & 3 \end{pmatrix}.$$

Note, that really you need only to compute 2 of the projections. If you pick an appropriate 2, the other 2 are easy to obtain from them (recall, how the projections onto E and $E^\perp$ are related).

5.3. Let A be an $m \times n$ matrix. Show that $\operatorname{Ker} A = \operatorname{Ker}(A^*A)$.

To do that you need to prove 2 inclusions, $\operatorname{Ker}(A^*A) \subset \operatorname{Ker} A$ and $\operatorname{Ker} A \subset \operatorname{Ker}(A^*A)$. One of the inclusions is trivial, for the other one use the fact that

$$\|A\mathbf{x}\|^2 = (A\mathbf{x}, A\mathbf{x}) = (A^*A\mathbf{x}, \mathbf{x}).$$

5.4. Use the equality $\operatorname{Ker} A = \operatorname{Ker}(A^*A)$ to prove that

SPECTRAL THEORY

Treil Chapter 4, Sections 1-2

Chapter 4

Introduction to spectral theory (eigenvalues and eigenvectors)

Spectral theory is the main tool that helps us to understand the structure of a linear operator. In this chapter we consider only operators acting from a vector space to itself (or, equivalently, $n \times n$ matrices). If we have such a linear transformation $A : V \to V$, we can multiply it by itself, take any power of it, or any polynomial.

The main idea of spectral theory is to split the operator into simple blocks and analyze each block separately.

To explain the main idea, let us consider *difference equations*. Many processes can be described by the equations of the following type

$$\mathbf{x}_{n+1} = A\mathbf{x}_n, \qquad n = 0, 1, 2, \dots,$$

where $A : V \to V$ is a linear transformation, and $\mathbf{x}_n$ is the state of the system at the time n. Given the initial state $\mathbf{x}_0$ we would like to know the state $\mathbf{x}_n$ at the time n, analyze the long time behavior of $\mathbf{x}_n$, etc. [1]

[1] The difference equations are discrete time analogues of the *differential equation* $\mathbf{x}'(t) = A\mathbf{x}(t)$. To solve the differential equation, one needs to compute $e^{tA} := \sum_{k=0}^{\infty} t^k A^n/k!$, and spectral theory also helps in doing this.

At the first glance the problem looks trivial: the solution $\mathbf{x}_n$ is given by the formula $\mathbf{x}_n = A^n\mathbf{x}_0$. But what if n is huge: thousands, millions? Or what if we want to analyze the behavior of $\mathbf{x}_n$ as $n \to \infty$?

Here the idea of *eigenvalues* and *eigenvectors* comes in. Suppose that $A\mathbf{x}_0 = \lambda\mathbf{x}_0$, where λ is some scalar. Then $A^2\mathbf{x}_0 = \lambda^2\mathbf{x}_0$, $A^3\mathbf{x}_0 = \lambda^3\mathbf{x}_0, \dots,$ $A^n\mathbf{x}_0 = \lambda^n\mathbf{x}_0$, so the behavior of the solution is very well understood

In this section we will consider only operators in finite-dimensional spaces. Spectral theory in infinitely many dimensions is significantly more complicated, and most of the results presented here fail in infinite-dimensional setting.

1. Main definitions

1.1. Eigenvalues, eigenvectors, spectrum. A scalar λ is called an *eigenvalue* of an operator $A : V \to V$ if there exists a *non-zero* vector $\mathbf{v} \in V$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is called the *eigenvector* of A (corresponding to the eigenvalue λ).

If we know that λ is an eigenvalue, the eigenvectors are easy to find: one just has to solve the equation $A\mathbf{x} = \lambda\mathbf{x}$, or, equivalently

$$(A - \lambda I)\mathbf{x} = \mathbf{0}.$$

So, finding *all* eigenvectors, corresponding to an eigenvalue λ is simply finding the nullspace of $A - \lambda I$. The nullspace $\operatorname{Ker}(A - \lambda I)$, i.e. the set of all eigenvectors and $\mathbf{0}$ vector, is called the *eigenspace*.

The set of all eigenvalues of an operator A is called *spectrum* of A, and is usually denoted $\sigma(A)$.

1.2. Finding eigenvalues: characteristic polynomials. A scalar λ is an eigenvalue if and only if the nullspace $\operatorname{Ker}(A - \lambda I)$ is non-trivial (so the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a non-trivial solution).

Let A act on $\mathbb{F}^n$ (i.e. $A : \mathbb{F}^n \to \mathbb{F}^n$). Since the matrix of A is square, $A - \lambda I$ has a non-trivial nullspace if and only if it is not invertible. We know that a square matrix is not invertible if and only if its determinant is 0. Therefore

$$\boxed{\lambda \in \sigma(A),\ \text{i.e. } \lambda \text{ is an eigenvalue of } A \quad \Longleftrightarrow \quad \det(A - \lambda I) = 0}$$

If A is an $n \times n$ matrix, the determinant $\det(A - \lambda I)$ is a polynomial of degree n of the variable λ. This polynomial is called the *characteristic polynomial* of A. So, to find all eigenvalues of A one just needs to compute the characteristic polynomial and find all its roots.

This method of finding the spectrum of an operator is not very practical in higher dimensions. Finding roots of a polynomial of high degree can be a very difficult problem, and it is impossible to solve the equation of degree higher than 4 in radicals. So, in higher dimensions different numerical methods of finding eigenvalues and eigenvectors are used.

1.3. Finding characteristic polynomial and eigenvalues of an abstract operator. So we know how to find the spectrum of a matrix. But how do we find eigenvalues of an operator acting in an abstract vector space? The recipe is simple:

Take an arbitrary basis, and compute eigenvalues of the matrix of the operator in this basis.

But how do we know that the result does not depend on a choice of the basis?

There can be several possible explanations. One is based on the notion of *similar matrices*. Let us recall that square matrices A and B are called similar if there exist an invertible matrix S such that

$$A = SBS^{-1}.$$

Note, that determinants of similar matrices coincide. Indeed

$$\det A = \det(SBS^{-1}) = \det S \det B \det S^{-1} = \det B$$

because $\det S^{-1} = 1/\det S$. Note that if $A = SBS^{-1}$ then

$$A - \lambda I = SBS^{-1} - \lambda SIS^{-1} = S(BS^{-1} - \lambda IS^{-1}) = S(B - \lambda I)S^{-1},$$

so the matrices $A - \lambda I$ and $B - \lambda I$ are similar. Therefore

$$\det(A - \lambda I) = \det(B - \lambda I),$$

i.e.

characteristic polynomials of similar matrices coincide.

If $T : V \to V$ is a linear transformation, and $\mathcal{A}$ and $\mathcal{B}$ are two bases in V, then

$$[T]_{\mathcal{A}\mathcal{A}} = [I]_{\mathcal{A}\mathcal{B}}[T]_{\mathcal{B}\mathcal{B}}[I]_{\mathcal{B}\mathcal{A}}$$

and since $[I]_{\mathcal{B}\mathcal{A}} = ([I]_{\mathcal{A}\mathcal{B}})^{-1}$ the matrices $[T]_{\mathcal{A}\mathcal{A}}$ and $[T]_{\mathcal{B}\mathcal{B}}$ are similar.

In other words, matrices of a linear transformation in different bases are similar.

Therefore, we can define the characteristic polynomial of an operator as the characteristic polynomial of its matrix in some basis. As we have discussed above, the result does not depend on the choice of the basis, so characteristic polynomial of an operator is well defined.

1.4. Complex vs real spaces. The fundamental theorem of algebra asserts that any polynomial (of degree at least 1) has a complex root. That implies that an operator in a finite-dimensional *complex* vector space has at least one eigenvalue, so its spectrum is non-empty.

On the other hand it is easy to construct a linear transformation in a real vector space without *real* eigenvalues, the rotation R_α, $\alpha \neq \pi n$ in $\mathbb{R}^2$ being one of examples. Since it is usually assumed that eigenvalues should belong to the field of scalars (if an operator acts in a vector space over a field $\mathbb{F}$ the eigenvalues should be in $\mathbb{F}$), such operators have empty spectrum.

Thus, the complex case (i.e. operators acting in complex vector spaces) seems to be the most natural setting for the spectral theory. Since $\mathbb{R} \subset \mathbb{C}$, we can always treat a real $n \times n$ matrix as an operator in $\mathbb{C}^n$ to allow complex eigenvalues. Treating real matrices as operators in $\mathbb{C}^n$ is typical in the spectral theory, and we will follow this agreement. Finding eigenvalues of a matrix (unless otherwise specified) will always mean finding all *complex* eigenvalues and not restricting oneself only to real ones.

Note that an operator in an abstract real vector space also can be interpreted as an operator in a complex space. A naïve approach would be to fix a basis (recall that all spaces in this chapter are finite-dimensional), and then work with coordinates in this basis allowing complex coordinates: that will be essentially move from operators in $\mathbb{R}^n$ to operators $\mathbb{C}^n$ described above.

This construction describes what is known as the *complexification* of a real vector space, and the result does not depend on the choice of a basis. A "high brow" abstract construction of the complexification, explaining why the result does not depend on the choice of a basis is described below in Section 8.2 of Chapter 5.

1.5. Multiplicities of eigenvalues. Let us remind the reader, that if p is a polynomial, and λ is its root (i.e. $p(\lambda) = 0$) then $z - \lambda$ divides $p(z)$, i.e. p can be represented as $p(z) = (z - \lambda)q(z)$, where q is some polynomial. If $q(\lambda) = 0$, then q also can be divided by $z - \lambda$, so $(z - \lambda)^2$ divides p and so on.

The largest positive integer k such that $(z - \lambda)^k$ divides $p(z)$ is called the multiplicity of the root λ.

If λ is an eigenvalue of an operator (matrix) A, then it is a root of the characteristic polynomial $p(z) = \det(A - zI)$. The multiplicity of this root is called the *(algebraic) multiplicity* of the eigenvalue λ.

Any polynomial $p(z) = \sum_{k=0}^{n} a_k z^k$ of degree n has exactly n complex roots, counting multiplicity. The words *counting multiplicities* mean that if a root has multiplicity d we have to list (count) it d times. In other words,

p can be represented as

$$p(z) = a_n(z - \lambda_1)(z - \lambda_2)\dots(z - \lambda_n).$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are its complex roots, counting multiplicities.

There is another notion of multiplicity of an eigenvalue: the dimension of the eigenspace $\operatorname{Ker}(A - \lambda I)$ is called *geometric multiplicity* of the eigenvalue λ.

Geometric multiplicity is not as widely used as algebraic multiplicity. So, when people say simply "multiplicity" they usually mean *algebraic multiplicity.*

Let us mention, that algebraic and geometric multiplicities of an eigenvalue can differ.

Proposition 1.1. *Geometric multiplicity of an eigenvalue cannot exceed its algebraic multiplicity.*

Proof. See Exercise 1.9 below. □

1.6. Trace and determinant.

Theorem 1.2. *Let A be $n \times n$ matrix, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be its (complex) eigenvalues (counting multiplicities). Then*

1. $\operatorname{trace} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$.
2. $\det A = \lambda_1 \lambda_2 \dots \lambda_n$.

Proof. See Exercises 1.10, 1.11 below. □

1.7. Eigenvalues of a triangular matrix. Computing eigenvalues is equivalent to finding roots of a characteristic polynomial of a matrix (or using some numerical method), which can be quite time consuming. However, there is one particular case, when we can just read eigenvalues off the matrix. Namely

eigenvalues of a triangular matrix (counting multiplicities) are exactly the diagonal entries $a_{1,1}, a_{2,2}, \dots, a_{n,n}$

By triangular here we mean either upper or lower triangular matrix. Since a diagonal matrix is a particular case of a triangular matrix (it is both upper and lower triangular

the eigenvalues of a diagonal matrix are its diagonal entries

The proof of the statement about triangular matrices is trivial: we need to subtract λ from the diagonal entries of A, and use the fact that determinant of a triangular matrix is the product of its diagonal entries. We get

the characteristic polynomial

$$\det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda)\dots(a_{n,n} - \lambda)$$

and its roots are exactly $a_{1,1}, a_{2,2}, \dots, a_{n,n}$.

Exercises.

1.1. True or false:

a) Every linear operator in an n-dimensional vector space has n distinct eigenvalues;

b) If a matrix has one eigenvector, it has infinitely many eigenvectors;

c) There exists a square real matrix with no real eigenvalues;

d) There exists a square matrix with no (complex) eigenvectors;

e) Similar matrices always have the same eigenvalues;

f) Similar matrices always have the same eigenvectors;

g) A non-zero sum of two eigenvectors of a matrix A is always an eigenvector;

h) A non-zero sum of two eigenvectors of a matrix A corresponding to the same eigenvalue λ is always an eigenvector.

1.2. Find characteristic polynomials, eigenvalues and eigenvectors of the following matrices:

$$\begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}.$$

1.3. Compute eigenvalues and eigenvectors of the rotation matrix

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

Note, that the eigenvalues (and eigenvectors) do not need to be real.

1.4. Compute characteristic polynomials and eigenvalues of the following matrices:

$$\begin{pmatrix} 1 & 2 & 5 & 67 \\ 0 & 2 & 3 & 6 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 1 & 0 & 2 \\ 0 & \pi & 43 & 2 \\ 0 & 0 & 16 & 1 \\ 0 & 0 & 0 & 54 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 2 & 4 & e & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix}.$$

Do not expand the characteristic polynomials, leave them as products.

1.5. Prove that eigenvalues (counting multiplicities) of a triangular matrix coincide with its diagonal entries

1.6. An operator A is called *nilpotent* if $A^k = \mathbf{0}$ for some k. Prove that if A is nilpotent, then $\sigma(A) = \{0\}$ (i.e. that 0 is the only eigenvalue of A).

1.7. Show that characteristic polynomial of a block triangular matrix

$$\begin{pmatrix} A & * \\ \mathbf{0} & B \end{pmatrix},$$

where A and B are square matrices, coincides with $\det(A-\lambda I)\det(B-\lambda I)$. (Use Exercise 3.11 from Chapter 3).

1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a basis in a vector space V. Assume also that the first k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ of the basis are eigenvectors of an operator A, corresponding to an eigenvalue λ (i.e. that $A\mathbf{v}_j = \lambda\mathbf{v}_j$, $j = 1, 2, \dots, k$). Show that in this basis the matrix of the operator A has block triangular form

$$\begin{pmatrix} \lambda I_k & * \\ \mathbf{0} & B \end{pmatrix},$$

where I_k is $k \times k$ identity matrix and B is some $(n-k) \times (n-k)$ matrix.

1.9. Use the two previous exercises to prove that geometric multiplicity of an eigenvalue cannot exceed its algebraic multiplicity.

1.10. Prove that determinant of a matrix A is the product of its eigenvalues (counting multiplicities).

Hint: first show that $\det(A-\lambda I) = (\lambda_1-\lambda)(\lambda_2-\lambda)\dots(\lambda_n-\lambda)$, where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues (counting multiplicities). Then compare the free terms (terms without λ) or plug in $\lambda = 0$ to get the conclusion.

1.11. Prove that the trace of a matrix equals the sum of eigenvalues in three steps. First, compute the coefficient of λ^{n-1} in the right side of the equality

$$\det(A-\lambda I) = (\lambda_1-\lambda)(\lambda_2-\lambda)\dots(\lambda_n-\lambda).$$

Then show that $\det(A-\lambda I)$ can be represented as

$$\det(A-\lambda I) = (a_{1,1}-\lambda)(a_{2,2}-\lambda)\dots(a_{n,n}-\lambda) + q(\lambda)$$

where $q(\lambda)$ is polynomial of degree at most $n-2$. And finally, comparing the coefficients of λ^{n-1} get the conclusion.

2. Diagonalization.

One of the application of the spectral theory is the diagonalization of operators, which means given an operator to find a basis in which the matrix of the operator is diagonal. Such basis does not always exists, i.e not all operators can be diagonalized (are diagonalizable). Importance of diagonalizable operators comes from the fact that the powers, and more general function of diagonal matrices are easy to compute, so if we diagonalize an operator we can easily compute functions of it.

We will explain how to compute functions of diagonalizable operators in this section. We also give a necessary and sufficient condition for an operator to be diagonalizable, as well as some simple sufficient conditions.

Note also that for operators in $\mathbb{F}^n$ (matrices) the diagonalizability of A means that it can be represented as $A = SDS^{-1}$, where D is a diagonal matrix (and S, of course, is invertible); we will explain this shortly.

Unless otherwise specified, all results in this section hold for both complex and real vector spaces (and even for spaces over arbitrary fields).

2.1. Preliminaries. Suppose an operator A in a vector space V is such that V has a basis $\mathcal{B} = \mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ of eigenvectors of A, with $\lambda_1, \lambda_2, \dots, \lambda_n$ being the corresponding eigenvalues. Then the matrix of A in this basis is the diagonal matrix with $\lambda_1, \lambda_2, \dots, \lambda_n$ on the diagonal

$$[A]_{\mathcal{B}\mathcal{B}} = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} = \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix}. \tag{2.1}$$

On the other hand, if the matrix of an operator A in a basis $\mathcal{B} = \mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ is given by (2.1) then trivially $A\mathbf{b}_k = \lambda_k \mathbf{b}_k$, i.e λ_k are eigenvalues and $\mathbf{b}_k$ are corresponding eigenvectors.

Note that the above reasoning holds for both complex and real vector spaces (and even for vector spaces over arbitrary fields)

Applying the above reasoning to operators in $\mathbb{F}^n$ (matrices) we immediately get the following theorem. Note, that while in this book $\mathbb{F}$ is either $\mathbb{C}$ or $\mathbb{R}$, this theorem hods for an arbitrary field $\mathbb{F}$.

Theorem 2.1. *A matrix A (with values in $\mathbb{F}$) admits a representation $A = SDS^{-1}$, where D is a diagonal matrix and S is an invertible one (both with entries in $\mathbb{F}$) if and only if there exists a basis in $\mathbb{F}^n$ of eigenvectors of A.*

Moreover, in this case diagonal entries of D are the eigenvalues and the columns of S are the corresponding eigenvectors (column number k corresponds to kth diagonal entry of D).

Proof. Let $D = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$, and let $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ be the columns of S (note that since S is invertible its columns form a basis in $\mathbb{F}^n$). Then the identity $A = SDS^{-1}$ means that $D = [A]_{\mathcal{B},\mathcal{B}}$.

Indeed, $S = [I]_{\mathcal{S},\mathcal{B}}$ is the change of the coordinates matrix from $\mathcal{B}$ to the standard basis $\mathcal{S}$, so we get from $A = SDS^{-1}$ that $D = S^{-1}AS = [I]_{\mathcal{B},\mathcal{S}} A [I]_{\mathcal{S},\mathcal{B}}]$, which means exactly that $D = [A]_{\mathcal{B},\mathcal{B}}$.

And as we just discussed above, $[A]_{\mathcal{B},\mathcal{B}} = D = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ if and only if λ_k are the eigenvalues and $\mathbf{b}_k$ are the corresponding eigenvectors of A. □

Remark. Note if a matrix admits the representation $A = SDS^{-1}$ with a diagonal matrix D, then a simple direct calculation shows that the columns of S are eigenvectors of A and diagonal entries of D are corresponding eigenvalues. This gives an alternative proof of the corresponding statement in Theorem 2.1.

As we discussed above, a diagonalizable operator $A : V \to V$ has exactly $n = \dim V$ eigenvalues (counting multiplicities). Any operator in a complex vector space V has n eigenvalues (counting multiplicities); an operator in a real space, on the other hand, could have no real eigenvalues.

We will, as it is customary in the spectral theory, treat real matrices as operators in the complex space $\mathbb{C}^n$, thus allowing complex eigenvalues and eigenvectors. Unless otherwise specified we will mean by the diagonalization of a matrix its *complex* diagonalization, i.e. a representation $A = SDS^{-1}$ where matrices S and D can have complex entries.

The question when a real matrix admits a real diagonalization ($A = SDS^{-1}$ with real matrices S and D) is in fact a very simple one, see Theorem 2.9 below.

2.2. Some motivations: functions of operators. Let the matrix of an operator A in a basis $\mathcal{B} = \mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$ is a diagonal one given by (2.1). Then it is easy to find an Nth power of the operator A. Namely, the matrix of A^N in the basis $\mathcal{B}$ is

$$[A^N]_{\mathcal{B}\mathcal{B}} = \operatorname{diag}\{\lambda_1^N, \lambda_2^N, \dots, \lambda_n^N\} = \begin{pmatrix} \lambda_1^N & & & 0 \\ & \lambda_2^N & & \\ & & \ddots & \\ 0 & & & \lambda_n^N \end{pmatrix}.$$

Moreover, functions of the operator are also very easy to compute: for example the operator (matrix) exponent e^{tA} is defined as $e^{tA} = I + tA + \dfrac{t^2A^2}{2!} + \dfrac{t^3A^3}{3!} + \ldots = \displaystyle\sum_{k=0}^{\infty} \frac{t^kA^k}{k!}$, and its matrix in the basis $\mathcal{B}$ is

$$[e^{tA}]_{\mathcal{B}\mathcal{B}} = \operatorname{diag}\{e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}\} = \begin{pmatrix} e^{\lambda_1 t} & & & 0 \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ 0 & & & e^{\lambda_n t} \end{pmatrix}.$$

Let now A be an operator in $\mathbb{F}^n$. To find the matrices of the operators A^N and e^{tA} in the standard basis $\mathcal{S}$, we need to recall that the change of coordinate matrix $[I]_{\mathcal{S}\mathcal{B}}$ is the matrix with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n$. Let us

call this matrix S, then according to the change of coordinates formula we have

$$A = [A]_{\mathcal{SS}} = S \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix} S^{-1} = SDS^{-1},$$

where we use D for the diagonal matrix in the middle.

Similarly

$$A^N = SD^N S^{-1} = S \begin{pmatrix} \lambda_1^N & & & 0 \\ & \lambda_2^N & & \\ & & \ddots & \\ 0 & & & \lambda_n^N \end{pmatrix} S^{-1} ,$$

and similarly for e^{tA}.

Another way of thinking about powers (or other functions) of diagonalizable operators is to see that if operator A can be represented as $A = SDS^{-1}$, then

$$A^N = \underbrace{(SDS^{-1})(SDS^{-1})\dots(SDS^{-1})}_{N \text{ times}} = SD^N S^{-1}$$

and it is easy to compute the Nth power of a diagonal matrix.

2.3. The case of n distinct eigenvalues. We now present very simple *sufficient* condition for an operator to be diagonalizable, see Corollary 2.3 below.

Theorem 2.2. *Let $\lambda_1, \lambda_2, \dots, \lambda_r$ be distinct eigenvalues of A, and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ be the corresponding eigenvectors. Then vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are linearly independent.*

Proof. We will use induction on r. The case $r = 1$ is trivial, because by the definition an eigenvector is non-zero, and a system consisting of one non-zero vector is linearly independent.

Suppose that the statement of the theorem is true for $r - 1$. Suppose there exists a non-trivial linear combination

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_r\mathbf{v}_r = \sum_{k=1}^{r} c_k\mathbf{v}_k = \mathbf{0}. \tag{2.2}$$

Applying $A - \lambda_r I$ to (2.2) and using the fact that $(A - \lambda_r I)\mathbf{v}_r = \mathbf{0}$ we get

$$\sum_{k=1}^{r-1} c_k(\lambda_k - \lambda_r)\mathbf{v}_k = \mathbf{0}.$$

By the induction hypothesis vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ are linearly independent, so $c_k(\lambda_k - \lambda_r) = 0$ for $k = 1, 2, \dots, r-1$. Since $\lambda_k \neq \lambda_r$ we can conclude that $c_k = 0$ for $k < r$. Then it follows from (2.2) that $c_r = 0$, i.e. we have the trivial linear combination. □

Corollary 2.3. *If an operator $A : V \to V$ has exactly $n = \dim V$ distinct eigenvalues, then it is diagonalizable.*

While very simple, this is a very important statement, and it will be used a lot! Please remember it.

Proof. For each eigenvalue λ_k let $\mathbf{v}_k$ be a corresponding eigenvector (just pick one eigenvector for each eigenvalue). By Theorem 2.2 the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is linearly independent, and since it consists of exactly $n = \dim V$ vectors it is a basis. □

2.4. Bases of subspaces (AKA direct sums of subspaces). To describe diagonalizable operators we need to introduce some new definitions.

Let $V_1, V_2, \dots, V_p$ be subspaces of a vector space V. We say that the system of subspaces is a basis in V if any vector $\mathbf{v} \in V$ admits a unique representation as a sum

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \ldots + \mathbf{v}_p = \sum_{k=1}^{p} \mathbf{v}_k, \qquad \mathbf{v}_k \in V_k. \tag{2.3}$$

We also say, that a system of subspaces $V_1, V_2, \dots, V_p$ is linearly independent if the equation

$$\mathbf{v}_1 + \mathbf{v}_2 + \ldots + \mathbf{v}_p = \mathbf{0}, \qquad \mathbf{v}_k \in V_k$$

has only trivial solution ($\mathbf{v}_k = \mathbf{0}\ \forall k = 1, 2, \dots, p$).

Another way to phrase that is to say that a system of subspaces $V_1, V_2, \dots, V_p$ is linearly independent if and only if any system of non-zero vectors $\mathbf{v}_k$, where $\mathbf{v}_k \in V_k$, is linearly independent.

We say that the system of subspaces $V_1, V_2, \dots, V_p$ is generating (or complete, or spanning) if any vector $\mathbf{v} \in V$ admits representation as (2.3) (not necessarily unique).

Remark 2.4. From the above definition one can immediately see that Theorem 2.2 in fact states that the system of eigenspaces E_k of an operator A

$$E_k := \operatorname{Ker}(A - \lambda_k I), \qquad \lambda_k \in \sigma(A),$$

is linearly independent.

Remark 2.5. It is easy to see that similarly to the bases of vectors, a system of subspaces $V_1, V_2, \dots, V_p$ is a basis if and only if it is generating and linearly independent. We leave the proof of this fact as an exercise for the reader.

There is a simple example of a basis of subspaces. Let V be a vector space with a basis $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$. Split the set of indices $1, 2, \ldots, n$ into p subsets $\Lambda_1, \Lambda_2, \ldots, \Lambda_p$, and define subspaces $V_k := \operatorname{span}\{\mathbf{v}_j : j \in \Lambda_k\}$. Clearly the subspaces V_k form a basis of V.

The following theorem shows that in the finite-dimensional case it is essentially the only possible example of a basis of subspaces.

Theorem 2.6. *Let $V_1, V_2, \ldots, V_p$ be a basis of subspaces, and let us have in each subspace V_k a basis (of vectors) $\mathcal{B}_k$[2]. Then the union $\cup_k \mathcal{B}_k$ of these bases is a basis in V.*

To prove the theorem we need the following lemma

Lemma 2.7. *Let $V_1, V_2, \ldots, V_p$ be a linearly independent family of subspaces, and let us have in each subspace V_k a linearly independent system $\mathcal{B}_k$ of vectors [3] Then the union $\mathcal{B} := \cup_k \mathcal{B}_k$ is a linearly independent system.*

Proof. The proof of the lemma is almost trivial, if one thinks a bit about it. The main difficulty in writing the proof is a choice of a appropriate notation. Instead of using two indices (one for the number k and the other for the number of a vector in $\mathcal{B}_k$, let us use "flat" notation.

Namely, let n be the number of vectors in $\mathcal{B} := \cup_k \mathcal{B}_k$. Let us order the set $\mathcal{B}$, for example as follows: first list all vectors from $\mathcal{B}_1$, then all vectors in $\mathcal{B}_2$, etc, listing all vectors from $\mathcal{B}_p$ last.

This way, we index all vectors in $\mathcal{B}$ by integers $1, 2, \ldots, n$, and the set of indices $\{1, 2, \ldots, n\}$ splits into the sets $\Lambda_1, \Lambda_2, \ldots, \Lambda_p$ such that the set $\mathcal{B}_k$ consists of vectors $\mathbf{b}_j : j \in \Lambda_k$.

Suppose we have a non-trivial linear combination

$$c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \ldots + c_n\mathbf{b}_n = \sum_{j=1}^{n} c_j\mathbf{b}_j = \mathbf{0}. \tag{2.4}$$

Denote

$$\mathbf{v}_k = \sum_{j \in \Lambda_k} c_j\mathbf{b}_j.$$

Then (2.4) can be rewritten as

$$\mathbf{v}_1 + \mathbf{v}_2 + \ldots + \mathbf{v}_p = \mathbf{0}.$$

[2] We do not list the vectors in $\mathcal{B}_k$, one just should keep in mind that each $\mathcal{B}_k$ consists of finitely many vectors in V_k

[3] Again, here we do not name each vector in $\mathcal{B}_k$ individually, we just keep in mind that each set $\mathcal{B}_k$ consists of finitely many vectors.

Since $\mathbf{v}_k \in V_k$ and the system of subspaces V_k is linearly independent, $\mathbf{v}_k = \mathbf{0}$ $\forall k$. Than means that for every k

$$\sum_{j\in\Lambda_k} c_j \mathbf{b}_j = \mathbf{0},$$

and since the system of vectors $\mathbf{b}_j : j \in \Lambda_k$ (i.e. the system $\mathcal{B}_k$) are linearly independent, we have $c_j = 0$ for all $j \in \Lambda_k$. Since it is true for all Λ_k, we can conclude that $c_j = 0$ for all j. □

Proof of Theorem 2.6. To prove the theorem we will use the same notation as in the proof of Lemma 2.7, i.e. the system $\mathcal{B}_k$ consists of vectors $\mathbf{b}_j$, $j \in \Lambda_k$.

Lemma 2.7 asserts that the system of vectors $\mathbf{b}_j$, $j = 1, 2, \dots, n$ is linearly independent, so it only remains to show that the system is complete.

Since the system of subspaces $V_1, V_2, \dots, V_p$ is a basis, any vector $\mathbf{v} \in V$ can be represented as

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \ldots + \mathbf{v}_p = \sum_{k=1}^{p} \mathbf{v}_k, \qquad \mathbf{v}_k \in V_k.$$

Since the vectors $\mathbf{b}_j$, $j \in \Lambda_k$ form a basis in V_k, the vectors $\mathbf{v}_k$ can be represented as

$$\mathbf{v}_k = \sum_{j\in\Lambda_k} c_j \mathbf{b}_j,$$

and therefore $\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{b}_j$. □

2.5. Criterion of diagonalizability. First of all let us recall a simple necessary condition. Since the eigenvalues (counting multiplicities) of a diagonal matrix $D = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ are exactly $\lambda_1, \lambda_2, \dots, \lambda_n$, we see that if an operator $A : V \to V$ is diagonalizable, it has exactly $n = \dim V$ eigenvalues (counting multiplicities).

Theorem below holds for both real and complex vector spaces (and even for spaces over genera fields).

Theorem 2.8. *Let an operator $A : V \to V$ has exactly $n = \dim V$ eigenvalues (counting multiplicities)[4]. Then A is diagonalizable if and only if for each eigenvalue λ the dimension of the eigenspace* $\operatorname{Ker}(A - \lambda I)$ *(i.e. the geometric multiplicity of λ) coincides with the algebraic multiplicity of λ.*

Proof. First of all let us note, that for a diagonal matrix, the algebraic and geometric multiplicities of eigenvalues coincide, and therefore the same holds for the diagonalizable operators.

[4]Since any operator in a complex vector space has exactly n eigenvalues (counting multiplicities), this assumption is moot in the complex case.

Let us now prove the other implication. Let $\lambda_1, \lambda_2, \dots, \lambda_p$ be eigenvalues of A, and let $E_k := \operatorname{Ker}(A - \lambda_k I)$ be the corresponding eigenspaces. According to Remark 2.4, the subspaces E_k, $k = 1, 2, \dots, p$ are linearly independent.

Let $\mathcal{B}_k$ be a basis in E_k. By Lemma 2.7 the system $\mathcal{B} = \cup_k \mathcal{B}_k$ is a linearly independent system of vectors.

We know that each $\mathcal{B}_k$ consists of $\dim E_k$(= multiplicity of λ_k) vectors. So the number of vectors in $\mathcal{B}$ equal to the sum of multiplicities of eigenvalues λ_k. But the sum of multiplicities of the eigenvalues is the number of eigenvalues counting multiplicities, which is exactly $n = \dim V$. So, we have a linearly independent system of $n = \dim V$ eigenvectors, which means it is a basis. □

2.6. Real factorization. The theorem below is, in fact, already proven (iT is essentially Theorem 2.8 for real spaces). We state it here to summarize the situation with real diagonalization of real matrices.

Theorem 2.9. *A real $n \times n$ matrix A admits a real factorization (i.e. representation $A = SDS^{-1}$ where S and D are real matrices, D is diagonal and S is invertible) if and only if it admits complex factorization and all eigenvalues of A are real.*

2.7. Some example.

2.7.1. *Real eigenvalues.* Consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 8 & 1 \end{pmatrix}.$$

Its characteristic polynomial is equal to

$$\begin{vmatrix} 1-\lambda & 2 \\ 8 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 16$$

and its roots (eigenvalues) are $\lambda = 5$ and $\lambda = -3$. For the eigenvalue $\lambda = 5$

$$A - 5I = \begin{pmatrix} 1-5 & 2 \\ 8 & 1-5 \end{pmatrix} = \begin{pmatrix} -4 & 2 \\ 8 & -4 \end{pmatrix}$$

A basis in its nullspace consists of one vector $(1, 2)^T$, so this is the corresponding eigenvector.

Similarly, for $\lambda = -3$

$$A - \lambda I = A + 3I = \begin{pmatrix} 4 & 2 \\ 8 & 4 \end{pmatrix}$$

and the eigenspace $\mathrm{Ker}(A+3I)$ is spanned by the vector $(1,-2)^T$. The matrix A can be diagonalized as

$$A = \begin{pmatrix} 1 & 2 \\ 8 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}^{-1}$$

2.7.2. *Complex eigenvalues.* Consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}.$$

Its characteristic polynomial is

$$\begin{vmatrix} 1-\lambda & 2 \\ -2 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 + 2^2$$

and the eigenvalues (roots of the characteristic polynomial are $\lambda = 1 \pm 2i$. For $\lambda = 1+2i$

$$A - \lambda I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}$$

This matrix has rank 1, so the eigenspace $\mathrm{Ker}(A-\lambda T)$ is spanned by one vector, for example by $(1,i)^T$.

Since the matrix A is real, we do not need to compute an eigenvector for $\lambda = 1-2i$: we can get it for free by taking the complex conjugate of the above eigenvector, see Exercise 2.2 below. So, for $\lambda = 1-2i$ a corresponding eigenvector is $(1,-i)^T$, and so the matrix A can be diagonalized as

$$A = \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \begin{pmatrix} 1+2i & 0 \\ 0 & 1-2i \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}^{-1}.$$

2.7.3. *A non-diagonalizable matrix.* Consider the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Its characteristic polynomial is

$$\begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^2,$$

so A has an eigenvalue 1 of multiplicity 2. But, it is easy to see that $\dim \mathrm{Ker}(A-I) = 1$ (1 pivot, so $2-1=1$ free variable). Therefore, the geometric multiplicity of the eigenvalue 1 is different from its algebraic multiplicity, so A is not diagonalizable.

There is also an explanation which does not use Theorem 2.8. Namely, we got that the eigenspace $\mathrm{Ker}(A-1I)$ is one dimensional (spanned by the vector $(1,0)^T$). If A were diagonalizable, it would have a diagonal form

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ in some basis,[5] and so the dimension of the eigenspace wold be 2. Therefore A cannot be diagonalized.

Exercises.

2.1. Let A be $n \times n$ matrix. True or false:

a) A^T has the same eigenvalues as A.

b) A^T has the same eigenvectors as A.

c) If A is is diagonalizable, then so is A^T.

Justify your conclusions.

2.2. Let A be a square matrix with real entries, and let λ be its complex eigenvalue. Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$ is a corresponding eigenvector, $A\mathbf{v} = \lambda\mathbf{v}$. Prove that the $\overline{\lambda}$ is an eigenvalue of A and $A\overline{\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}}$. Here $\overline{\mathbf{v}}$ is the complex conjugate of the vector $\mathbf{v}$, $\overline{\mathbf{v}} := (\overline{v}_1, \overline{v}_2, \dots, \overline{v}_n)^T$.

2.3. Let

$$A = \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix}.$$

Find A^{2004} by diagonalizing A.

2.4. Construct a matrix A with eigenvalues 1 and 3 and corresponding eigenvectors $(1, 2)^T$ and $(1, 1)^T$. Is such a matrix unique?

2.5. Diagonalize the following matrices, if possible:

a) $\begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$.

b) $\begin{pmatrix} -1 & -1 \\ 6 & 4 \end{pmatrix}$.

c) $\begin{pmatrix} -2 & 2 & 6 \\ 5 & 1 & -6 \\ -5 & 2 & 9 \end{pmatrix}$ ($\lambda = 2$ is one of the eigenvalues)

2.6. Consider the matrix

$$A = \begin{pmatrix} 2 & 6 & -6 \\ 0 & 5 & -2 \\ 0 & 0 & 4 \end{pmatrix}.$$

a) Find its eigenvalues. Is it possible to find the eigenvalues without computing?

b) Is this matrix diagonalizable? Find out without computing anything.

c) If the matrix is diagonalizable, diagonalize it.

---

[5] Note, that the only linear transformation having matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ in some basis is the identity transformation I. Since A is definitely not the identity, we can immediately conclude that A cannot be diagonalized, so counting dimension of the eigenspace is not necessary.

2.7. Diagonalize the matrix

$$\begin{pmatrix} 2 & 0 & 6 \\ 0 & 2 & 4 \\ 0 & 0 & 4 \end{pmatrix}.$$

2.8. Find all square roots of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ -3 & 0 \end{pmatrix}$$

i.e. find all matrices B such that $B^2 = A$. **Hint:** Finding a square root of a diagonal matrix is easy. You can leave your answer as a product.

2.9. Let us recall that the famous Fibonacci sequence:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$$

is defined as follows: we put $\varphi_0 = 0$, $\varphi_1 = 1$ and define

$$\varphi_{n+2} = \varphi_{n+1} + \varphi_n.$$

We want to find a formula for φ_n. To do this

a) Find a 2×2 matrix A such that

$$\begin{pmatrix} \varphi_{n+2} \\ \varphi_{n+1} \end{pmatrix} = A \begin{pmatrix} \varphi_{n+1} \\ \varphi_n \end{pmatrix}$$

Hint: Combine the trivial equation $\varphi_{n+1} = \varphi_{n+1}$ with the Fibonacci relation $\varphi_{n+2} = \varphi_{n+1} + \varphi_n$.

b) Diagonalize A and find a formula for A^n.

c) Noticing that

$$\begin{pmatrix} \varphi_{n+1} \\ \varphi_n \end{pmatrix} = A^n \begin{pmatrix} \varphi_1 \\ \varphi_0 \end{pmatrix} = A^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

find a formula for φ_n. (You will need to compute an inverse and perform multiplication here).

d) Show that the vector $(\varphi_{n+1}/\varphi_n, 1)^T$ converges to an eigenvector of A. What do you think, is it a coincidence?

2.10. Let A be a 5×5 matrix with 3 eigenvalues (not counting multiplicities). Suppose we know that one eigenspace is three-dimensional. Can you say if A is diagonalizable?

2.11. Give an example of a 3×3 matrix which cannot be diagonalized. After you constructed the matrix, can you make it "generic", so no special structure of the matrix could be seen?

2.12. Let a non-zero matrix A satisfy $A^5 = \mathbf{0}$. Prove that A cannot be diagonalized. More generally, any non-zero nilpotent matrix, i.e. a non-zero matrix satisfying $A^N = \mathbf{0}$ for some N cannot be diagonalized.

2.13. Eigenvalues of a transposition:

a) Consider the transformation T in the space $M_{2\times 2}$ of 2×2 matrices, $T(A) = A^T$. Find all its eigenvalues and eigenvectors. Is it possible to diagonalize this transformation? **Hint:** While it is possible to write a matrix of this linear transformation in some basis, compute characteristic polynomial, and so on, it is easier to find eigenvalues and eigenvectors directly from the definition.

b) Can you do the same problem but in the space of $n \times n$ matrices?

2.14. Prove that two subspaces V_1 and V_2 are linearly independent if and only if $V_1 \cap V_2 = \{\mathbf{0}\}$.

QUADRATIC FORMS

Treil Chapter 7, Sections 1-4

Chapter 7

Bilinear and quadratic forms

While the study of *real* quadratic forms (i.e. real homogeneous polynomials of degree 2) was probably the initial motivation for the subject of this chapter, complex quadratic forms $(A\mathbf{x}, \mathbf{x})$, $\mathbf{x} \in \mathbb{C}^n$, $A = A^*$ are also of significant interest. So, unless otherwise specified, result and calculations hold in both real and complex case.

To avoid writing twice essentially the same formulas, we use the notation adapted to the complex case: in particular, in the real case the notation A^* is used instead of A^T.

1. Main definition

1.1. Bilinear forms on $\mathbb{R}^n$. A bilinear form on $\mathbb{R}^n$ is a function $L = L(\mathbf{x}, \mathbf{y})$ of two arguments $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ which is linear in each argument, i.e. such that

1. $L(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2, \mathbf{y}) = \alpha L(\mathbf{x}_1, \mathbf{y}) + \beta L(\mathbf{x}_2, \mathbf{y})$;
2. $L(\mathbf{x}, \alpha\mathbf{y}_1 + \beta\mathbf{y}_2) = \alpha L(\mathbf{x}, \mathbf{y}_1) + \beta L(\mathbf{x}, \mathbf{y}_2)$.

One can consider bilinear form whose values belong to an arbitrary vector space, but in this book we only consider forms that take real values.

If $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ and $\mathbf{y} = (y_1, y_2, \dots, y_n)^T$, a bilinear form can be written as

$$L(\mathbf{x}, \mathbf{y}) = \sum_{j,k=1}^{n} a_{j,k} x_k y_j,$$

or in matrix form

$$L(\mathbf{x},\mathbf{y}) = (A\mathbf{x},\mathbf{y})$$

where

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,n} \end{pmatrix}.$$

The matrix A is determined uniquely by the bilinear form L.

1.2. Quadratic forms on $\mathbb{R}^n$. There are several equivalent definition of a quadratic form.

One can say that a quadratic form on $\mathbb{R}^n$ is the *"diagonal"* of a bilinear form L, i.e. that any quadratic form Q is defined by $Q[\mathbf{x}] = L(\mathbf{x},\mathbf{x}) = (A\mathbf{x},\mathbf{x})$.

Another, more algebraic way, is to say that a quadratic form is a *homogeneous polynomial of degree 2*, i.e. that $Q[\mathbf{x}]$ is a polynomial of n variables $x_1, x_2, \dots, x_n$ having only terms of degree 2. That means that only terms ax_k^2 and cx_jx_k are allowed.

There many ways (in fact, infinitely many) to write a quadratic form $Q[\mathbf{x}]$ as $Q[\mathbf{x}] = (A\mathbf{x},\mathbf{x})$. For example, the quadratic form $Q[\mathbf{x}] = x_1^2 + x_2^2 - 4x_1x_2$ on $\mathbb{R}^2$ can be represented as $(A\mathbf{x},\mathbf{x})$ where A can be any of the matrices

$$\begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}.$$

In fact, any matrix A of form

$$\begin{pmatrix} 1 & a-4 \\ -a & 1 \end{pmatrix}$$

will work.

But if we require the matrix A to be symmetric, then such a matrix is unique:

Any quadratic form $Q[\mathbf{x}]$ on $\mathbb{R}^n$ admits unique representation $Q[\mathbf{x}] = (A\mathbf{x},\mathbf{x})$ where A is a (real) symmetric matrix.

For example, for the quadratic form

$$Q[\mathbf{x}] = x_1^2 + 3x_2^2 + 5x_3^2 + 4x_1x_2 - 16x_1x_3 + 7x_2x_3$$

on $\mathbb{R}^3$, the corresponding symmetric matrix A is

$$\begin{pmatrix} 1 & 2 & -8 \\ 2 & 3 & 3.5 \\ -8 & 3.5 & 5 \end{pmatrix}.$$

1.3. Quadratic forms on $\mathbb{C}^n$. One can also define a *quadratic form* on $\mathbb{C}^n$ (or any complex inner product space) by taking a self-adjoint transformation $A = A^*$ and defining Q by $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$. While our main examples will be in $\mathbb{R}^n$, all the theorems are true in the setting of $\mathbb{C}^n$ as well. Bearing this in mind, we will always use A^* instead of A^T

The only essential difference with the real case is that in the complex case we do not have any freedom of choice: if the quadratic form is real, the corresponding matrix has to be Hermitian (self-adjoint).

Note that if $A = A^*$ then

$$(A\mathbf{x}, \mathbf{x}) = (\mathbf{x}, A^*\mathbf{x}) = (\mathbf{x}, A\mathbf{x}) = \overline{(A\mathbf{x}, \mathbf{x})},$$

so $(A\mathbf{x}, \mathbf{x}) \in \mathbb{R}$.

The converse is also true.

Lemma 1.1. *Let $(A\mathbf{x}, \mathbf{x})$ be real for all $\mathbf{x} \in \mathbb{C}^n$. Then $A = A^*$.*

We leave the proof as an exercise for the reader, see Problem 1.4 below.

One of the possible ways to prove Lemma 1.1 is to use the following version of polarization identities.

Lemma 1.2. *Let A be an operator in an inner product space X.*

1. *If X is a complex space, then for any $\mathbf{x}, \mathbf{y} \in X$*

$$(A\mathbf{x}, \mathbf{y}) = \frac{1}{4} \sum_{\alpha \in \mathbb{C}: \alpha^4 = 1} \alpha (A(\mathbf{x} + \alpha\mathbf{y}), \mathbf{x} + \alpha\mathbf{y}).$$

2. *If X is a real space and $A = A^*$, then any $\mathbf{x}, \mathbf{y} \in X$*

$$(A\mathbf{x}, \mathbf{y}) = \frac{1}{4} \Big[(A(\mathbf{x} + \mathbf{y}), \mathbf{x} + \mathbf{y}) - (A(\mathbf{x} - \mathbf{y}), \mathbf{x} - \mathbf{y}) \Big].$$

For the proof of Lemma 1.2 see Exercise 6.3 in Chapter 5 above.

Exercises.

1.1. Find the matrix of the bilinear form L on $\mathbb{R}^3$,

$L(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_1y_2 + 14x_1y_3 - 5x_2y_1 + 2x_2y_2 - 3x_2y_3 + 8x_3y_1 + 19x_3y_2 - 2x_3y_3$.

1.2. Define the bilinear form L on $\mathbb{R}^2$ by

$$L(\mathbf{x}, \mathbf{y}) = \det[\mathbf{x}, \mathbf{y}],$$

i.e. to compute $L(\mathbf{x}, \mathbf{y})$ we form a 2×2 matrix with columns $\mathbf{x}, \mathbf{y}$ and compute its determinant.

Find the matrix of L.

1.3. Find the matrix of the quadratic form Q on $\mathbb{R}^3$

$$Q[\mathbf{x}] = x_1^2 + 2x_1x_2 - 3x_1x_3 - 9x_2^2 + 6x_2x_3 + 13x_3^2.$$

1.4. Prove Lemma 1.1 above.

Hint: Use the polarization identity, see Lemma 1.2. Alternatively, you can consider the expression $(A(\mathbf{x}+z\mathbf{y}), \mathbf{x}+z\mathbf{y})$ and show that if it is real for all $z \in \mathbb{C}$ then $(A\mathbf{x}, \mathbf{y}) = \overline{(\mathbf{y}, A^*\mathbf{x})}$.

2. Diagonalization of quadratic forms

You have probably met quadratic forms before, when you studied second order curves in the plane. Maybe you even studied the second order surfaces in $\mathbb{R}^3$.

We want to present a unified approach to classification of such objects. Suppose we are given a set in $\mathbb{R}^n$ defined by the equation $Q[\mathbf{x}] = 1$, where Q is some quadratic form. If Q has some simple form, for example if the corresponding matrix is diagonal, i.e. if $Q[x] = a_1x_1^2 + a_2x_2^2 + \ldots + a_nx_n^2$, we can easily visualize this set, especially if $n = 2, 3$. In higher dimensions, it is also possible, if not to visualize, then to understand the structure of the set very well.

So, if we are given a general, complicated quadratic form, we want to simplify it as much as possible, for example to make it diagonal. The standard way of doing that is the change of variables.

2.1. Orthogonal diagonalization. Let us have a quadratic form $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$ in $\mathbb{F}^n$ ($\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$). Introduce new variables $\mathbf{y} = (y_1, y_2, \ldots, y_n)^T \in \mathbb{F}^n$, with $\mathbf{y} = S^{-1}\mathbf{x}$, where S is some invertible $n \times n$ matrix, so $\mathbf{x} = S\mathbf{y}$.

Then,

$$Q[\mathbf{x}] = Q[S\mathbf{y}] = (AS\mathbf{y}, S\mathbf{y}) = (S^*AS\mathbf{y}, \mathbf{y}),$$

so in the new variables $\mathbf{y}$ the quadratic form has matrix S^*AS.

So, we want to find an invertible matrix S such that the matrix S^*AS is diagonal. Note, that it is different from the diagonalization of matrices we had before: we tried to represent a matrix A as $A = SDS^{-1}$, so the matrix $D = S^{-1}AS$ was diagonal. However, for unitary matrices U, we have $U^* = U^{-1}$, and we can orthogonally diagonalize symmetric matrices. So we can apply the orthogonal diagonalization we studied before to the quadratic forms.

Namely, we can represent the matrix A as $A = UDU^* = UDU^{-1}$. Recall, that D is a diagonal matrix with eigenvalues of A on the diagonal, and U is the matrix of eigenvectors (we need to pick an orthogonal basis of eigenvectors). Then $D = U^*AU$, so in the variables $\mathbf{y} = U^{-1}\mathbf{x}$ the quadratic form has diagonal matrix.

Let us analyze the geometric meaning of the orthogonal diagonalization. The columns $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ of the unitary matrix U form an orthonormal basis in $\mathbb{F}^n$, let us call this basis $\mathcal{B}$. The change of coordinate matrix $[I]_{\mathcal{S},\mathcal{B}}$ from this basis to the standard one is exactly U. We know that $\mathbf{y} = (y_1, y_2, \dots, y_n)^T = U^{-1}\mathbf{x}$, so the coordinates $y_1, y_2, \dots, y_n$ can be interpreted as coordinates of the vector $\mathbf{x}$ in the new basis $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$.

So, orthogonal diagonalization allows us to visualize very well the set $Q[\mathbf{x}] = 1$, or a similar one, as long as we can visualize it for diagonal matrices.

Example. Consider the quadratic form of two variables (i.e. quadratic form on $\mathbb{R}^2$), $Q(x, y) = 2x^2 + 2y^2 + 2xy$. Let us describe the set of points $(x, y)^T \in \mathbb{R}^2$ satisfying $Q(x, y) = 1$.

The matrix A of Q is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Orthogonally diagonalizing this matrix we can represent it as

$$A = U \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} U^*, \qquad \text{where} \quad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

or, equivalently

$$U^* A U = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} =: D.$$

The set $\{\mathbf{y} : (D\mathbf{y}, \mathbf{y}) = 1\}$ is the ellipse with half-axes $1/\sqrt{3}$ and 1. Therefore the set $\{\mathbf{x} \in \mathbb{R}^2 : (A\mathbf{x}, \mathbf{x}) = 1\}$, is the same ellipse only in the basis $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$, $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$, or, equivalently, the same ellipse, rotated $\pi/4$.

2.2. Non-orthogonal diagonalization. Orthogonal diagonalization involves computing eigenvalues and eigenvectors, so it may be difficult to do without computers for large n. On the other hand, the non-orthogonal diagonalization, i.e. finding an invertible S (without requiring $S^{-1} = S^*$) such that $D = S^* A S$ is diagonal, is much easier computationally and can be done using only algebraic operations (addition, subtraction, multiplication, division).

Below we present two most used methods of non-orthogonal diagonalization.

2.2.1. *Diagonalization by completion of squares.* The first methods is based on completion of squares. We will illustrate this method on real quadratic forms (forms on $\mathbb{R}^n$). After simple modifications this method could be used in the complex case, but we will not discuss it here. If necessary, an interested reader should be able to to make the appropriate modifications.

Let us again consider the quadratic form of two variables, $Q[\mathbf{x}] = 2x_1^2 + 2x_1x_2 + 2x_2^2$ (it is the same quadratic form as in the above example, only

here we call variables not x, y but x_1, x_2). Since

$$2\left(x_1+\frac{1}{2}x_2\right)^2 = 2\left(x_1^2+2x_1\frac{1}{2}x_2+\frac{1}{4}x_2^2\right) = 2x_1^2+2x_1x_2+\frac{1}{2}x_2^2$$

(note, that the first two terms coincide with the first two terms of Q), we get

$$2x_1^2+2x_1x_2+2x_2^2 = 2\left(x_1+\frac{1}{2}x_2\right)^2+\frac{3}{2}x_2^2 = 2y_1^2+\frac{3}{2}y_2^2,$$

where $y_1 = x_1+\frac{1}{2}x_2$ and $y_2 = x_2$.

The same method can be applied to quadratic form of more than 2 variables. Let us consider, for example, a form $Q[\mathbf{x}]$ in $\mathbb{R}^3$,

$$Q[\mathbf{x}] = x_1^2-6x_1x_2+4x_1x_3-6x_2x_3+8x_2^2-3x_3^2.$$

Considering all terms involving the first variable x_1 (the first 3 terms in this case), let us pick a full square or a multiple of a full square which has exactly these terms (plus some other terms).

Since

$$(x_1-3x_2+2x_3)^2 = x_1^2-6x_1x_2+4x_1x_3-12x_2x_3+9x_2^2+4x_3^2$$

we can rewrite the quadratic form as

$$(x_1-3x_2+2x_3)^2-x_2^2+6x_2x_3-7x_3^2.$$

Note, that the expression $-x_2^2+6x_2x_3-7x_3^2$ involves only variables x_2 and x_3. Since

$$-(x_2-3x_3)^2 = -(x_2^2-6x_2x_3+9x_3^2) = -x_2^2+6x_2x_3-9x_3^2$$

we have

$$-x_2^2+6x_2x_3-7x_3^2 = -(x_2-3x_3)^2+2x_3^2.$$

Thus we can write the quadratic form Q as

$$Q[\mathbf{x}] = (x_1-3x_2+2x_3)^2-(x_2-3x_3)^2+2x_3^2 = y_1^2-y_2^2+2y_3^2$$

where

$$y_1 = x_1-3x_2+2x_3, \qquad y_2 = x_2-3x_3, \qquad y_3 = x_3.$$

Finally, let us address the question that an attentive reader is probably already asking: what to do if at some point we do have a product of two variables, but no corresponding squares? For example, how to diagonalize the form x_1x_2? The answer follows immediately from the identity

$$4x_1x_2 = (x_1+x_2)^2-(x_1-x_2)^2, \tag{2.1}$$

which gives us the representation

$$Q[\mathbf{x}] = y_1^2-y_2^2, \qquad y_1 = (x_1+x_2)/2,\ y_2 = (x_1-x_2)/2.$$

2.2.2. *Diagonalization using row/column operations.* There is another way of performing non-orthogonal diagonalization of a quadratic form. The idea is to perform row operations on the matrix A of the quadratic form. The difference with the row reduction (Gauss–Jordan elimination) is that after each row operation we need to perform the same column operation, the reason for that being that we want to make the matrix S^*AS diagonal.

Let us explain how everything works on an example. Suppose we want to diagonalize a quadratic form with matrix

$$A = \begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix}.$$

We augment the matrix A by the identity matrix, and perform on the augmented matrix $(A|I)$ row/column operations. After each row operation we have to perform on the matrix A the same column operation.[1] We get

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 3 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \begin{matrix} \\ +R_1 \\ \\ \end{matrix} \sim \left(\begin{array}{ccc|ccc} 1 & -1 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 1 & 0 \\ 3 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \sim$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 1 & 0 \\ 3 & 4 & 1 & 0 & 0 & 1 \end{array}\right) \begin{matrix} \\ \\ -3R_1 \end{matrix} \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 1 & 0 \\ 0 & 4 & -8 & -3 & 0 & 1 \end{array}\right) \sim$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 4 & 1 & 1 & 0 \\ 0 & 4 & -8 & -3 & 0 & 1 \end{array}\right) \begin{matrix} \\ \\ -4R_2 \end{matrix} \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -24 & -7 & -4 & 1 \end{array}\right) \cdot S^{-1}$$

Diagonal matrix ← → Not orthonormal

Note, that we perform column operations only on the left side of the augmented matrix

We get the diagonal D matrix on the left, and the matrix S^* on the right, so $D = S^*AS$,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -24 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -7 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix}.$$

Let us explain why the method works. A row operation is a left multiplication by an elementary matrix. The corresponding column operation is the right multiplication by the transposed elementary matrix. Therefore,

[1] In the case of complex Hermitian matrices we perform for each row operation the conjugate o the corresponding column operation, see Remark 2.1 below

performing row operations $E_1, E_2, \dots, E_N$ and the same column operations we transform the matrix A to

$$E_N \dots E_2 E_1 A E_1^* E_2^* \dots E_N^* = EAE^*. \tag{2.2}$$

As for the identity matrix in the right side, we performed only row operations on it, so the identity matrix is transformed to

$$E_N \dots E_2 E_1 I = EI = E.$$

If we now denote $E^* = S$ we get that S^*AS is a diagonal matrix, and the matrix $E = S^*$ is the right half of the transformed augmented matrix.

In the above example we got lucky, because we did not need to interchange two rows. This operation is a bit tricker to perform. It is quite simple if you know what to do, but it may be hard to guess the correct row operations. Let us consider, for example, a quadratic form with the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

If we want to diagonalize it by row and column operations, the simplest idea would be to interchange rows 1 and 2. But we also must to perform the same column operation, i.e. interchange columns 1 and 2, so we will end up with the same matrix.

So, we need something more non-trivial. The identity (2.1), for example, can be used to diagonalize this quadratic form. However, a simpler idea also works: use row operations to get a non-zero entry on the diagonal! For example, if we start with making $a_{1,1}$ non-zero, the following series of row (and the corresponding column) operations is one of the possible choices:

$$\left(\begin{array}{cc|cc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right) {\scriptstyle +\frac{1}{2}R_2} \sim \left(\begin{array}{cc|cc} 1/2 & 1 & 1 & 1/2 \\ 1 & 0 & 0 & 1 \end{array}\right) \sim$$

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 1/2 \\ 1 & 0 & 0 & 1 \end{array}\right) {\scriptstyle -R_1} \sim \left(\begin{array}{cc|cc} 1 & 1 & 1 & 1/2 \\ 0 & -1 & -1 & 1/2 \end{array}\right) \sim$$

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & 1/2 \\ 0 & -1 & -1 & 1/2 \end{array}\right).$$

Remark. Non-orthogonal diagonalization gives us a simple description of a set $Q[\mathbf{x}] = 1$ in a non-orthogonal basis. It is harder to visualize, than the representation given by the orthogonal diagonalization. However, if we are not interested in the details, for example if it is sufficient for us just to know that the set is an ellipsoid (or hyperboloid, etc), then the non-orthogonal diagonalization is an easier way to get the answer.

Remark 2.1. For quadratic forms with complex entries (i.e. for forms $(A\mathbf{x}, \mathbf{x})$, $A = A^*$), the non-orthogonal diagonalization works the same way

as in the real case, with the only difference, that the corresponding "column operations" have the complex conjugate coefficients.

The reason for that is that if a row operation is given by left multiplication by an elementary matrix E_k, then the corresponding column operation is given by the right multiplication by E_k^*, see (2.2).

Note that formula (2.2) works in both complex and reals cases: in real case we could write E_k^T instead of E_k^*, but using Hermitian adjoint allows us to have the same formula in both cases.

Exercises.

2.1. Diagonalize the quadratic form with the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{pmatrix}.$$

Use two methods: completion of squares and row operations. Which one do you like better?

Can you say if the matrix A is positive definite or not?

2.2. For the matrix A

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

orthogonally diagonalize the corresponding quadratic form, i.e. find a diagonal matrix D and a unitary matrix U such that $D = U^*AU$.

3. Sylvester's Law of Inertia

As we discussed above, there are many ways to diagonalize a quadratic form. Note, that a resulting diagonal matrix is not unique. For example, if we got a diagonal matrix

$$D = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\},$$

we can take a diagonal matrix

$$S = \operatorname{diag}\{s_1, s_2, \dots, s_n\}, \qquad s_k \in \mathbb{R}, \quad s_k \neq 0$$

and transform D to

$$S^*DS = \operatorname{diag}\{s_1^2\lambda_1, s_2^2\lambda_2, \dots, s_n^2\lambda_n\}.$$

This transformation changes the diagonal entries of D. However, it does not change the *signs* of the diagonal entries. And this is always the case!

Namely, the famous Sylvester's Law of Inertia states that:

> For a Hermitian matrix A (i.e. for a quadratic form $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$) and any of its diagonalization $D = S^*AS$, the number of positive (negative, zero) diagonal entries of D depends only on A, but not on a particular choice of diagonalization.

Here we of course assume that S is an invertible matrix, and D is a diagonal one.

The idea of the proof of the Sylvester's Law of Inertia is to express the number of positive (negative, zero) diagonal entries of a diagonalization $D = S^*AS$ in terms of A, not involving S or D.

We will need the following definition.

Definition. Given an $n \times n$ Hermitian matrix $A = A^*$ (a quadratic form $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$ on $\mathbb{F}^n$) we call a subspace $E \subset \mathbb{F}^n$ *positive* (resp. *negative*, resp. *neutral*) if

$$(A\mathbf{x}, \mathbf{x}) > 0 \qquad (\text{resp. } (A\mathbf{x}, \mathbf{x}) < 0, \quad \text{resp. } (A\mathbf{x}, \mathbf{x}) = 0)$$

for all $\mathbf{x} \in E$, $\mathbf{x} \neq \mathbf{0}$.

Sometimes, to emphasize the role of A we will say A-positive (A negative, A-neutral).

Theorem 3.1. *Let A be an $n \times n$ Hermitian matrix, and let $D = S^*AS$ be its diagonalization by an invertible matrix S. Then the number of positive (resp. negative) diagonal entries of D coincides with the maximal dimension of an A-positive (resp. A-negative) subspace.*

The above theorem says that if r_+ is the number of positive diagonal entries of D, then there exists an A-positive subspace E of dimension r_+, but it is impossible to find a positive subspace E with $\dim E > r_+$.

We will need the following lemma, which can be considered a particular case of the above theorem.

Lemma 3.2. *Let D be a diagonal matrix $D = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$. Then the number of positive (resp. negative) diagonal entries of D coincides with the maximal dimension of a D-positive (resp. D-negative) subspace.*

Proof. By rearranging the standard basis in $\mathbb{F}^n$ (changing the numeration) we can always assume without loss of generality that the positive diagonal entries of D are the first r_+ diagonal entries.

Consider the subspace E_+ spanned by the first r_+ coordinate vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{r_+}$. Clearly E_+ is a D-positive subspace, and $\dim E_+ = r_+$.

Let us now show that for any other D-positive subspace E we have $\dim E \leq r_+$. Consider the orthogonal projection $P = P_{E_+}$,

$$P\mathbf{x} = (x_1, x_2, \dots, x_{r_+}, 0 \dots, 0)^T, \qquad \mathbf{x} = (x_1, x_2, \dots, x_n)^T.$$

For a D-positive subspace E define an operator $T : E \to E_+$ by

$$T\mathbf{x} = P\mathbf{x}, \qquad \forall \mathbf{x} \in E.$$

In other words, T is the *restriction* of the projection P: P is defined on the whole space, but we restricted its domain to E and target space to E_+. We got an operator acting from E to E_+, and we use a different letter to distinguish it from P.

Note, that $\operatorname{Ker} T = \{\mathbf{0}\}$. Indeed, let for $\mathbf{x} = (x_1, x_2, \dots, x_n)^T \in E$ we have $T\mathbf{x} = P\mathbf{x} = \mathbf{0}$. Then, by the definition of P

$$x_1 = x_2 = \ldots = x_{r_+} = 0,$$

and therefore

$$(D\mathbf{x},\mathbf{x}) = \sum_{k=r_++1}^{n} \lambda_k x_k^2 \le 0 \qquad (\lambda_k \le 0 \text{ for } k > r_+).$$

But $\mathbf{x}$ belongs to a D-positive subspace E, so the inequality $(D\mathbf{x},\mathbf{x}) \le 0$ holds only for $\mathbf{x} = \mathbf{0}$.

Let us now apply the Rank Theorem (Theorem 7.1 from Chapter 2). First of all, $\operatorname{rank} T = \dim \operatorname{Ran} T \le \dim E_+ = r_+$ because $\operatorname{Ran} T \subset E_+$. By the Rank Theorem, $\dim \operatorname{Ker} T + \operatorname{rank} T = \dim E$. But we just proved that $\operatorname{Ker} T = \{\mathbf{0}\}$, i.e. that $\dim \operatorname{Ker} T = 0$, so

$$\dim E = \operatorname{rank} T \le \dim E_+ = r_+.$$

To prove the statement about negative entries, we just apply the above reasoning to the matrix $-D$. □

Proof of Theorem 3.1. Let $D = S^*AS$ be a diagonalization of A. Since

$$(D\mathbf{x},\mathbf{x}) = (S^*AS\mathbf{x},\mathbf{x}) = (AS\mathbf{x},S\mathbf{x})$$

it follows that for any D-positive subspace E, the subspace SE is an A-positive subspace. The same identity implies that for any A-positive subspace F the subspace $S^{-1}F$ is D-positive.

Since S and S^{-1} are invertible transformations, $\dim E = \dim SE$ and $\dim F = \dim S^{-1}F$. Therefore, for any D positive subspace E we can find an A-positive subspace (namely SE) of the same dimension, and vice versa: for any A-positive subspace F we can find a D-positive subspace (namely $S^{-1}F$) of the same dimension. Therefore the maximal possible dimensions of a A-positive and a D-positive subspace coincide, and the theorem is proved.

The case of negative diagonal entries treated similarly, we leave the details as an exercise to the reader. □

4. Positive definite forms. Minimax characterization of eigenvalues and the Sylvester's criterion of positivity

Definition. A quadratic form Q is called

- Positive definite if $Q[\mathbf{x}] > 0$ for all $\mathbf{x} \ne \mathbf{0}$.
- Positive semidefinite if $Q[\mathbf{x}] \ge 0$ for all $\mathbf{x}$.
- Negative definite if $Q[\mathbf{x}] < 0$ for all $\mathbf{x} \ne \mathbf{0}$.
- Negative semidefinite if $Q[\mathbf{x}] \le 0$ for all $\mathbf{x}$.
- Indefinite if it take both positive and negative values, i.e. if there exist vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ such that $Q[\mathbf{x}_1] > 0$ and $Q[\mathbf{x}_2] < 0$.

Definition. A Hermitian matrix $A = A^*$ is called positive definite (negative definite, etc...) if the corresponding quadratic form $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$ is positive definite (negative definite, etc...).

Theorem 4.1. *Let $A = A^*$. Then*

1. *A is positive definite iff all eigenvalues of A are positive.*
2. *A is positive semidefinite iff all eigenvalues of A are non-negative.*
3. *A is negative definite iff all eigenvalues of A are negative.*
4. *A is negative semidefinite iff all eigenvalues of A are non-positive.*
5. *A is indefinite iff it has both positive and negative eigenvalues.*

Proof. The proof follows trivially from the orthogonal diagonalization. Indeed, there is an orthonormal basis in which matrix of A is diagonal, and for diagonal matrices the theorem is trivial. □

Remark. Note, that to find whether a matrix (a quadratic form) is positive definite (negative definite, etc) one does not have to compute eigenvalues. By Sylvester's Law of Inertia it is sufficient to perform an arbitrary, not necessarily orthogonal diagonalization $D = S^*AS$ and look at the diagonal entries of D.

4.1. Sylvester's criterion of positivity. It is an easy exercise to see that a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ \overline{b} & c \end{pmatrix}$$

is positive definite if and only if

$$a > 0 \qquad \text{and } \det A = ac - |b|^2 > 0 \tag{4.1}$$

Indeed, if $a > 0$ and $\det A = ac - |b|^2 > 0$, then $c > 0$, so $\operatorname{trace} A = a + c > 0$. So we know that if λ_1, λ_2 are eigenvalues of A then $\lambda_1\lambda_2 > 0$ ($\det A > 0$) and $\lambda_1 + \lambda_2 = \operatorname{trace} A > 0$. But that only possible if both eigenvalues are positive. So we have proved that conditions (4.1) imply that A is positive definite. The opposite implication is quite simple, we leave it as an exercise for the reader.

This result can be generalized to the case of $n \times n$ matrices. Namely, for a matrix A

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,n} \end{pmatrix}$$

let us consider its all *upper left submatrices*

$$A_1 = (a_{1,1}),\ A_2 = \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix},\ A_3 = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix}, \dots, A_n = A$$

Theorem 4.2 (Sylvester's Criterion of Positivity)**.** *A matrix $A = A^*$ is positive definite if and only if*

$$\det A_k > 0 \qquad \text{for all } k = 1, 2, \dots, n.$$

First of all let us notice that if $A > 0$ then $A_k > 0$ also (can you explain why?). Therefore, since all eigenvalues of a positive definite matrix are positive, see Theorem 4.1, $\det A_k > 0$ for all k.

One can show that if $\det A_k > 0\ \forall k$ then all eigenvalues of A are positive by analyzing diagonalization of a quadratic form using row and column operations, which was described in Section 2.2. The key here is the observation that if we perform row/column operations in natural order (i.e. first subtracting the first row/column from all other rows/columns, then subtracting the second row/columns from the rows/columns $3, 4, \dots, n$, and so on...), and if we are not doing any row interchanges, then we automatically diagonalize quadratic forms A_k as well. Namely, after we subtract first and second rows and columns, we get diagonalization of A_2; after we subtract the third row/column we get the diagonalization of A_3, and so on.

Since we are performing only row replacement we do not change the determinant. Moreover, since we are not performing row exchanges and performing the operations in the correct order, we preserve determinants of A_k. Therefore, the condition $\det A_k > 0$ guarantees that each new entry in the diagonal is positive.

Of course, one has to be sure that we can use only row replacements, and perform the operations in the correct order, i.e. that we do not encounter any pathological situation. If one analyzes the algorithm, one can see that the only bad situation that can happen is the situation where at some step we have zero in the pivot place. In other words, if after we subtracted the first k rows and columns and obtained a diagonalization of A_k, the entry in the $k+1$st row and $k+1$st column is 0. We leave it as an exercise for the reader to show that this is impossible. □

The proof we outlined above is quite simple. However, let us present, in more detail, another one, which can be found in more advanced textbooks. I personally prefer this second proof, for it demonstrates some important connections.

We will need the following characterization of eigenvalues of a hermitian matrix.

4.2. Minimax characterization of eigenvalues. Let us recall that the *codimension* of a subspace $E \subset X$ is by the definition the dimension of its orthogonal complement, $\operatorname{codim} E = \dim(E^\perp)$. Since for a subspace $E \subset X$, $\dim X = n$ we have $\dim E + \dim E^\perp = n$, we can see that $\operatorname{codim} E = \dim X - \dim E$.

Recall that the trivial subspace $\{\mathbf{0}\}$ has dimension zero, so the whole space X has codimension 0.

Theorem 4.3 (Minimax characterization of eigenvalues)**.** *Let $A = A^*$ be an $n \times n$ matrix, and let $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ be its eigenvalues taken in the decreasing order. Then*

$$\lambda_k = \max_{\substack{E:\\ \dim E = k}} \min_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \min_{\substack{F:\\ \operatorname{codim} F = k-1}} \max_{\substack{\mathbf{x}\in F\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}).$$

Let us explain in more details what the expressions like $\max\min$ and $\min\max$ mean. To compute the first one, we need to consider all subspaces E of dimension k. For each such subspace E we consider the set of all $\mathbf{x} \in E$ of norm 1, and find the minimum of $(A\mathbf{x}, \mathbf{x})$ over all such $\mathbf{x}$. Thus for each subspace we obtain a number, and we need to pick a subspace E such that the number is maximal. That is the $\max\min$.

The $\min\max$ is defined similarly.

Remark. A sophisticated reader may notice a problem here: why do the maxima and minima exist? It is well known, that maximum and minimum have a nasty habit of not existing: for example, the function $f(x) = x$ has neither maximum nor minimum on the open interval $(0, 1)$.

However, in this case maximum and minimum do exist. There are two possible explanations of the fact that $(A\mathbf{x}, \mathbf{x})$ attains maximum and minimum. The first one requires some familiarity with basic notions of analysis: one should just say that the unit sphere in E, i.e. the set $\{\mathbf{x} \in E : \|\mathbf{x}\| = 1\}$ is compact, and that a continuous function ($Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$ in our case) on a compact set attains its maximum and minimum.

Another explanation will be to notice that the function $Q[\mathbf{x}] = (A\mathbf{x}, \mathbf{x})$, $\mathbf{x} \in E$ is a quadratic form on E. It is not difficult to compute the matrix of this form in some orthonormal basis in E, but let us only note that this matrix is not A: it has to be a $k \times k$ matrix, where $k = \dim E$.

It is easy to see that for a quadratic form the maximum and minimum over a unit sphere is the maximal and minimal eigenvalues of its matrix.

As for optimizing over all subspaces, we will prove below that the maximum and minimum do exist.

Proof of Theorem 4.3. First of all, by picking an appropriate orthonormal basis, we can assume without loss of generality that the matrix A is diagonal, $A = \operatorname{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

Pick subspaces E and F, $\dim E = k$, $\operatorname{codim} F = k-1$, i.e. $\dim F = n-k+1$. Since $\dim E + \dim F > n$, there exists a non-zero vector $\mathbf{x}_0 \in E \cap F$. By normalizing it we can assume without loss of generality that $\|\mathbf{x}_0\| = 1$. We can always arrange the eigenvalues in decreasing order, so let us assume that $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$.

Since $\mathbf{x}$ belongs to the both subspaces E and F

$$\min_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) \leq (A\mathbf{x}_0, \mathbf{x}_0) \leq \max_{\substack{\mathbf{x}\in F\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}).$$

We did not assume anything except dimensions about the subspaces E and F, so the above inequality

$$\min_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) \leq \max_{\substack{\mathbf{x}\in F\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}). \tag{4.2}$$

holds for all pairs of E and F of appropriate dimensions.

Define

$$E_0 := \operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k\}, \qquad F_0 := \operatorname{span}\{\mathbf{e}_k, \mathbf{e}_{k+1}, \mathbf{e}_{k+2}, \dots, \mathbf{e}_n\}.$$

Since for a self-adjoint matrix B, the maximum and minimum of $(B\mathbf{x}, \mathbf{x})$ over the unit sphere $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$ are the maximal and the minimal eigenvalue respectively (easy to check on diagonal matrices), we get that

$$\min_{\substack{\mathbf{x}\in E_0\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \max_{\substack{\mathbf{x}\in F_0\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \lambda_k.$$

It follows from (4.2) that for any subspace E, $\dim E = k$

$$\min_{\substack{\mathbf{x}\in E\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) \leq \max_{\substack{\mathbf{x}\in F_0\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \lambda_k$$

and similarly, for any subspace F of codimension $k-1$,

$$\max_{\substack{\mathbf{x}\in F\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) \geq \min_{\substack{\mathbf{x}\in E_0\\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \lambda_k.$$

But on subspaces E_0 and F_0 both maximum and minimum are λ_k, so $\min\max = \max\min = \lambda_k$. □

Corollary 4.4 (Intertwining of eigenvalues)**.** *Let $A = A^* = \{a_{j,k}\}_{j,k=1}^n$ be a self-adjoint matrix, and let $\widetilde{A} = \{a_{j,k}\}_{j,k=1}^{n-1}$ be its submatrix of size $(n-1)\times(n-1)$. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ and $\mu_1, \mu_2, \dots, \mu_{n-1}$ be the eigenvalues of A and $\widetilde{A}$ respectively, taken in decreasing order. Then*

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \ldots \geq \lambda_{n-1} \geq \mu_{n-1} \geq \lambda_n.$$

i.e.

$$\lambda_k \geq \mu_k \geq \lambda_{k+1}, \qquad k = 1, 2, \dots, n-1$$

Proof. Let $\widetilde{X} \subset \mathbb{F}^n$ be the subspace spanned by the first $n-1$ basis vectors, $\widetilde{X} = \operatorname{span}\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{n-1}\}$. Since $(\widetilde{A}\mathbf{x}, \mathbf{x}) = (A\mathbf{x}, \mathbf{x})$ for all $\mathbf{x} \in \widetilde{X}$, Theorem 4.3 implies that

$$\mu_k = \max_{\substack{E \subset \widetilde{X} \\ \dim E = k}} \min_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}).$$

To get λ_k we need to get maximum over the set of all subspaces E of $\mathbb{F}^n$, $\dim E = k$, i.e. take maximum over a bigger set (any subspace of $\widetilde{X}$ is a subspace of $\mathbb{F}^n$). Therefore

$$\mu_k \leq \lambda_k.$$

(the maximum can only increase, if we increase the set).

On the other hand, any subspace $E \subset \widetilde{X}$ of codimension $k-1$ (here we mean codimension in $\widetilde{X}$) has dimension $n-1-(k-1) = n-k$, so its codimension in $\mathbb{F}^n$ is k. Therefore

$$\mu_k = \min_{\substack{E \subset \widetilde{X} \\ \dim E = n-k}} \max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) \geq \min_{\substack{E \subset \mathbb{F}^n \\ \dim E = n-k}} \max_{\substack{\mathbf{x} \in E \\ \|\mathbf{x}\|=1}} (A\mathbf{x}, \mathbf{x}) = \lambda_{k+1}$$

(minimum over a bigger set can only be smaller). □

Proof of Theorem 4.2. If $A > 0$, then $A_k > 0$ for $k = 1, 2, \dots, n$ as well (can you explain why?). Since all eigenvalues of a positive definite matrix are positive (see Theorem 4.1), $\det A_k > 0$ for all $k = 1, 2, \dots, n$.

Let us now prove the other implication. Let $\det A_k > 0$ for all k. We will show, using induction in k, that all A_k (and so $A = A_n$) are positive definite.

Clearly A_1 is positive definite (it is 1×1 matrix, so $A_1 = \det A_1$). Assuming that $A_{k-1} > 0$ (and $\det A_k > 0$) let us show that A_k is positive definite. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ and $\mu_1, \mu_2, \dots, \mu_{k-1}$ be eigenvalues of A_k and A_{k-1} respectively. By Corollary 4.4

$$\lambda_j \geq \mu_j > 0 \qquad \text{for } j = 1, 2, \dots, k-1.$$

Since $\det A_k = \lambda_1 \lambda_2 \dots \lambda_{k-1} \lambda_k > 0$, the last eigenvalue λ_k must also be positive. Therefore, since all its eigenvalues are positive, the matrix A_k is positive definite. □

4.3. Some remarks. First of all notice, that Sylvester Criterion of Positivity does not generalize to positive semidefinite matrices if $n \geq 3$, meaning that for $n \times n$ matrices, $n \geq 3$, the conditions $\det A_k \geq 0$ do not imply that A is positive semidefinite, see Problem 4.6 below.

REFERENCE - DETERMINANTS

Treil Chapter 3, Sections 1-6

Chapter 3

Determinants

1. Introduction.

The reader probably already met determinants in calculus or algebra, at least the determinants of 2×2 and 3×3 matrices. For a 2×2 matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

the determinant is simply $ad - bc$; the determinant of a 3×3 matrix can be found by the "Star of David" rule.

In this chapter we would like to introduce determinants for $n \times n$ matrices. I don't want just to give a formal definition. First I want to give some motivation, and then derive some properties the determinant should have. Then if we want to have these properties, we do not have any choice, and arrive to several equivalent definitions of the determinant.

It is more convenient to start not with the determinant of a matrix, but with determinant of a system of vectors. There is no real difference here, since we always can join vectors together (say as columns) to form a matrix.

Let us have n vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in $\mathbb{R}^n$ (notice that the number of vectors coincides with dimension), and we want to find the n-dimensional volume of the parallelepiped determined by these vectors.

The parallelepiped determined by the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ can be defined as the collection of all vectors $\mathbf{v} \in \mathbb{R}^n$ that can be represented as

$$\mathbf{v} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \ldots + t_n\mathbf{v}_n, \qquad 0 \le t_k \le 1 \quad \forall k = 1, 2, \ldots, n.$$

It can be easily visualized when $n = 2$ (parallelogram) and $n = 3$ (parallelepiped). So, what is the n-dimensional volume?

If $n = 2$ it is area; if $n = 3$ it is indeed the volume. In dimension 1 is it just the length.

Finally, let us introduce some notation. For a system of vectors (columns) $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ we will denote its determinant (that we are going to construct) as $D(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$. If we join these vectors in a matrix A (column number k of A is $\mathbf{a}_k$), then we will use the notation $\det A$,

$$\det A = D(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$$

Also, for a matrix

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{n,1} & a_{n,2} & \ldots & a_{n,n} \end{pmatrix}$$

its determinant is often is denoted by

$$\begin{vmatrix} a_{1,1} & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1} & a_{2,2} & \ldots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{n,1} & a_{n,2} & \ldots & a_{n,n} \end{vmatrix}.$$

2. What properties determinant should have.

We know, that for dimensions 2 and 3 "volume" of a parallelepiped is determined by the *base times height* rule: if we pick one vector, then height is the distance from this vector to the subspace spanned by the remaining vectors, and the base is the $(n-1)$-dimensional volume of the parallelepiped determined by the remaining vectors.

Now let us generalize this idea to higher dimensions. For a moment we do not care about how exactly to determine height and base. We will show, that if we assume that the base and the height satisfy some natural properties, then we do not have any choice, and the volume (determinant) is uniquely defined.

2.1. Linearity in each argument. First of all, if we multiply vector $\mathbf{v}_1$ by a positive number a, then the height (i.e. the distance to the linear span $\mathcal{L}(\mathbf{v}_2, \ldots, \mathbf{v}_n)$) is multiplied by a. If we admit negative heights (and negative volumes), then this property holds for all scalars a, and so the determinant $D(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$ of the system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ should satisfy

$$D(\alpha\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n) = \alpha D(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n).$$

Of course, there is nothing special about vector $\mathbf{v}_1$, so for any index k

$$D(\mathbf{v}_1, \dots, \underset{k}{\alpha\mathbf{v}_k}, \dots, \mathbf{v}_n) = \alpha D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) \tag{2.1}$$

To get the next property, let us notice that if we add 2 vectors, then the "height" of the result should be equal the sum of the "heights" of summands, i.e. that

$$D(\mathbf{v}_1, \dots, \underbrace{\mathbf{u}_k + \mathbf{v}_k}_{k}, \dots, \mathbf{v}_n) = D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{u}_k}, \dots, \mathbf{v}_n) + D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) \tag{2.2}$$

In other words, the above two properties say that the determinant of n vectors is *linear in each argument (vector)*, meaning that if we fix $n-1$ vectors and interpret the remaining vector as a variable (argument), we get a linear function.

Remark. We already know that *linearity* is a very nice property, that helps in many situations. So, admitting negative heights (and therefore negative volumes) is a very small price to pay to get linearity, since we can always put on the absolute value afterwards.

In fact, by admitting negative heights, we did not sacrifice anything! To the contrary, we even gained something, because the sign of the determinant contains some information about the system of vectors (orientation).

2.2. Determinant is 0 if two vectors coincide. If $\mathbf{v}_j = \mathbf{v}_k$ for some $j \neq k$, then the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent, so

$$\dim \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} < n.$$

But then the volume must be 0 (for example, because the "height", i.e. the distance from v_j to $\operatorname{span}\{\mathbf{v}_r : r \neq j\}$ is 0). Thus, it is natural to assume that

$$D(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = 0 \qquad \text{if } \mathbf{v}_j = \mathbf{v}_k \quad \text{for some } j \neq k. \tag{2.3}$$

In what follows that will be the property of the determinant we will use.

Remark. A stronger property that for a linearly dependent system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ there holds $D(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = 0$ is also quite natural.

However, this property easily follows from a weaker assumption (2.3) and the linearity (2.1), (2.2), see Proposition 3.1 for details.

2.3. Preservation under "column replacement". The next property also seems natural. Namely, if we take a vector, say $\mathbf{v}_j$, and add to it a multiple of another vector $\mathbf{v}_k$, the "height" does not change, so

$$D(\mathbf{v}_1, \dots, \underbrace{\mathbf{v}_j + \alpha\mathbf{v}_k}_{j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) = D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n). \tag{2.4}$$

In other words, if we apply the *column operation* of the third type, the determinant does not change.

However, this property is not an independent one, it is an easy corollary of (2.3) and the linearity (2.1), (2.2). Namely

$$\begin{aligned} D(\mathbf{v}_1, \dots, \underbrace{\mathbf{v}_j + \alpha\mathbf{v}_k}_{j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) \\ = D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) + \alpha D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_k}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) \\ = D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n); \end{aligned}$$

the first equality here follows from the linearity (2.1), (2.2), and the second one is just (2.3).

Functions of several variables that change sign when one interchanges any two arguments are called *antisymmetric*.

2.4. Antisymmetry. The next property the determinant should have, is that if we interchange 2 vectors, the determinant changes sign:

$$D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_k}, \dots, \underset{k}{\mathbf{v}_j}, \dots, \mathbf{v}_n) = -D(\mathbf{v}_1, \dots, \underset{j}{\mathbf{v}_j}, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n). \tag{2.5}$$

At first sight this property does not look natural, but it can be deduced from the previous ones. Namely, applying property (2.4) three times, and

then using (2.1) we get

$$\begin{aligned}
&D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_j},\dots,\underset{k}{\mathbf{v}_k},\dots,\mathbf{v}_n) = \\
&\quad = D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_j},\dots,\underbrace{\mathbf{v}_k-\mathbf{v}_j}_{k},\dots,\mathbf{v}_n) \\
&\quad = D(\mathbf{v}_1,\dots,\underbrace{\mathbf{v}_j+(\mathbf{v}_k-\mathbf{v}_j)}_{j},\dots,\underbrace{\mathbf{v}_k-\mathbf{v}_j}_{k},\dots,\mathbf{v}_n) \\
&\quad = D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_k},\dots,\underbrace{\mathbf{v}_k-\mathbf{v}_j}_{k},\dots,\mathbf{v}_n) \\
&\quad = D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_k},\dots,\underbrace{(\mathbf{v}_k-\mathbf{v}_j)-\mathbf{v}_k}_{k},\dots,\mathbf{v}_n) \\
&\quad = D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_k},\dots,\underset{k}{-\mathbf{v}_j},\dots,\mathbf{v}_n) \\
&\quad = -D(\mathbf{v}_1,\dots,\underset{j}{\mathbf{v}_k},\dots,\underset{k}{\mathbf{v}_j},\dots,\mathbf{v}_n).
\end{aligned}$$

Recall, that the property (2.4) follows from (2.3) and the linearity (2.1), (2.2), so the antisymmetry (2.5) also follows from these properties.

2.5. Normalization. The last property is the easiest one. For the standard basis $\mathbf{e}_1,\mathbf{e}_2,\dots,\mathbf{e}_n$ in $\mathbb{R}^n$ the corresponding parallelepiped is the n-dimensional unit cube, so

$$D(\mathbf{e}_1,\mathbf{e}_2,\dots,\mathbf{e}_n)=1. \tag{2.6}$$

In matrix notation this can be written as

$$\det(I)=1$$

3. Constructing the determinant.

The plan of the game is now as follows: using the properties that as we decided in Section 2 the determinant should have, we derive other properties of the determinant, some of them highly non-trivial. We will show how to use these properties to compute the determinant using our old friend—row reduction.

Later, in Section 4, we will show that the determinant, i.e. a function with the desired properties exists and unique. After all we have to be sure that the object we are computing and studying exists.

While our initial geometric motivation for determinant and its properties came from considering vectors in the real vector space $\mathbb{R}^n$, so they relate only to matrices with real entries, all the constructions below use only algebraic operations (addition, multiplication, division) and are applicable to matrices with complex entries, and even with entries in an arbitrary field.

So in what follows we are constructing determinant not just for real matrices, but for complex matrices as well (and also for matrices with entries in an arbitrary field). The nice geometric motivation for the properties works only in the real case, but after we decided on the properties of the determinant (see properties 1–3 below) everything works in the general case.

3.1. Basic properties. We will use the following basic properties of the determinant:

1. Determinant is linear in each column, i.e. in vector notation for every index k

$$D(\mathbf{v}_1, \dots, \underbrace{\alpha\mathbf{u}_k + \beta\mathbf{v}_k}_{k}, \dots, \mathbf{v}_n) = \\ \alpha D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{u}_k}, \dots, \mathbf{v}_n) + \beta D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n)$$

   for all scalars α, β.
2. Determinant is *antisymmetric*, i.e. if one interchanges two columns, the determinant changes sign.
3. Normalization property: $\det I = 1$.

All these properties were discussed above in Section 2. The first property is just the (2.1) and (2.2) combined. The second one is (2.5), and the last one is the normalization property (2.6). Note, that we did not use properties (2.3) and (2.4): they can be deduced from the above three. These three properties completely define determinant!

3.2. Properties of determinant deduced from the basic properties. In what follows, $\det A = D(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ is a function satisfying properties 1–3 in Section 3.1 above.

Proposition 3.1. *For a square matrix A the following statements hold:*

1. *If A has a zero column, then* $\det A = 0$.
2. *If A has two equal columns, then* $\det A = 0$*;*
3. *If one column of A is a multiple of another, then* $\det A = 0$*;*
4. *If columns of A are linearly dependent, i.e. if the matrix is not invertible, then* $\det A = 0$.

Proof. Statement 1 follows immediately from linearity. If we multiply the zero column by zero, we do not change the matrix and its determinant. But by the property 1 above, we should get 0.

The fact that determinant is antisymmetric, implies statement 2. Indeed, if we interchange two equal columns, we change nothing, so the determinant remains the same. On the other hand, interchanging two columns changes sign of determinant, so

$$\det A = -\det A,$$

which is possible only if $\det A = 0$.

Statement 3 is immediate corollary of statement 2 and linearity.

To prove the last statement, let us first suppose that the first vector $\mathbf{v}_1$ is a linear combination of the other vectors,

$$\mathbf{v}_1 = \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 + \ldots + \alpha_n\mathbf{v}_n = \sum_{k=2}^{n} \alpha_k\mathbf{v}_k.$$

Then by linearity we have (in vector notation)

$$D(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n) = D\left(\big(\sum_{k=2}^{n} \alpha_k\mathbf{v}_k\big), \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_n \right) = \sum_{k=2}^{n} \alpha_k D(\mathbf{v}_k, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_n)$$

and each determinant in the sum is zero because of two equal columns.

Let us now consider general case, i.e. let us assume that the system $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is linearly dependent. Then one of the vectors, say $\mathbf{v}_k$ can be represented as a linear combination of the others. Interchanging this vector with $\mathbf{v}_1$ we arrive to the situation we just treated, so

$$D(\mathbf{v}_1, \ldots, \underset{k}{\mathbf{v}_k}, \ldots, \mathbf{v}_n) = -D(\mathbf{v}_k, \ldots, \underset{k}{\mathbf{v}_1}, \ldots, \mathbf{v}_n) = -0 = 0,$$

so the determinant in this case is also 0. □

The next proposition generalizes property (2.4). As we already have said above, this property can be deduced from the three "basic" properties of the determinant, we are using in this section.

Proposition 3.2. *The determinant does not change if we add to a column a linear combination of the other columns (leaving the other columns intact). In particular, the determinant is preserved under "column replacement" (column operation of third type).*

Note, that adding to a column a multiple of itself is prohibited here. We can only add multiples of the other columns.

Proof. Fix a vector $\mathbf{v}_k$, and let $\mathbf{u}$ be a linear combination of the other vectors,

$$\mathbf{u} = \sum_{j\neq k} \alpha_j\mathbf{v}_j.$$

Then by linearity

$$D(\mathbf{v}_1, \dots, \underbrace{\mathbf{v}_k + \mathbf{u}}_{k}, \dots, \mathbf{v}_n) = D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{v}_k}, \dots, \mathbf{v}_n) + D(\mathbf{v}_1, \dots, \underset{k}{\mathbf{u}}, \dots, \mathbf{v}_n),$$

and by Proposition 3.1 the last term is zero. □

3.3. Determinants of diagonal and triangular matrices. Now we are ready to compute determinant for some important special classes of matrices. The first class is the so-called *diagonal* matrices. Let us recall that a square matrix $A = \{a_{j,k}\}_{j,j=1}^{n}$ is called *diagonal* if all entries *off the main diagonal* are zero, i.e. if $a_{j,k} = 0$ for all $j \neq k$. We will often use the notation $\operatorname{diag}\{a_1, a_2, \dots, a_n\}$ for the diagonal matrix

$$\begin{pmatrix} a_1 & 0 & \dots & 0 \\ 0 & a_2 & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & a_n \end{pmatrix}.$$

Since a diagonal matrix $\operatorname{diag}\{a_1, a_2, \dots, a_n\}$ can be obtained from the identity matrix I by multiplying column number k by a_k,

Determinant of a diagonal matrix equal the product of the diagonal entries, $$\det(\operatorname{diag}\{a_1, a_2, \dots, a_n\}) = a_1 a_2 \dots a_n.$$

The next important class is the class of so-called *triangular* matrices. A square matrix $A = \{a_{j,k}\}_{j,j=1}^{n}$ is called *upper triangular* if all entries *below* the main diagonal are 0, i.e. if $a_{j,k} = 0$ for all $k < j$. A square matrix is called *lower triangular* if all entries *above* the main are 0, i.e if $a_{j,k} = 0$ for all $j < k$. We call a matrix *triangular*, if it is either lower or upper triangular matrix.

It is easy to see that

Determinant of a *triangular* matrix equals to the product of the diagonal entries, $$\det A = a_{1,1} a_{2,2} \dots a_{n,n}.$$

Indeed, if a triangular matrix has zero on the main diagonal, it is not invertible (this can easily be checked by column operations) and therefore both sides equal zero. If all diagonal entries are non-zero, then using column replacement (column operations of third type) one can transform the matrix into a diagonal one with the same diagonal entries: For upper triangular matrix one should first subtract appropriate multiples of the first column from the columns number $2, 3, \dots, n$, "killing" all entries in the first row,

then subtract appropriate multiples of the second column from columns number $3, \dots, n$, and so on.

To treat the case of lower triangular matrices one has to do "column reduction" from the left to the right, i.e. first subtract appropriate multiples of the last column from columns number $n-1, \dots, 2, 1$, and so on.

3.4. Computing the determinant. Now we know how to compute determinants, using their properties: one just needs to do column reduction (i.e. row reduction for A^T) keeping track of column operations changing the determinant. Fortunately, the most often used operation—row replacement, i.e. operation of third type does not change the determinant. So we only need to keep track of interchanging of columns and of multiplication of column by a scalar.

If an echelon form of A^T does not have pivots in every column (and row), then A is not invertible, so $\det A = 0$. If A is invertible, we arrive at a triangular matrix, and $\det A$ is the product of diagonal entries times the correction from column interchanges and multiplications.

The above algorithm implies that $\det A$ can be zero only if a matrix A is not invertible. Combining this with the last statement of Proposition 3.1 we get

Proposition 3.3. $\det A = 0$ *if and only if* A *is not invertible. An equivalent statement:* $\det A \neq 0$ *if and only if* A *is invertible.*

Note, that although we now know how to compute determinants, the determinant is still not defined. One can ask: why don't we define it as the result we get from the above algorithm? The problem is that formally this result is not well defined: that means we did not prove that different sequences of column operations yield the same answer.

3.5. Determinants of a transpose and of a product. Determinants of elementary matrices. In this section we prove two important theorems.

Theorem 3.4 (Determinant of a transpose)**.** *For a square matrix* A*,*

$$\det A = \det(A^T).$$

This theorem implies that for all statement about columns we discussed above, the corresponding statements about rows are also true. In particular, determinants behave under *row operations* the same way they behave under *column operations*. So, we can use row operations to compute determinants.

Theorem 3.5 (Determinant of a product)**.** *For* $n \times n$ *matrices* A *and* B

$$\det(AB) = (\det A)(\det B)$$

In other words

Determinant of a product equals product of determinants.

To prove both theorems we need the following lemma.

Lemma 3.6. *For a square matrix A and an elementary matrix E (of the same size)*

$$\det(AE) = (\det A)(\det E)$$

Proof. The proof can be done just by direct checking: determinants of special matrices are easy to compute; right multiplication by an elementary matrix is a column operation, and effect of column operations on the determinant is well known.

This can look like a lucky coincidence, that the determinants of elementary matrices agree with the corresponding column operations, but it is not a coincidence at all.

Namely, for a column operation the corresponding elementary matrix can be obtained from the identity matrix I by this column operation. So, its determinant is 1 (determinant of I) times the effect of the column operation.

And that is all! It may be hard to realize at first, but the above paragraph is a *complete and rigorous* proof of the lemma! □

Applying N times Lemma 3.6 we get the following corollary.

Corollary 3.7. *For any matrix A and any sequence of elementary matrices $E_1, E_2, \dots, E_N$ (all matrices are $n \times n$)*

$$\det(AE_1E_2 \dots E_N) = (\det A)(\det E_1)(\det E_2) \dots (\det E_N)$$

Lemma 3.8. *Any invertible matrix is a product of elementary matrices.*

Proof. We know that any invertible matrix is row equivalent to the identity matrix, which is its reduced echelon form. So

$$I = E_N E_{N-1} \dots E_2 E_1 A,$$

and therefore any invertible matrix can be represented as a product of elementary matrices,

$$A = E_1^{-1} E_2^{-1} \dots E_{N-1}^{-1} E_N^{-1} I = E_1^{-1} E_2^{-1} \dots E_{N-1}^{-1} E_N^{-1}$$

(the inverse of an elementary matrix is an elementary matrix). □

Proof of Theorem 3.4. First of all, it can be easily checked, that for an elementary matrix E we have $\det E = \det(E^T)$. Notice, that it is sufficient to prove the theorem only for invertible matrices A, since if A is not invertible then A^T is also not invertible, and both determinants are zero.

By Lemma 3.8 matrix A can be represented as a product of elementary matrices,

$$A = E_1 E_2 \dots E_N,$$

and by Corollary 3.7 the determinant of A is the product of determinants of the elementary matrices. Since taking the transpose just transposes each elementary matrix and reverses their order, Corollary 3.7 implies that $\det A = \det A^T$. □

Proof of Theorem 3.5. Let us first suppose that the matrix B is invertible. Then Lemma 3.8 implies that B can be represented as a product of elementary matrices

$$B = E_1 E_2 \dots E_N,$$

and so by Corollary 3.7

$$\det(AB) = (\det A)[(\det E_1)(\det E_2)\dots(\det E_N)] = (\det A)(\det B).$$

If B is not invertible, then the product AB is also not invertible, and the theorem just says that $0 = 0$.

To check that the product $AB = C$ is not invertible, let us assume that it is invertible. Then multiplying the identity $AB = C$ by C^{-1} from the left, we get $C^{-1}AB = I$, so $C^{-1}A$ is a left inverse of B. So B is left invertible, and since it is square, it is invertible. We got a contradiction. □

3.6. Summary of properties of determinant. First of all, let us say once more, that *the determinant is defined only for square matrices!* Since we now know that $\det A = \det(A^T)$, the statements that we knew about columns are true for rows too.

1. Determinant is linear in each row (column) when the other rows (columns) are fixed.
2. If one interchanges two rows (columns) of a matrix A, the determinant changes sign.
3. For a triangular (in particular, for a diagonal) matrix its determinant is the product of the diagonal entries. In particular, $\det I = 1$.
4. If a matrix A has a zero row (or column), $\det A = 0$.
5. If a matrix A has two equal rows (columns), $\det A = 0$.
6. If one of the rows (columns) of A is a linear combination of the other rows (columns), i.e. if the matrix is not invertible, then $\det A = 0$;
   More generally,
7. $\det A = 0$ if and only if A is not invertible, or equivalently
8. $\det A \neq 0$ if and only if A is invertible.

9. $\det A$ does not change if we add to a row (column) a linear combination of the other rows (columns). In particular, the determinant is preserved under the row (column) replacement, i.e. under the row (column) operation of the third kind.
10. $\det A^T = \det A$.
11. $\det(AB) = (\det A)(\det B)$.

And finally,

12. If A is an $n \times n$ matrix, then $\det(aA) = a^n \det A$.

The last property follows from the linearity of the determinant, if we recall that to multiply a matrix A by a we have to multiply each row by a, and that each multiplication multiplies the determinant by a.

Exercises.

3.1. If A is an $n \times n$ matrix, how are the determinants $\det A$ and $\det(5A)$ related?
Remark: $\det(5A) = 5 \det A$ only in the trivial case of 1×1 matrices

3.2. How are the determinants $\det A$ and $\det B$ related if

a)
$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2a_1 & 3a_2 & 5a_3 \\ 2b_1 & 3b_2 & 5b_3 \\ 2c_1 & 3c_2 & 5c_3 \end{pmatrix};$$

b)
$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3a_1 & 4a_2 + 5a_1 & 5a_3 \\ 3b_1 & 4b_2 + 5b_1 & 5b_3 \\ 3c_1 & 4c_2 + 5c_1 & 5c_3 \end{pmatrix}.$$

3.3. Using column or row operations compute the determinants

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix}, \qquad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}, \qquad \begin{vmatrix} 1 & 0 & -2 & 3 \\ -3 & 1 & 1 & 2 \\ 0 & 4 & -1 & 1 \\ 2 & 3 & 0 & 1 \end{vmatrix}, \qquad \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix}.$$

3.4. A square $(n \times n)$ matrix is called skew-symmetric (or antisymmetric) if $A^T = -A$. Prove that if A is skew-symmetric and n is odd, then $\det A = 0$. Is this true for even n?

3.5. A square matrix is called *nilpotent* if $A^k = \mathbf{0}$ for some positive integer k. Show that for a nilpotent matrix A $\det A = 0$.

3.6. Prove that if the matrices A and B are similar, than $\det A = \det B$.

3.7. A real square matrix Q is called orthogonal if $Q^T Q = I$. Prove that if Q is an orthogonal matrix then $\det Q = \pm 1$.

3.8. Show that
$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (z - x)(z - y)(y - x).$$

This is a particular case of the so-called Vandermonde determinant.

3.9. Let points A, B and C in the plane $\mathbb{R}^2$ have coordinates (x_1, y_1), (x_2, y_2) and (x_3, y_3) respectively. Show that the area of triangle ABC is the absolute value of

$$\frac{1}{2}\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

Hint: use row operations and geometric interpretation of 2×2 determinants (area).

3.10. Let A be a square matrix. Show that block triangular matrices

$$\begin{pmatrix} I & * \\ \mathbf{0} & A \end{pmatrix}, \qquad \begin{pmatrix} A & * \\ \mathbf{0} & I \end{pmatrix}, \qquad \begin{pmatrix} I & \mathbf{0} \\ * & A \end{pmatrix}, \qquad \begin{pmatrix} A & \mathbf{0} \\ * & I \end{pmatrix}$$

all have determinant equal to $\det A$. Here $*$ can be anything.

The following problems illustrate the power of block matrix notation.

3.11. Use the previous problem to show that if A and C are square matrices, then

$$\det \begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \det A \det C.$$

Hint: $\begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \begin{pmatrix} I & B \\ \mathbf{0} & C \end{pmatrix}\begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}$.

3.12. Let A be $m \times n$ and B be $n \times m$ matrices. Prove that

$$\det \begin{pmatrix} 0 & A \\ -B & I \end{pmatrix} = \det(AB).$$

Hint: While it is possible to transform the matrix by row operations to a form where the determinant is easy to compute, the easiest way is to right multiply the matrix by $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$.

4. Formal definition. Existence and uniqueness of the determinant.

In this section we arrive to the formal definition of the determinant. We show that a function, satisfying the basic properties 1, 2, 3 from Section 3 exists, and moreover, such function is unique, i.e. we do not have any choice in constructing the determinant.

Consider an $n \times n$ matrix $A = \{a_{j,k}\}_{j,k=1}^n$, and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be its columns, i.e.

$$\mathbf{v}_k = \begin{pmatrix} a_{1,k} \\ a_{2,k} \\ \vdots \\ a_{n,k} \end{pmatrix} = a_{1,k}\mathbf{e}_1 + a_{2,k}\mathbf{e}_2 + \ldots + a_{n,k}\mathbf{e}_n = \sum_{j=1}^{n} a_{j,k}\mathbf{e}_j.$$

Using linearity of the determinant we expand it in the first column $\mathbf{v}_1$:

$$D(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = D(\sum_{j=1}^{n} a_{j,1}\mathbf{e}_j, \mathbf{v}_2, \dots, \mathbf{v}_n) = \sum_{j=1}^{n} a_{j,1} D(\mathbf{e}_j, \mathbf{v}_2, \dots, \mathbf{v}_n). \tag{4.1}$$

Then we expand it in the second column, then in the third, and so on. We get

$$D(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = \sum_{j_1=1}^{n} \sum_{j_2=1}^{n} \dots \sum_{j_n=1}^{n} a_{j_1,1} a_{j_2,2} \dots a_{j_n,n} D(\mathbf{e}_{j_1}.\mathbf{e}_{j_2}, \dots \mathbf{e}_{j_n}).$$

Notice, that we have to use a different index of summation for each column: we call them $j_1, j_2, \dots, j_n$; the index j_1 here is the same as the index j in (4.1).

It is a huge sum, it contains n^n terms. Fortunately, some of the terms are zero. Namely, if any 2 of the indices $j_1, j_2, \dots, j_n$ coincide, the determinant $D(\mathbf{e}_{j_1}.\mathbf{e}_{j_2}, \dots \mathbf{e}_{j_n})$ is zero, because there are two equal columns here.

So, let us rewrite the sum, omitting all zero terms. The most convenient way to do that is using the notion of a *permutation*. Informally, a permutation of an ordered set $\{1, 2, \dots, n\}$ is a rearrangement of its elements. A convenient formal way to represent such a rearrangement is by using a function

$$\sigma : \{1, 2, \dots, n\} \to \{1, 2, \dots, n\},$$

where $\sigma(1), \sigma(2), \dots, \sigma(n)$ gives the new order of the set $1, 2, \dots, n$. In other words, the permutation σ rearranges the ordered set $1, 2, \dots, n$ into $\sigma(1), \sigma(2), \dots, \sigma(n)$.

Such function σ has to be one-to-one (different values for different arguments) and onto (assumes all possible values from the target space). The functions which are one-to-one and onto are called *bijections*, and they give one-to-one correspondence between the domain and the target space.[1]

Although it is not directly relevant here, let us notice, that it is well-known in combinatorics, that the number of different permutations of the set $\{1, 2, \dots, n\}$ is exactly $n!$. The set of all permutations of the set $\{1, 2, \dots, n\}$ will be denoted $\mathrm{Perm}(n)$.

[1] There is another canonical way to represent permutation by a bijection σ, namely in this representation $\sigma(k)$ gives new position of the element number k. In this representation σ rearranges $\sigma(1), \sigma(2), \dots, \sigma(n)$ into $1, 2, \dots, n$.

While in the first representation it is easy to write the function if you know the rearrangement of the set $1, 2, \dots, n$, the second one is more adapted to the composition of permutations: it coincides with the composition of functions. Namely if we first perform the permutation that correspond to a function σ and then one that correspond to τ, the resulting permutation will correspond to $\tau \circ \sigma$.

Using the notion of a permutation, we can rewrite the determinant as

$$D(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = \sum_{\sigma \in \mathrm{Perm}(n)} a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} D(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}).$$

The matrix with columns $\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}$ can be obtained from the identity matrix by finitely many column interchanges, so the determinant

$$D(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)})$$

is 1 or -1 depending on the number of column interchanges.

To formalize that, we (informally) define the *sign* (denoted $\operatorname{sign}\sigma$) of a permutation σ to be 1 if an even number of interchanges is necessary to rearrange the n-tuple $1, 2, \dots, n$ into $\sigma(1), \sigma(2), \dots, \sigma(n)$, and $\operatorname{sign}(\sigma) = -1$ if the number of interchanges is odd.

It is a well-known fact from the combinatorics, that the sign of permutation is well defined, i.e. that although there are infinitely many ways to get the n-tuple $\sigma(1), \sigma(2), \dots, \sigma(n)$ from $1, 2, \dots, n$, the number of interchanges is either always odd or always even.

One of the ways to show that is to introduce an alternative definition. Let $K = K(\sigma)$ be the number of *disorders* of σ, i.e. the number of pairs (j, k), $j, k \in \{1, 2, \dots, n\}$, $j < k$ such that $\sigma(j) > \sigma(k)$, and see if the number is even or odd. We call the permutation σ *odd* if K is odd and *even* if K is even. Then define $\operatorname{sign}\sigma := (-1)^{K(\sigma)}$; note that this way $\operatorname{sign}\sigma$ is well defined.

We want to show that $\operatorname{sign}\sigma = (-1)^{K(\sigma)}$ can indeed be computed by rearranging the n-tuple $1, 2, \dots, n$ into $\sigma(1), \sigma(2), \dots, \sigma(n)$ and counting the number of interchanges, as was described above.

If $\sigma(k) = k\ \forall k$, then the number of *disorders* $K(\sigma)$ is 0, so *sign* of such *identity* permutation is 1. Note also, that any elementary transpose, which interchange two neighbors, changes the sign of a permutation, because it changes (increases or decreases) the number of *disorders* exactly by 1. So, to get from a permutation to another one always needs an even number of elementary transposes if the permutations have the same sign, and an odd number if the signs are different.

Finally, any interchange of two entries can be achieved by an odd number of elementary transposes. This implies that *sign* changes under an interchange of two entries. So, to get from $1, 2, \dots, n$ to an even permutation (positive sign) one always need even number of interchanges, and odd number of interchanges is needed to get an odd permutation (negative sign).

So, if we want determinant to satisfy basic properties 1–3 from Section 3, we must define it as

$$\det A = \sum_{\sigma \in \mathrm{Perm}(n)} a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} \operatorname{sign}(\sigma), \tag{4.2}$$

where the sum is taken over all permutations of the set $\{1, 2, \dots, n\}$.

If we define the determinant this way, it is easy to check that it satisfies the basic properties 1–3 from Section 3. Indeed, it is linear in each column, because for each column every term (product) in the sum contains exactly one entry from this column.

Interchanging two columns of A just adds an extra interchange to the permutation, so right side in (4.2) changes sign. Finally, for the identity matrix I, the right side of (4.2) is 1 (it has one non-zero term).

Exercises.

4.1. Suppose the permutation σ takes $(1, 2, 3, 4, 5)$ to $(5, 4, 1, 2, 3)$.

a) Find sign of σ;

b) What does $\sigma^2 := \sigma \circ \sigma$ do to $(1, 2, 3, 4, 5)$?

c) What does the inverse permutation σ^{-1} do to $(1, 2, 3, 4, 5)$?

d) What is the sign of σ^{-1}?

4.2. Let P be a *permutation matrix*, i.e. an $n \times n$ matrix consisting of zeroes and ones and such that there is exactly one 1 in every row and every column.

a) Can you describe the corresponding linear transformation? That will explain the name.

b) Show that P is invertible. Can you describe P^{-1}?

c) Show that for some $N > 0$

$$P^N := \underbrace{PP \dots P}_{N \text{ times}} = I.$$

Use the fact that there are only finitely many permutations.

4.3. Why is there an even number of permutations of $(1, 2, \dots, 9)$ and why are exactly half of them odd permutations? **Hint:** This problem can be hard to solve in terms of permutations, but there is a very simple solution using determinants.

4.4. If σ is an odd permutation, explain why σ^2 is even but σ^{-1} is odd.

4.5. How many multiplications and additions is required to compute the determinant using formal definition (4.2) of the determinant of an $n \times n$ matrix? Do not count the operations needed to compute $\operatorname{sign} \sigma$.

5. Cofactor expansion.

For an $n \times n$ matrix $A = \{a_{j,k}\}_{j,k=1}^{n}$ let $A_{j,k}$ denotes the $(n-1) \times (n-1)$ matrix obtained from A by crossing out row number j and column number k.

Theorem 5.1 (Cofactor expansion of determinant)**.** *Let A be an $n \times n$ matrix. For each j, $1 \le j \le n$, determinant of A can be expanded in the row number j as*

$$\det A = a_{j,1}(-1)^{j+1} \det A_{j,1} + a_{j,2}(-1)^{j+2} \det A_{j,2} + \ldots + a_{j,n}(-1)^{j+n} \det A_{j,n} = \sum_{k=1}^{n} a_{j,k}(-1)^{j+k} \det A_{j,k}.$$

Similarly, for each k, $1 \le k \le n$, the determinant can be expanded in the column number k,

$$\det A = \sum_{j=1}^{n} a_{j,k}(-1)^{j+k} \det A_{j,k}.$$

Proof. Let us first prove the formula for the expansion in row number 1. The formula for expansion in row number 2 then can be obtained from it by interchanging rows number 1 and 2. Interchanging then rows number 2 and 3 we get the formula for the expansion in row number 3, and so on.

Since $\det A = \det A^T$, column expansion follows automatically.

Let us first consider a special case, when the first row has one non-zero term $a_{1,1}$. Performing column operations on columns $2, 3, \ldots, n$ we transform A to the lower triangular form. The determinant of A then can be computed as

$$\boxed{\begin{array}{l}\text{the product of diagonal}\\ \text{entries of the triangular}\\ \text{matrix}\end{array}} \times \boxed{\begin{array}{l}\text{correcting factor from}\\ \text{the column operations}\end{array}}.$$

But the product of all diagonal entries except the first one (i.e. without $a_{1,1}$) times the correcting factor is exactly $\det A_{1,1}$, so in this particular case $\det A = a_{1,1} \det A_{1,1}$.

Let us now consider the case when all entries in the first row except $a_{1,2}$ are zeroes. This case can be reduced to the previous one by interchanging columns number 1 and 2, and therefore in this case $\det A = (-1)a_{1,2} \det A_{1,2}$.

The case when $a_{1,3}$ is the only non-zero entry in the first row, can be reduced to the previous one by interchanging rows 2 and 3, so in this case $\det A = a_{1,3} \det A_{1,3}$.

Repeating this procedure we get that in the case when $a_{1,k}$ is the only non-zero entry in the first row $\det A = (-1)^{1+k} a_{1,k} \det A_{1,k}$. [2]

In the general case, linearity of the determinant in each row implies that

$$\det A = \det A^{(1)} + \det A^{(2)} + \ldots + \det A^{(n)} = \sum_{k=1}^{n} \det A^{(k)}$$

where the matrix $A^{(k)}$ is obtained from A by replacing all entries in the first row except $a_{1,k}$ by 0. As we just discussed above

$$\det A^{(k)} = (-1)^{1+k} a_{1,k} \det A_{1,k},$$

so

$$\det A = \sum_{k=1}^{n} (-1)^{1+k} a_{1,k} \det A_{1,k}.$$

To get the cofactor expansion in the second row, we can interchange the first and second rows and apply the above formula. The row exchange changes the sign, so we get

$$\det A = -\sum_{k=1}^{n} (-1)^{1+k} a_{2,k} \det A_{2,k} = \sum_{k=1}^{n} (-1)^{2+k} a_{2,k} \det A_{2,k}.$$

Exchanging rows 3 and 2 and expanding in the second row we get formula

$$\det A = \sum_{k=1}^{n} (-1)^{3+k} a_{3,k} \det A_{3,k},$$

and so on.

To expand the determinant $\det A$ in a column one need to apply the row expansion formula for A^T. □

Definition. The numbers

$$C_{j,k} = (-1)^{j+k} \det A_{j,k}$$

are called *cofactors*.

---

[2] In the case when $a_{1,k}$ is the only non-zero entry in the first row it may be tempting to exchange columns number 1 and number k, to reduce the problem to the case $a_{1,1} \neq 0$. However, when we exchange columns 1 and k we change the order of other columns: if we just cross out column number k, then column number 1 will be the first of the remaining columns. But, if we exchange columns 1 and k, and then cross out column k (which is now the first one), then the column 1 will be now column number $k-1$. To avoid the complications of keeping track of the order of columns, we can, as we did above, exchange columns number k and $k-1$, reducing everything to the situation we treated on the previous step. Such an operation does not change the order for the rest of the columns.

Using this notation, the formula for expansion of the determinant in the row number j can be rewritten as

$$\det A = a_{j,1}C_{j,1} + a_{j,2}C_{j,2} + \ldots + a_{j,n}C_{j,n} = \sum_{k=1}^{n} a_{j,k}C_{j,k}.$$

Similarly, expansion in the column number k can be written as

$$\det A = a_{1,k}C_{1,k} + a_{2,k}C_{2,k} + \ldots + a_{n,k}C_{n,k} = \sum_{j=1}^{n} a_{j,k}C_{j,k}$$

Very often the cofactor expansion formula is used as the definition of determinant.

Remark. Very often the cofactor expansion formula is used as the definition of determinant. It is not difficult to show that the quantity given by this formula satisfies the basic properties of the determinant: the normalization property is trivial, the proof of antisymmetry is easy. However, the proof of linearity is a bit tedious (although not too difficult).

Remark. Although it looks very nice, the cofactor expansion formula is not suitable for computing determinant of matrices bigger than 3×3.

As one can count it requires more than $n!$ multiplications (to be precise it requires $\sum_{k=1}^{n-1} n!/k!$ multiplications), and $n!$ grows very rapidly. For example, cofactor expansion of a 20×20 matrix require more than $20! \approx 2.433\cdot10^{18}$ multiplications. It would take a computer performing a *billion* multiplications per second over 77 years to perform 20! multiplications; performing the multiplications required for the cofactor expansion of the determinant of a 20×20 matrix will require more than 132 years.[3]

On the other hand, computing the determinant of an $n \times n$ matrix using *row reduction* requires $(n^3 + 2n - 3)/3$ multiplications (and about the same number of additions). It would take a computer performing a *million* operations per second (very slow, by today's standards) a fraction of a second to compute the determinant of a 100×100 matrix by row reduction.

It can only be practical to apply the cofactor expansion formula in higher dimensions if a row (or a column) has a lot of zero entries.

However, the cofactor expansion formula is of great theoretical importance, as the next section shows.

5.1. Cofactor formula for the inverse matrix. The matrix $C = \{C_{j,k}\}_{j,k=1}^{n}$ whose entries are cofactors of a given matrix A is called the *cofactor matrix* of A.

Theorem 5.2. *Let A be an invertible matrix and let C be its cofactor matrix. Then*

$$A^{-1} = \frac{1}{\det A}\, C^T.$$

[3]The reader can check the numbers using, for example, WolframAlpha

Proof. Let us find the product AC^T. The diagonal entry number j is obtained by multiplying jth row of A by jth column of A (i.e. jth row of C), so

$$(AC^T)_{j,j} = a_{j,1}C_{j,1} + a_{j,2}C_{j,2} + \ldots + a_{j,n}C_{j,n} = \det A,$$

by the cofactor expansion formula.

To get the off diagonal terms we need to multiply kth row of A by jth column of C^T, $j \neq k$, to get

$$a_{k,1}C_{j,1} + a_{k,2}C_{j,2} + \ldots + a_{k,n}C_{j,n}.$$

It follows from the cofactor expansions formula (expanding in jth row) that this is the determinant of the matrix obtained from A by replacing row number j by the row number k (and leaving all other rows as they were). But the rows j and k of this matrix coincide, so the determinant is 0. So, all off-diagonal entries of AC^T are zeroes (and all diagonal ones equal $\det A$), thus

$$AC^T = (\det A)\, I.$$

That means that the matrix $\frac{1}{\det A}\, C^T$ is a right inverse of A, and since A is square, it is the inverse. □

Recalling that for an invertible matrix A the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det A} C^T\mathbf{b},$$

we get the following corollary of the above theorem.

Corollary 5.3 (Cramer's rule)**.** *For an invertible matrix A the entry number k of the solution of the equation $A\mathbf{x} = \mathbf{b}$ is given by the formula*

$$x_k = \frac{\det B_k}{\det A},$$

where the matrix B_k is obtained from A by replacing column number k of A by the vector $\mathbf{b}$.

5.2. Some applications of the cofactor formula for the inverse.

Example (Inverting 2×2 matrices)**.** The cofactor formula really shines when one needs to invert a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The cofactors are just entries (1×1 matrices), the cofactor matrix is

$$\begin{pmatrix} d & -c \\ -b & a \end{pmatrix},$$

so the inverse matrix A^{-1} is given by the formula

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

While the cofactor formula for the inverse does not look practical for dimensions higher than 3, it has a great theoretical value, as the examples below illustrate.

Example (Matrix with integer inverse)**.** Suppose that we want to construct a matrix A with integer entries, such that its inverse also has integer entries (inverting such a matrix would make a nice homework problem: no messing with fractions). If $\det A = 1$ and its entries are integer, the cofactor formula for inverses implies that A^{-1} also have integer entries.

Note, that it is easy to construct an integer matrix A with $\det A = 1$: one should start with a triangular matrix with 1 on the main diagonal, and then apply several row or column replacements (operations of the third type) to make the matrix look generic.

Example (Inverse of a polynomial matrix)**.** Another example is to consider a *polynomial matrix* $A(x)$, i.e. a matrix whose entries are not numbers but polynomials $a_{j,k}(x)$ of the variable x. If $\det A(x) \equiv 1$, then the inverse matrix $A^{-1}(x)$ is also a polynomial matrix.

If $\det A(x) = p(x) \not\equiv 0$, it follows from the cofactor expansion that $p(x)$ is a polynomial, so $A^{-1}(x)$ has rational entries: moreover, $p(x)$ is a multiple of each denominator.

Exercises.

5.1. Evaluate the determinants using any method

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & -5 \\ 6 & -4 & 3 \end{vmatrix}, \qquad \begin{vmatrix} 1 & -2 & 3 & -12 \\ -5 & 12 & -14 & 19 \\ -9 & 22 & -20 & 31 \\ -4 & 9 & -14 & 15 \end{vmatrix}.$$

5.2. Use row (column) expansion to evaluate the determinants. Note, that you don't need to use the first row (column): picking row (column) with many zeroes will simplify your calculations.

$$\begin{vmatrix} 1 & 2 & 0 \\ 1 & 1 & 5 \\ 1 & -3 & 0 \end{vmatrix}, \qquad \begin{vmatrix} 4 & -6 & -4 & 4 \\ 2 & 1 & 0 & 0 \\ 0 & -3 & 1 & 3 \\ -2 & 2 & -3 & -5 \end{vmatrix}.$$

5.3. For the $n \times n$ matrix

$$A = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & a_0 \\ -1 & 0 & 0 & \dots & 0 & a_1 \\ 0 & -1 & 0 & \dots & 0 & a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & a_{n-2} \\ 0 & 0 & 0 & \dots & -1 & a_{n-1} \end{pmatrix}$$

compute $\det(A + tI)$, where I is $n \times n$ identity matrix. You should get a nice expression involving $a_0, a_1, \dots, a_{n-1}$ and t. Row expansion and induction is probably the best way to go.

5.4. Using cofactor formula compute inverses of the matrices

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 19 & -17 \\ 3 & -2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

5.5. Let D_n be the determinant of the $n \times n$ tridiagonal matrix

$$\begin{pmatrix} 1 & -1 & & & 0 \\ 1 & 1 & -1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 1 & -1 \\ 0 & & & 1 & 1 \end{pmatrix}.$$

Using cofactor expansion show that $D_n = D_{n-1} + D_{n-2}$. This yields that the sequence D_n is the Fibonacci sequence $1, 2, 3, 5, 8, 13, 21, \dots$

5.6. Vandermonde determinant revisited. Our goal is to prove the formula

$$\begin{vmatrix} 1 & c_0 & c_0^2 & \dots & c_0^n \\ 1 & c_1 & c_1^2 & \dots & c_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & c_n & c_n^2 & \dots & c_n^n \end{vmatrix} = \prod_{0 \le j < k \le n} (c_k - c_j)$$

for the $(n+1) \times (n+1)$ Vandermonde determinant.

We will apply induction. To do this

a) Check that the formula holds for $n = 1$, $n = 2$.

b) Call the variable c_n in the last row x, and show that the determinant is a polynomial of degree n, $A_0 + A_1 x + A_2 x^2 + \dots + A_n x^n$, with the coefficients A_k depending on $c_0, c_1, \dots, c_{n-1}$.

c) Show that the polynomial has zeroes at $x = c_0, c_1, \dots, c_{n-1}$, so it can be represented as $A_n \cdot (x - c_0)(x - c_1) \dots (x - c_{n-1})$, where A_n as above.

d) Assuming that the formula for the Vandermonde determinant is true for $n - 1$, compute A_n and prove the formula for n.

5.7. How many multiplication is needed to compute the determinant of an $n \times n$ matrix using the cofactor expansion? Prove the formula.

6. Minors and rank.

For a matrix A let us consider its $k \times k$ *submatrix*, obtained by taking k rows and k columns. The determinant of this matrix is called a *minor* of order k. Note, that an $m \times n$ matrix has $\binom{m}{k} \cdot \binom{n}{k}$ different $k \times k$ submatrices, and so it has $\binom{m}{k} \cdot \binom{n}{k}$ minors of order k.

Theorem 6.1. *For a non-zero matrix A its rank equals to the maximal integer k such that there exists a non-zero minor of order k.*

Proof. Let us first show, that if $k > \operatorname{rank} A$ then all minors of order k are 0. Indeed, since the dimension of the column space $\operatorname{Ran} A$ is $\operatorname{rank} A < k$, any k columns of A are linearly dependent. Therefore, for any $k \times k$ submatrix of A its columns are linearly dependent, and so all minors of order k are 0.

To complete the proof we need to show that there exists a non-zero minor of order $k = \operatorname{rank} A$. There can be many such minors, but probably the easiest way to get such a minor is to take pivot rows and pivot columns (i.e. rows and columns of the original matrix, containing a pivot). This $k \times k$ submatrix has the same pivots as the original matrix, so it is invertible (pivot in every column and every row) and its determinant is non-zero. □

This theorem does not look very useful, because it is much easier to perform row reduction than to compute all minors. However, it is of great theoretical importance, as the following corollary shows.

Corollary 6.2. *Let $A = A(x)$ be an $m \times n$ polynomial matrix (i.e. a matrix whose entries are polynomials of x). Then* $\operatorname{rank} A(x)$ *is constant everywhere, except maybe finitely many points, where the rank is smaller.*

Proof. Let r be the largest integer such that $\operatorname{rank} A(x) = r$ for some x. To show that such r exists, we first try $r = \min\{m, n\}$. If there exists x such that $\operatorname{rank} A(x) = r$, we have found r. If not, we replace r by $r - 1$ and try again. After finitely many steps we either stop or hit 0. So, r exists.

Let x_0 be a point such that $\operatorname{rank} A(x_0) = r$, and let M be a minor of order k such that $M(x_0) \neq 0$. Since $M(x)$ is the determinant of a $k \times k$ polynomial matrix, $M(x)$ is a polynomial. Since $M(x_0) \neq 0$, it is not identically zero, so it can be zero only at finitely many points. So, everywhere except maybe finitely many points $\operatorname{rank} A(x) \geq r$. But by the definition of r, $\operatorname{rank} A(x) \leq r$ for all x. □

7. Review exercises for Chapter 3.

7.1. True or false